

Psych 220 Big Ideas in Psychology Part 1

Numbered Transcript

PSYC 220 • Dr. Roy Baumeister

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Lecture 1: The Need to Belong

At a glance — Baumeister on the need to belong.

- 1.1 All right. Well, welcome everyone to psychology's big ideas. I'm teaching this. Most scientists are specialists. They focus on one topic for their career. I discovered if I could write literature reviews where I go and read all the published work on some other topic, I could be a generalist. So I've roamed around in my career to a lot of different topics and the ideas. I'll summarize some of the things I've gotten from this in the coming lectures.
- 1.2 So today we'll start with one of the fundamental questions about the human mind. What are its basic driving forces? What are the fundamental motivations that shaped the mind? A century ago, Sigmund Freud said there were two—sex and aggression. I think he overestimated aggression. Sex is probably right, but maybe not exactly the way, and it's certainly not the main thing. I want to bring up a different one today, which is what I call the need to belong. That's a drive to form and connect, form relationships with other people and to maintain those relationships.
- 1.3 The idea is that this is deeply rooted in the mind, that everybody has this drive to connect with others, that we don't like being alone. It drives a lot of the other things that happen in the mind. Now, we use the term need, which is, in psychology, people talk about needs and wants. Need is a higher bar than, say, just a want. Need has to be something that you really have to have. I mean, to go back to Freud's example of sex, well, sex is a want. It's a very strong one, in many cases, but you don't really need sex.
- 1.4 It's possible to live a long, happy life without having any sex at all. Not that a lot of people would want to, but some do for religious or other reasons. You can live without sex, but without belonging, this, not so well. When we first published this work, we had to prove that it was not just something that would make people unhappy if they didn't get, because sure enough, lonely people tend to be unhappy, but it was actually bad for them.
- 1.5 And yes, it underlies lots of health effects, that people who are alone in the world, in one early review said, mortality from all causes of death are higher among people who are alone in the world. The medical establishment resisted the idea that that matters that much, even, let's say, people who visit you in the hospital, the more people that visit you, the faster you recover and the more you get healthy. But that was definitely confirmed in lots of research.
- 1.6 So, yes, it affects how people think, how people feel, their emotions. Whenever you form a stronger relationship or a new relationship, that's often a big burst of positive emotions, positive feelings. And conversely, things that threaten or damage or end a relationship bring a lot of negative emotions. So it's one of the bases of emotional life. As I said, people think about it a lot. Even when they're by themselves, they're thinking

about, "What am I going to say to so-and-so, or what am I going to do with so-and-so tomorrow?"

- 1.7 And again, mental and physical health, very strongly tied to it as well. In terms of what exactly it is, this is one of those things the scientists push you and say, "OK, what do you mean? What is belonging? What is it exactly that people need?" So we thought, well, you need to be with the other person to have some interpersonal contact, and maybe you need a framework that you care about each other. That's not enough.
- 1.8 Now, we looked at the interactions and, sure, having interactions goes with belonging, but there's some negative interactions. It turns out a bad marriage is worse than nothing. So if you're fighting all the time or having conflict, that's not really good for you and not really healthy, either. So maybe we said, maybe it's positive interactions. But you see people who just enjoy being together without a lot of positive interaction. Sometimes look at an old married couple in a restaurant, they're not even talking to each other that much, but they're glad to be there together, or they watch television together, whatever.
- 1.9 So it came around to, it's positive or neutral; in other words, not negative interactions. But then the interactions aren't enough. There has to be this framework that you care about each other that extends into the past and the future. And we looked at cases where people have one of these and not the other, and it was very much half a loaf. So, for example, people in long-distance relationships, well, they valued the relationship, but they missed being together.
- 1.10 It's hard to imagine how it is now, because we have such good technology for talking to people we love who are far away. But in the past, it was often very difficult. Sailors on the submarines would go off, leave their wives and families, and there'd be no contact for months, because there was no telephoning or, once in a while, they might be able to send a letter. So they were glad to have the connection, but they missed the daily contact.
- 1.11 And then we looked for people who had the opposite, lots of contact but without the framework of care. I thought probably the classic example would be sex workers, prostitutes, very intimate interactions with a lot of different people. And they enjoyed that part. They like meeting lots of new people and so on, but they missed the connection.
- 1.12 In fact, economists, you know how economists are, they look at these things and say, "Well, a given woman could make more money working in a brothel because there are lots of customers coming through, so should I have more of those." But a lot of them prefer to move out into private practice, so to speak, so they could have repeat customers and have something more akin to an ongoing relationship. So again, either without the other is good, but the ideal is to have both the framework of we care about each other and positive interactions.

- 1.13** I wanted to move on then and start studying this. We embarked on lab studies to see, well, what would happen when people are rejected? If the need to belong is so basic and widespread, well, rejection means that that's not happening. You're not able to satisfy it, so that's got to be upsetting. If we look at the bottom line there, the theory was, rejection would cause a wave of emotional distress, which would then change behavior in a variety of ways. As you'll see, the theory turned out to be wrong, but it was a pretty plausible place to start. It made a lot of intuitive and theoretical sense.
- 1.14** How do we study a rejection in the laboratory? We devised a bunch of procedures. One is, we'd have group of people come in and talk to each other for 10 or 15 minutes. We'd give them a list of questions, so they get to know each other a bit. Then we'd say, well, for the next part, we need you to pair off. Everybody write down the names of the two people in the group you'd most like to work with. And so we collected all that information and threw it away.
- 1.15** And then we went around to each person randomly in a separate room and said it was a problem that nobody picked you or everybody picked you, then said we wanted to have you work with one person who you chose and one person you didn't. But since nobody or everybody chose you, we got to have you do something else. So that was the pretext for it. So that was one rejection manipulation. You met a bunch of people, and none of them apparently liked you.
- 1.16** Another is, we give people a personality test and give them some accurate feedback about their scores on this or that. And at the end, we'd say, by random assignment, either, "Oh, something weird, two people with your personality profile tend to be alone in life as they get older. And you might have a lot of friends now, but once you're past the age where people are constantly meeting new people, your friends will drift away, and you'll find you're spending time more and more alone."
- 1.17** We had several control groups for that. One is, you'll always be surrounded by people who love you and care about you. So that's future belonging. That's good news. And of course, being told you're going to end up alone and not have friends, that's bad news. So we wanted to have another bad news that wasn't about belonging. They say, "Well, as you get older, you'll be accident-prone. You'll be spending time in emergency rooms, broken bones, things like that."
- 1.18** And then also just a no-feedback control. So again, that way we could manipulate feeling; it's not rejection now, but potentially there in the future. Another one we used a couple of times was, we have people come in and collect information, say, "Well, there's another person. You're going to get to know each other. Tell a little bit about yourself in the questionnaire. We'll take it to him, and we'll bring you his."
- 1.19** And then the experiment comes back and says either, "Oh, the other person suddenly had to leave after he looked at your questionnaire. He didn't want to meet you, and I can't force him to do that." So it's a personal rejection. Or, "He suddenly got a phone

call and had to rush off for a family problem or a medical emergency or something like that." So you're not going to meet him in either case, but in the one, it's something about you, other, something not about you.

- 1.20** And the last sometimes we did is just have people remember previous experiences when they were growing up, of being rejected or being accepted. So it gets them into that framework. I don't like that one as much because personal memories from a long time ago, you've probably rationalized it or figured it out or explained it in your own mind. It's not the same as suddenly finding out, "Wow, nobody picked you." It's not the immediate rejection. But anyway, they all work fairly similarly.
- 1.21** So, what happened when we started doing experiments on this? Well, we found plenty of behavioral effects, but we couldn't find much emotion. So that was a surprise. Again, the theory was rejection, emotion, behavior. So we got rejection and behavior, but where was the emotion? Maybe it's a delayed thing. Maybe we were doing it wrong, measuring it wrong. The reviewers and the journals couldn't believe that we didn't, that people weren't feeling bad, so it gave us all these different ways to try to measure emotion, but it just never showed up.
- 1.22** So, anyway, the change in behavior is good news in terms of advancing science. Let me describe some of the effects, but just remember, the emotion wasn't apparently part of it. For one thing, rejected people became more aggressive. They're more aggressive toward a new person who insults them, because that's the typical way you study aggression in psychology. It's hard to get people to be aggressive toward someone who doesn't provoke them. But they'd write an essay and somebody would criticize it and say, "This is really stupid."
- 1.23** Then the rejected people would be extra aggressive when they had a chance to deliver shocks or blasts of unpleasant noise or whatever. We did find, once or twice, they were even more aggressive toward a neutral person. Very unusual in aggression research. But if they'd been rejected, they were mad at the world, and a new person came along, they would be extra aggressive toward them. If a new person came along and praised them, then they were not aggressive. So they're open to making friends, but to anyone negative or even neutral, aggression goes up.
- 1.24** And again, no sign of increased frustration, anger, or anything like that. Just the change in the behavior. OK, aggression is antisocial. Let's look at prosocial behavior. Again, the rejected people showed big drops in how prosocial behavior—and this is paradoxical because you'd think if you were rejected, you'd want to become more of a nice guy to attract other people to come and be with you, form relations. But no, they turned less prosocial.
- 1.25** So we asked them for donations for the student emergency funds. These are students. No, they donated less. You usually have to give them some money. Say, "OK, you earned \$2.50 from your performance in the experiment, and would you donate any of

it to this fund?" "No, not today." So they were less likely to donate. I would ask them, "Would you do me a favor," the experimenter, "would you do another experiment after this?" No, they couldn't do that either.

- 1.26** I thought, all these things are somewhat costly. Maybe they're somehow being, conserving their resources or something. So let's pick up something totally easy. The experimenter was explaining the results and had a cup full of pencils on there and then accidentally knocked it over. So the pencils fell down on the floor. And then the experimenter got down and started picking them up, and they said, does the subject, the research participant, get down and help them pick up phones, the pencils, too? But no, the rejected people just sat there.
- 1.27** The other ones, the future belonging ones, they got down on their hands and knees and grabbed up a whole bunch of pencils—and the neutral, no-feedback ones, the same thing. But the rejected ones, no, they're not going to help. A prisoner's dilemma is another classic measure of people cooperating with others or mainly out for themselves. And there, too, the rejected people were more out for themselves.
- 1.28** So, again, behavioral effects. We looked at self-destructive behavior, how people screw themselves up. And one of the future lectures will go into much more detail because that's an interesting paradox. Why do people do things that mess themselves up? But one pattern is foolish risk-taking. So to go for high-risk, high-payoff things, that the expected gain is pretty low. Well, the rejected people are more likely to go for that. They'll take stupid risks.
- 1.29** We can measure choices. Would they take the easy way out or do something that's more healthy for them? We found they procrastinated rather than studying for an upcoming test in the lab. So all these things showed an odd increase in self-defeating, self-destructive behavior. So we got lots of behavioral effects of rejection. But why? Psychologists always want to know what's going on inside their mind. It should be emotion, but we're not finding any sign of emotion. So, there must be some inner processes. What are they?
- 1.30** We started in the next phase to look at what goes on in the mind among these people who've had one of these rejection manipulations. Well, one, it reduces intelligent thought. This, we came up very well, that being rejected substantially reduced their performance on an IQ test. So, suddenly, exclusion makes you stupid is a little bit of a strong way to put it, but you're not nearly as smart as you usually are. And it's worth knowing, if you take a test after you've broken up with your loved one or something like that, you're probably going to score worse at it.
- 1.31** Now, IQ tests have different kinds of tests, and not all of them were affected. The ones that were affected seemed to have to do with a controlled mental process as the automatic parts of intelligence weren't affected. For example, most intelligence tests have a vocabulary test. It's valid because smart people know more words than stupid

people. But rejection didn't affect your vocabulary. You still knew the meaning of the words that you knew. That didn't change.

- 1.32** But logical reasoning, any kind of thing where you're given some information and have to extrapolate or infer or impute or another way move on to something else, those things were interfered with. Those things went worse. So somehow, you sort the mind into automatic processes and unconscious ones, as the controlled processes that seem to be affected. And again, no signs of emotion are coming with this.
- 1.33** That made us think, all right, maybe it's the executive function of the mind that somehow shuts down a little bit when you're rejected. Well, self-control is a big part of that. But that would sort of explain why they're both selfish, as in not doing prosocial stuff, but also self-destructive and not really taking care of themselves. That could be something wrong with the executive function, the mind that's not working well, the free will, if you like that term.
- 1.34** Could we find evidence of that? Sure enough. One study, after people got the manipulation in the group, they were in the room and they said, "Oh, you taste these cookies." And we just really wanted to count how many cookies they ate. You can't tell people we're going to count how many cookies you eat because then they get very nervous. But if you say, "Well, we'd like you to— we're going to use these for some future experiment, so rate them on this questionnaire." And they can eat all they want. "I'm just trying to do a good job on the questionnaire."
- 1.35** Anyway, look at it. They ate almost twice as many cookies after they were rejected as opposed to the ones that everybody chose. I remember one person said, "I don't really like those cookies, but I can't stop eating them." There's some automatic response there. Another study we tried, having people drink something that tastes bad that was good for them. I think this is taking-your-medicine kind of study. It was actually a vinegar kind of drink, because vinegar is good for you.
- 1.36** And I think we offered them a small amount of money for each small cup that they would drink. Anyway, as you can see from the table, the rejected people did a lot fewer, drank a lot fewer of these. Again, they didn't taste good, but they were good for you. So they're not doing what's good for them, which is one of the things self-control is, is make yourself do stuff that benefits you.
- 1.37** Another is, how long do you keep trying before you give up on a difficult problem? Secretly, we made the problems unsolvable. They didn't know that. They thought they were solvable, and we said, "Well, keep working till you get them." Look, the other three conditions, they worked for half an hour, practically, on these problems, before they said, "I just can't get them." But the ones who were rejected, who were told you'd be alone in life in the future, they gave up after a little over 20 minutes, a substantial drop in their willingness to use self-control to push themselves to keep trying to keep getting better at this.

- 1.38** Another test. The great researcher, John Cacioppo, taught me about this. He had studied loneliness, and he said lonely people have some deficit in attention. They can't concentrate as well. It's a peculiar thing. The way you test it is, you put on headphones on the subject, and you have different information coming in one ear. One ear, you have somebody giving a speech. The other has a list of words, and they're supposed to ignore the speech and monitor the words and count how many words have the letter M or P in them.
- 1.39** So you've just got to keep your attention focused. It's self-control of your attention. And as you can see from the results here, too, the rejected people, the ones who thought they're going to be alone in the future, they got fewer right. They looked, again, like lonely people, even though they were just randomly chosen from the student population, that they were not able to focus their attention as well and ignore the irrelevant stuff and stay focused. So, self-control, self-regulation is impaired among rejected people. We found it thus across multiple studies and procedures. Again, emotion didn't show up.
- 1.40** All right, so why does self-regulation get worse? There are a variety of ways that self-control or self-regulation can go wrong. One, it depends on self-awareness. Maybe you're not as self-aware. Maybe it somehow uses up your willpower and makes you less able to do things. Maybe it somehow just disables the whole system by pulling the plug. Maybe it's somehow just disorienting, being rejected is like getting hit on the head with a piece of wood, and you don't know where you are for a moment or two.
- 1.41** And then there's one more theory we came up with, we call the implicit bargain. It's a little more complicated theory, but we thought we'd give it a try. So the idea is, why do we have self control? It's so that we can live together with other people in society. Well, it makes sacrifices. You use self-control to deny yourself things you want to do, to make yourself obey the rules, getting out of bed in the morning and going to work when you don't want to and so on.
- 1.42** So there's sacrifices to the self, but you're rewarded for it. It's the bargain. You get to belong to society, that society likes people who obey the rules and control themselves and so forth. So the sacrifices are essentially worth it. But then that bargain can break down on either side. People who don't control themselves, they tend to get rejected from society. They're more likely to get divorced or lose their romantic relationships, to lose their jobs if they screw up, even sent to prison if they break the law. So people who don't use self-control properly get themselves in trouble.
- 1.43** And by the same token, the people who feel that they're not going to be accepted, then they may say, "Well, what the heck? I'm not going to bother exerting self-control. Why should I push myself to be good if society doesn't appreciate me?" So it could break down on either side.

- 1.44 In fact, I had some colleagues who were sociologists, and they thought this work was really interesting because they say, well, out in society—they don't work in the lab one-on-one like we do; they look at the big patterns—but out in society, the groups of people who feel like they're not accepted and so on are more likely to break the laws and show poor self-control because, again, it isn't worth it to them. The bargain isn't there. They're not getting the rewards of behaving properly.
- 1.45 The point about this theory is, oh, this means they could if they wanted to. They just lose their motive to be good. Why should I bother being good if I get rejected? So I said, well, let's see. Let's give them a reason to exert self-control and see if they can suddenly do it. And sure enough, when in a condition when we offered them money, if they would get those words right, then they could, even though they'd been rejected. So it's not that the rejection made them incapable. It somehow reduced their push, their inner drive, to conform to the rules and to perform well.
- 1.46 So implication is, excluded people are still capable of self-regulation if they have a reason for doing so. So they're unwilling, not unable. So we thought, well, what if the self-control test, what if we tie that to being accepted again? Would they want to be accepted if being rejected was what upset them? That's possible. So for that, we use the Stroop task, the classic lab test of self-control. You see the two columns of words there.
- 1.47 And the assignment, of course, is, you're supposed to read off the color of the font, not what the word says. So it's very easy to do on the left column, green, yellow, red, because the word is the same as the color. But on the right, try to go down that and just read off the color. Well, you see red and you're supposed to say green. When you read the word red, it automatically activates the meaning of red. So you have to override that—that's the self-control part—and instead, substitute saying green. So it's a test of self-control.
- 1.48 And we told some people that this test is relevant to your future physical health. Why that would be, I don't know, but they believe it. And we told other people that it's a test of your ability to get along with other people, to understand them, read between the lines, and so on. And you'll notice in the results now on this, the low scores are good. So, it's the people who were rejected, and it was only about physical health, they got the usual effect, that they were worse at self-control.
- 1.49 But that's wiped out if they were told that this is a test of your ability to get along with others. They were really motivated, in other words, to prove that they did have social skills. So, self-control regulation is impaired among rejected people. They're unwilling, but they're not unable to self-regulate. And if you give them a motivation, whether it's money or getting social acceptance, then they could actually do it. So it fits that implicit bargain hypothesis.

- 1.50** Now, so they're rejected. They're being not so prosocial, more aggressive, and so on. But do they want to connect with others? If the need to belong is a basic drive, even though they seem to be behaving in counterproductive ways, they should still show some signs of wanting to connect with others. What we found over a series of studies is, rejected people are drawn to ways; they do want to reconnect, but they don't want to take a chance on being rejected again.
- 1.51** There's sort of two things going on in the mind. "Yeah, I want to be with people, but I don't want to have another rejection." So they'd be receptive to offers, but again, don't want to take anything where they put themselves out there. So, in one study, we were running this at Florida State, and after the rejection manipulation, we offered them the chance to buy a school spirit wristband that said they were a student at Florida State and show their pride in belonging to the university community.
- 1.52** Of course, they came to the experiment not expecting to buy anything, and they didn't really need it. But sure, the rejected people bought a fair number of them. Over half of them said, "Sure, I'll show my school spirit." Whereas the non-rejected people, no, not even 10% wanted to buy the school band. So that's one sign that, yeah, they're trying to connect, but notice it's a safe way. There's no way you're going to be rejected again. You're just showing your pride in belonging to the university.
- 1.53** Another one we did, this is a little fancier. We had to do this by asking people hypothetically, what would you do? Imagine this situation because you couldn't really have people snorting cocaine in the laboratory. So they imagined an experience. First, they recalled from their past either a rejection experience or an acceptance experience or a physical injury, sometime they'd gotten hurt, broken a leg, or something.
- 1.54** Then they said, all right, imagine you're at a party, and at the end, there's a chance for you to try cocaine. In one case, it's your friends are all snorting cocaine, and they're saying, "Come on, come on, join us and do it with us." In another, the friends left, and somebody forgot the cocaine, so you could try it just secretly, and nobody would know. But if you want to get high, it's there for the taking. Anyway, rates of wanting to try cocaine were pretty low in general. Of course, there's always somebody who says, "Yeah, yeah, yeah." But mostly, they didn't want to do it.
- 1.55** However, the rate went way up among the people who recalled a rejection experience, and they were imagining their friends are saying, "Come on, come on, join us. Try the cocaine." I mean, that's a dangerous illegal substance. And again, people are more willing to do it if there's a chance to reconnect with others.
- 1.56** All right. Still, what about emotion? The intuitive idea, people imagine these things, they think they would have a lot of emotion if they were rejected or if the future was going to be alone. We tried many different measures, couldn't sign it. If anything, there was less emotion after rejection than others. Then once in a while, we'd find a little bit of an emotion effect, but it would have no connection to the behavior, statistically. So

again, our theory, rejection, emotion, behavior, that wasn't working even when we did get emotion, which we hardly ever did.

- 1.57** Well, one thought was, maybe it goes to unconscious emotion. There are ways of measuring unconscious emotion that are kind of tricky, like have people group words together. Like, does puppy go with rhinoceros or parade? Well, they're both animals, so you could put it with the rhinoceros, but they're both happy, so you put it with parade; and so, you can see if people are grouping things by emotions or more by other features of the words.
- 1.58** Anyway, negative emotions showed no change there either. But we did find an uptick in positive emotions, positive unconscious emotional connections, which to me suggests how the coping process works. It's kind of intriguing that something bad happens, and your mind starts looking for happy thoughts, starts groping around for positive emotional connections, positive affect. And then so, at some point, you will feel better about the negative effects of rejection.
- 1.59** So maybe because, we all remember that being rejected, whenever it happened, was an unpleasant thing. But if you didn't feel anything right away, you wouldn't remember that. You'd just remember, after a while, it felt bad. But maybe it won't feel as bad if your unconscious has already been working to shore up some positive emotions.
- 1.60** The big paper did a meta-analysis of almost 200 experiments of looking at people's reports of emotion after exclusion. Well, there was a shift in the negative direction, but it didn't cross the midpoint. Most people come to a laboratory, and they're kind of in a good mood. Life is pretty good. And so there was less of that among the rejected people, but they didn't go over into reporting a negative. It's basically going just toward the zero point, as if your emotion system is shutting down.
- 1.61** We also found in that big study, rejection did not lower self-esteem, but acceptance raised it, which is a very unusual effect here in the next lecture, because bad things usually have stronger impact. But people have a lot of defenses for their self-esteem. So nobody in the group picked you. Well, they're all stupid, or something like that. Or I should have worn my other shirt. So people are good at protecting their self-esteem, but didn't show a downward effect, even from being rejected. So, that was leading us toward the idea that somehow it looks like the emotion system stops working when you get interpersonal rejection. Could that be?
- 1.62** And I went to a conference in Australia, and there were a couple of guys there presenting results from animal research, showing that animals that are excluded from the group, things like a rat will have a bunch of puppies, and one or two puppies get kicked out of the litter, and they become less sensitive to physical pain.
- 1.63** So there's some kind of analgesia. It's like the system reacts to it as it would to a physical shock, shutting down physical pain. So I wondered, well, could it be the same thing that's numbing out emotions as is numbing out physical pain? There was

this guy, Panksepp, very creative researcher. He said when animals evolved to be social—they already had organs, pleasure and pain and so on to deal with the physical environment—when they evolved to be social, they have to start dealing with the social world.

- 1.64** And instead of developing a new set of organs, they retrofitted the ones they already have to deal with social situations. So physical pain and interpersonal rejection might trigger the same kind of reaction in the system. It was at least plausible. So we thought, well, maybe rejection the way we're doing it in the lab, it sort of stuns you, and your body releases opioids or whatever that creates a numbness to both pain and emotion.
- 1.65** I mean, you can understand the evolutionary reason why it might numb out pain, as if an animal is fighting for its life in some desperate situation and gets hurt; it doesn't help it to feel "ow, ow, ow, ow." It's better to numb that out, at least for a while, so it can get to safety and then it'll hurt. Many of us, I think, have had the experience where you're playing a rough game and later on, you go, "What did I do to my leg?" But at the time, you didn't feel it. So that's what we started thinking might be what goes on, is that rejection creates this effect that numbs out both pain and emotion.
- 1.66** So I came back from Australia, and I told my lab people, "We got to go buy a pain machine." So we got one; it was something like this. No, it was actually a very small thing that you put on your finger, and it's snug at first, and then it just starts pressing harder and harder. And then you get two measures. They say, "At what point does it start to hurt?" That's the threshold. And, "At what point, turn it off. I can't take it anymore?" That's your maximum tolerance. We didn't have any clear theory about the difference between them, and we thought, all right, let's see if the rejection manipulations that we use affect those.
- 1.67** And they did. The people who were told, "Later in life, you're going to end up alone and spending more and more time by yourself," look, three or four times as much pressure before it started to hurt, before the initial registration. And in terms of tolerance, "I can't take it or turn it off," again, they took considerably more, over 100 kilocalories, or kilo whatever it is, pressure there before they couldn't take it anymore. So, yes, their pain tolerance went way up as a result of being rejected.
- 1.68** So then we thought, well, let's push this. Let's see, is the emotion system shut down in general? Does the rejection stop you from feeling anything? So we looked at a couple of the phenomena that are known in the research literature. One is predicting future emotions. There's a pretty good body of work that people predict how they're going to react to something in the future, and they tend to overpredict. They predict very strong, lasting reactions. In fact, what usually happens is, they have an initial reaction, but it goes away pretty fast.
- 1.69** So, I was at Florida State at the time, The big football rival was the University of Florida. Nobody outside the state, I think, really can tell the difference, but it was a big deal. It

was the last game of the season, and everybody gets psyched up for it. We had people predict, how happy would you be if our team wins, and how unhappy would you be if our team loses? It's on a 1-7 scale, where 7 is as happy as you could possibly be, and 1 is not at all happy.

- 1.70** If you look at the middle line there, the no-feedback control, that's the basic control. This is astonishing to me how much they think they care about football; 6.7 out of 7 maximum is the average. They're almost going to be as happy as they could possibly be, whereas 1.9, with 1 being the lowest, if we lose to the gators, it would just be miserable. But if you look at the bottom line, the future alone people, I mean, 4 is the midpoint, and their numbers are very close to 4. "Yeah, maybe it'd be a little bit nice if our team wins, but it's not a big deal at all."
- 1.71** So again, the emotional system is not working. They're not imagining the future and how they'd feel. They're saying, "I don't feel anything. Therefore, I don't really care." Some other findings here. Again, the reporting of emotion during the rejected state wasn't there either, as usual. And the pain tolerance and the threshold were correlated with their predictions. So the emotion system is not working. It cannot simulate how you will feel in the future.
- 1.72** So I thought, well, this might have interpersonal consequences. Maybe you can't empathize with other people. You can't mentally simulate how their emotions are going to be. So let's try that. So we had people read the story. It was adjusted for gender. So this is the one, if you were a woman, you saw this one; if you're a man, you saw the genders reversed.
- 1.73** It says, "Two days ago, I broke up with my boyfriend. We've been going together since our junior year in high school. We've been really close, and it's been great being at FSU together. I thought he felt the same, but things have changed. Now he wants to date other people. He says he still cares a lot about me, but doesn't want to be tied down to just one person. I've been real down. It's all I think about. My friends all tell me that I'll meet other guys, and they say all I need is for something good to happen to cheer me up. I guess they're right, but so far it hasn't happened."
- 1.74** So how much do you feel empathy for this person? Well, again, the future alone people felt a lot less. This is a measure for feeling sympathy, warmth, compassion, tender, and so on, toward the person who wrote the essay. So yes, clearly less empathy for this other person. Some of the actual responses, the no-feedback control, they said, "Oh, that's so sad." They said, "Oh, that happened to me, too." But the future alones said, "Tough shit." I mean, how clear can you be about not having any sympathy or empathy for that person?
- 1.75** And then, well, this being social science, well, maybe they don't want to empathize with the person because she was rejected, too, and you were rejected. And so you don't—that's a sensitive area. So we did another one. The story was, they broke their

leg and are having trouble getting her on campus and couldn't get a handicapped parking sticker and blah, blah, blah. So again, same thing, though. The people who were in the rejection condition reported less sympathy, less empathy, less compassion, and so on for this person.

- 1.76** Other findings, emotion again not reported there. They reported feeling nothing. Physical pain and the empathy were highly correlated, so rejection causes less feeling of pain and less empathy. And so you can't simulate another person's emotions. All right, well, this led to an even more ambitious project. The way we run our career in the lab was, when things are going well, we try crazier and crazier things. And then when the studies stop working, then we fall back on things that are more safe and sound.
- 1.77** So Nathan DeWall, one of my PhD students, came in and said, "Well, if it's the pain system and the rejection emotion system are connected, well, then maybe taking a painkiller will make rejection less painful." If you have other people mistreating you, breaking your heart or whatever, if you take Tylenol, that will reduce the pain. That's what I thought. That was a crazy idea. But go ahead and try it, and we put it together.
- 1.78** So we measured people over three weeks. They took one pill in the morning, one at bedtime. It was double-blind, so neither the experimenter nor the subject knew whether you were getting Tylenol or just a placebo pill that had no effect whatever. To measure their reactions, when people get rejected, they say, "My feelings are hurt." That's sort of a common expression. It doesn't mean necessarily you're having a wave of emotion, but you feel it in some way that your feelings were hurt. We had a daily measure of hurt feelings and a couple other measures.
- 1.79** So what happened? If you look at day one, of course, the two groups were the same. But by three weeks, the people who were getting Tylenol, who were getting a painkiller, reporting less hurt feelings. Their social self-esteem had also gone up, even though their academic self-esteem had not gone up. So yeah, the daily dose of Tylenol, twice-daily dose, seem to blunt their negative feelings about the minor rejections and slights that one suffers in the course of daily life.
- 1.80** Worth trying, find yourself in a position of romantic rejection or something like that, well, take a painkiller. It seems to reduce the distress to some degree. So no change in the placebo group. The hurt feelings declined over the three weeks. We also repeated this with a brain scan. I think people were playing one of these lab games where they got rejected by other people. Sure enough, the acetaminophen, the Tylenol, wiped out the activation of the pain centers in the brain. So again, taking the painkiller reduces the impact of interpersonal rejection.
- 1.81** Well, it turns out the marijuana has—the molecule has some resemblance to painkillers, which I guess is why many places are starting to legalize medical use of marijuana for people who have physical pain. DeWall, my student, went off and

became a professor at University of Kentucky. At one point, he was involved with this giant dormitory. So he had people give him questionnaires about their feelings over the course of the year.

- 1.82 Sure enough, he found out, this is no surprise, that the people who didn't have very many friends tended to feel lonely, but not if they smoked marijuana regularly. That seemed to prevent that effect—which some people say is useful information to know, if you go off to live in a foreign city for a while by yourself, well, you worry about being lonely, you manage to smoke marijuana, and don't get busted. But if it's legal, that reduces the negative lonely feelings, again, like Tylenol did.
- 1.83 So, being alone, being rejected, makes people become less sensitive to pain. We assume it's a temporary effect, slower to feel it in the first place, slower to say, "I can't stand it anymore." Large effects, and the pain medication will reduce this effect. So one more thing, does empathy mediate behavior? If your emotion system is shut down and you can't understand other people's feelings, maybe that's why these rejected people were curmudgeons and weren't getting along well with others.
- 1.84 If it interferes with your empathy, maybe that's why you might be more aggressive or less prosocial and so on. So, my group came up with this experiment. I really thought this would never work because, first of all, you had to be told you'll be alone in the future or you'll be surrounded by people who love you. And then you measured the empathy toward the romantic person dumped by her friend, and you're using the same essay from before. So you're going to be rejected by these.
- 1.85 Now, do you have empathy toward this person? And then will you donate money to the student emergency fund, after we gave them some money for what they had done thus far in the experiment? But it worked marvelously. Yes, the future alone people had less empathy for the person who had been dumped and donated substantially less money to the student fund. And those are statistically linked, so it was mediated. Those are part of the same process.
- 1.86 So, emotion does matter after all. There's a long quest for what's the role of emotion in rejection. It's not the way we thought; it doesn't set off an immediate wave of emotion to affect your behavior. But we need emotion to understand other people and connect with each other. When your emotion system is shut down because the rejection is numbing out its reactivity, then you don't connect with others as you normally would.
- 1.87 To conclude, what I've said thus far, the need to belong is one of the most basic and powerful drives in the psyche. We weren't the first to discover it, but everybody thought it was in the background. We moved it to the front and said, "This is one of the main things that drives human behavior, emotion, thought, and everything else." In the laboratory, we're mostly using unexpected social exclusion and rejections.
- 1.88 Big behavioral effects. More aggression, more antisocial behavior, less helping. Self-destructive behavior went up; controlled processes, reasoning, intelligent thought went

down, but we didn't find emotional distress. Self-regulation, on the other hand, does change if people don't control themselves as well; the executive function isn't working as well. Behaviors tend to be more short-sighted, impulsive, selfish.

- 1.89** It's not that they're unable to self-regulate. They just, with the implicit bargain idea, "If people are going to reject me, why should I bother trying to be a nice person, following the rules, doing my duty and all that?" And some of the behavioral effects may well come from this poor self-regulation. And then last, it makes the body less sensitive to physical pain when you're rejected. There's this, presumably, a release of opioids. We don't have the technology to measure what's going on inside the body.
- 1.90** But opioid release, like with a physical injury, that numbs it out so you don't feel the pain. And so you become insensitive to pain, and that's correlated with the insensitive to emotion. So the numbness in emotion goes along with it. And after rejection, that contributes to why people are not doing the prosocial things that you'd think they would if they want to go back and connect with others. That's it for now. Do we have any questions?
- 1.91** How does someone who's emotionally numb or shut off, what can trigger them being able to be open again to feel?
- 1.92** Well, these are temporary effects. I mean, we remember being rejected is unpleasant, but that came later. If it's a release of opioids, it's just a temporary reaction. I made the analogy of an animal that gets injured, breaks its leg or something, fighting for its life, and it doesn't hurt that much at the moment, but that night, it'll hurt a lot. So that does automatically come back. You don't have to work to make that happen.
- 1.93** Chronic rejection, it's sort of interesting what happens. We have one study that we never published, a bunch of different authors and huge data set and so on, that looked at people who had had rejection during childhood. They remembered being left out of things. And then compared how they felt as adults, how much pain from the same illnesses or injuries as other people at the same level. Obviously, statistically, it was very complicated.
- 1.94** But if anything, it looked like they felt worse as adulthood. So the long-term effect may be the opposite of the short. It's like, maybe you use up the system, the opioid system that gets you through the crises of being rejected as a child, but it ends up leaving you be more sensitive. The data weren't totally strong, but it really looked like that. Anyway, it's a good question, but it'll come back. Don't worry.

Lecture 2: The Weight of Negativity

At a glance — The weight of negativity.

- 2.1 All right. This lecture is going to focus on something I stumbled on in the middle of my career that I gradually came to think, this is one of the basic properties of the human mind and may apply to other animal minds, too. It may go back quite far in evolution. The basic idea is that the mind reacts to negative things more strongly than positive ones.
- 2.2 The scientific paper we published was called "Bad is stronger than good," because I like to be really plain. Other people talk about the effect as a negativity bias and things like that. That sounds like a psychological term, but "bad is stronger than good," you can remember pretty well. And the reactions will include behaviors, emotions, thoughts, any sort of reaction that it might have. And again, the idea is, those are intensified in response to negative things.
- 2.3 So, we're not alone in this in psychology. One of my good friends is a journalist, John Tierney. In fact, we wrote a book together about this. If you'd like to read it, *The Power of Bad*, he's a wonderful writer. But he said he noticed this all, right at the start in his journalism career, that everything had to be a crisis. If you wanted to get on the front page, it has to be huge disaster.
- 2.4 One of his first assignments was writing about a big traffic jam, and people are going to the beach, and it had to be presented as, "Oh, there's a traffic crisis happening there." The traditional line in journalism is, "If it bleeds, it leads." So, a story that's got blood in it gets featured. And the bigger perspective is somewhat surprising. Life generally gets better and better. If you compare where people were in the middle of the 20th century versus where they are today, they have a lot more money, better health. So many things are improved.
- 2.5 And yet, all during that time, there's a constant stream of books talking about this crisis or that crisis, or there's going to be this disaster. Very few are focusing on, really, the positive developments that have improved life. Also, I've been involved recently with the field of positive psychology, which says psychology shouldn't just study how to take mentally ill and other miserable people and bring them up to neutral. We should also help people feel good and to cultivate positive flourishing experiences.
- 2.6 Some people react to that and say, "Well, what took you so long?" Why did psychology spend its first century or whatever, focusing on the negative? And the focus is there. You can—studies look at textbooks and how many chapters are devoted to bad things versus good things, even how many words there are. It's abundantly clear there. But in a sense, you can understand why it took a while for positive psychology.

- 2.7 You can say, well, maybe psychologists just dwell on people's bad traits, or they enjoy reading about people being miserable or something like that. But a more positive spin is that it's harder to find good effects because the bad effects are stronger. If it's a new science developing new research methods, it's got to find big effects to shine through all the measurement error and error variants and all of those things. So, yeah, study the bad things first, and then when the methods get better, you could get around to come to the good things.
- 2.8 It's also a reason why we need more positive psychology, because the greater power of bad things for life to be good, we need a lot of good things to outnumber them. So there were some early signs of this. A couple of areas had noticed the greater power of bad things. In psychology, people who studied first impressions had gotten this, had noticed this. If you meet somebody new and you find out something bad about them, it contributes more to your overall impression than if you meet somebody new and you find out something correspondingly good about them.
- 2.9 So bad things, yes, they carry disproportionate weight there. They had done it. It was even looking more generally at how people look at each other. One way researchers have done this is, they did this immense amount of work of getting 500 different traits of adjectives about people, happy and friendly and dishonest and others, and rated them for how good or bad they were. Then they would say, OK, here's a person described by—they'd pick five of the words and say, "What do you think of this person?"
- 2.10 Well, you could average up the impact of those five traits, and that should give you exactly the impression of that person, right? That's totally logical—only it never worked that way. It was always more negative. The negative traits pulled the average down more than the positive ones pulled it up. So again, negative things had disproportionate weight in perceiving other people. So they had sort of noticed it in their domain.
- 2.11 Economists had also noticed what they called loss aversion. Basically, you're more unhappy about losing \$100 than you're happy about gaining \$100. The one is good, the other is bad, but the total effect is bigger for the negative one. And this affects how people make investments. You have to take some risk in order to get good returns, but people are more concerned about losing money than gaining, and so they don't invest optimally. Anyway, the economists had done that, too.
- 2.12 But those two had never put it together with each other; and then we were going to find a whole lot more once we got going on this. There we go. My own work, I talked already about the acceptance and rejection studies. When we had three conditions, acceptance, rejection, and neutral control, the control and the acceptance condition were often quite similar. The rejection was the one that was always very different. Somehow, again, acceptance, rejection should be opposite things, but the rejection has a much bigger effect than the acceptance, compared to the baseline.

- 2.13** And the same in other studies, people's success and failure, and look how they react to those. In the laboratory, again, failure would have a big effect, whereas a success, they would be pretty similar to the control condition. So there were signs there, but I hadn't really put it all together. But at one point, I started to wonder, is there a more general pattern? Would it apply to other things as well? So I said, well, let's start looking at other phenomena and seeing if this is true, if bad things have a bigger effect.
- 2.14** And I thought, ideally, we'll see where it's true and where it's not true. And that will tell us about why the effect is. That should make a more interesting theory. As it turned out, we couldn't find any places where it wasn't true, which was disappointing, but it was somewhat compensated by the idea that, this is so true everywhere, it must be one of the basic principles of how the mind works.
- 2.15** So, in research studies, some track, people keep diaries of how was your day and what went on and so on. And how do you feel? Well, a good day, you feel good; bad day, you feel bad. No surprise there. But the bad day carried over to the next day. You have a good day, it doesn't really statistically affect how you feel the next day, unless something really extraordinary happens. But usually, it's just gone.
- 2.16** But a bad day has a lingering emotional effect. If something bad happens, it carries over and so on. And bad events produce more emotions, adjustment, more changes in your thoughts and so on, based on these diary studies, than the corresponding good ones did. Another, instead of looking at simple day-to-day sorts of things, instead of looking at simple day-to-day sorts of things, look at major life events.
- 2.17** This was set in motion in a classic study published in the 1970s, where they looked at people who had really big things happen, and they'd won the lottery, and suddenly got a whole lot of money, or they'd been in some accident and gotten crippled. So big events, big life changes, and of course, the one felt good and the other felt bad. But over time, they sort of converged back to neutral, and to some extent, they were over it a year later.
- 2.18** So, it came up with the idea of the hedonic treadmill, that your happiness is kind of on a treadmill. You keep moving forward, but you end up in the same place. The point was also made later very persuasively to me by research on long-term happiness. If you ask people, "Why are you happy or why are you unhappy?", they'll give you an explanation of things that are going on. "My boss is a jerk, or my girlfriend is wonderful, or I found a new place to live." Good and bad things in their lives.
- 2.19** Well, OK, that makes sense. But the researchers tracked these people for 10 years, and so all those things have changed. You had a new boss and so on. But they were still at roughly the same happiness as they'd been 10 years ago. It was kind of in the person. So the people were happy about one thing. Ten years later, they were happy about something else; and the people who were miserable about one thing, they were miserable about something else 10 years later.

- 2.20** But it tends to be a fairly stable aspect of life, which is, by the way, important advice on your choosing whom to marry. You want somebody who could say, "That was a happy baby." That's, they're likely to be feeling good all the time, too, unless you don't want that, then find somebody who was a miserable baby. Anyway, the hedonic treadmill was a popular idea and influenced there.
- 2.21** Finally, some German researchers said, "Come on, being crippled—you're not going to get over that as fast as you'll get over winning the lottery. You'll spend the money and then it'll be gone, and you'll find you have all these relatives you didn't know you had, who are looking for a handout and other negative things." So they did a huge study and tracked people over time. And sure enough, you adjust to bad things much more slowly than to good things.
- 2.22** I mean, you do adjust to some degree over time, and you get there. But the negative things have a much more lasting and powerful and long-term impact on you. They both have change. They both affect you in the short run, of course, but the good things wear off. It's kind of unfortunate. I call it the Woody Allen line, that we know if God exists, he's an underachiever. I always think of that in connection with this. It would have been nicer if bad things were off right away and good things stayed with you a long time. But it seems to be, we're designed the other way.
- 2.23** So bad changes take longer to return to normal. Trauma is a concept in mental illness and so on, is a single bad event that has a lasting negative effect on you. My point is, there's no opposite concept. Psychology does not have a concept of a single good event that will change you for a long period of time. Now, some people have speculated that certain kinds of spiritual awakening experiences or transformations, and I like that idea. I haven't been able to see anybody get data on that yet. So there may be some, but still, that's going to be a highly specific and fairly rare.
- 2.24** Whereas trauma is a fairly common concept. And so a single negative experience will have a lasting negative effect on your mental health and well-being. There's actually research on what happens to people when they do have something really bad happen that's traumatic. There's post-traumatic stress, which we've heard of, I think, probably a lot. There's also post-traumatic growth. Some people say, "I became a better person. I learned, I got stronger," and so on.
- 2.25** You take the large data sets, it looks like the growth is more common than the stress, the negative reaction. I mean, they both are real, but the growth is more common, but all the media and everything emphasize the negative. Marty Seligman, my friend who's the founder of positive psychology, said this is kind of an unfortunate thing because, in our society, we're creating the expectation that, when something bad happens to you, you have to be miserable for a while.
- 2.26** It has to really affect you. It is going to affect you, and you'll want some support and all that. But a lot of people really do get something positive out from it. And if we created

more of an expectation and recognition of that, it would probably be better for people in general. Sex shows some of the same patterns. One bad sexual experience can really haunt people for a long time and interfere with their sexual pleasure and response and enjoyment. There's no correspondingly good thing.

- 2.27** And no matter how good you are, you're not going to give somebody such a wonderful sexual experience that it sets them up for a better sexual life the rest of their life or anything like that. So again, the bad things last much more than the good things. Indeed, when people study married couples and ongoing relationships, how much of the variance and how well they get along, how happy they are, well, good sex is good for the relationship, but it's a small plus. Bad sex is a really bad thing. It really counts for a lot of the negative in a marriage or other relationships.
- 2.28** So again, the good is good. I mean, let's not dismiss it. But the bad has a much more powerful lasting damaging effect. All right. And there's some recent research also that many couples grow apart in their sexual desires over the course of a marriage. So, one wants sex more than the other. And the mismatch tends to damage the marital relationship, the quality of the marriage, how satisfied they are with it. And it's really the decline in the one who wants it less, that that has the bigger statistical impact on the mutual happiness.
- 2.29** All right, relationships. Most people report to the researchers that their marriages have just gotten better and better, that their relationships are improving over time. The researchers look at the data, they can't really find any evidence of that happening. Now, I think it's partly because most relationships start out on a high note, right? You only get married if you're really liking each other and having a great time and so on.
- 2.30** And so what they find is, some relationships stay roughly at that level, and some go downhill, some get worse. So, in terms of your own love life and wanting to have good lasting relationships, the thing to do is to prevent the downward trends. You want to keep it going the same. Don't worry about ways to try to make it even better and even better. Focus instead on not letting the downward slide begin.
- 2.31** So, researchers have been studying relationships and what enables them to work. Couples do constructive things, and they do destructive things. You can say it like, if there's a conflict in the relationship, how do you deal with it? Constructive things. You can talk about it and compromise and deal with it that way, or you can at least hang in there and be loyal and not complain. Those are positive.
- 2.32** Negative things are more withdrawing, just not communicating, or going to look for somebody else, or picking fights or whatever. Those are negative ones. So, classify couples as to how much they do each of these different kinds of things. Well, looking at the health of the relationship, the destructive things have much bigger statistical effect on the relationship than the other. Yes, it's nice to compromise and nice to be loyal and

all that. But in terms of the total impact on the quality of the relationship, that's not nearly as important or powerful as the negative things that people do.

- 2.33** Sometimes I ask my students, "Why would someone marry you? What would make you a good partner, or a good husband or wife?" And they always pick up positive things. "I'm a good listener. I have a lot of empathy. I earn a lot of money. Good in bed, blah, blah, blah, sense of humor." But to be a good partner, the main thing is not to do the bad things. "I can hold my tongue and not say something mean when I'm angry, or I can not spend the family's money on some stupid thing and get us into trouble."
- 2.34** The destructive things have much more impact, and so abstaining from them should get top priority if you want to have a good quality relationship. Particularly powerful, statistically, is when one person does something bad and the other responds with something bad. Once they reciprocate it, then—obviously, that's two people doing something bad is worse than one person doing something bad. But it's not double. It's about five times as big, statistically.
- 2.35** So once you get into this spiral of one does this and another does something that's bad, that's going to get that downward progression in the quality of the relationship. That's very hard to pull out of, and it's certainly very hard to get it back to the good high where it was first. So, doing bad things is quite potent and negative; reciprocating bad things, even worse.
- 2.36** One of the great relationships researchers, Caryl Rusbult, that I talked to, told me once, "The reason marriage is hard is that, sooner or later, one is just going to be difficult and unpleasant for a period of time long enough that the other will start being that way, too." The other may start off being supportive, and come on, you get over it, and being nice and all that. But once the other starts going back, that really sets it on the downward spiral.
- 2.37** Back when relationships research was first getting going, there was a story of a bunch of researchers who were doing this thing, got together at a conference and one said, "Hey, I've got this great data set. I'm tracking couples over several years. We can see who stays together and who doesn't. So what do you think the strongest predictors are?" And they recorded all their predictors. What would make people get along well and have a good marriage?
- 2.38** And is it having the same political views, the same religious outlooks, the same hobbies, and so on? Well, they did it. One group suggested, which turned out to be the winner, was that it's the ratio of how often they fight to how often they have sex. It didn't matter the total amount of either one. It's just the relative proportion of them. This came to known as the Gottman Ratio. How many times do you have sex, divided by how many times do you fight?
- 2.39** The F rule for a polite company—I'll have to substitute one of the words. But it came to be known as five fornications for every fight. If you, in other words, have sex five times

as often as you have an argument, then that's a strong relationship that's likely to survive. If it's only twice as many, even though that's more sex, then it's pretty much doomed. So that's held up very well, and notice it's the ratio.

- 2.40** So it counts for couples who have sex five times a week and have an argument usually every week. Or it works for couples who have sex once a month and have two fights a year. But it's the same; it's the ratio that matters. When you put the focus on sex, it's not as if that's the magical key to a relationship, or maybe there's something magical about sex. But it's easy to count.
- 2.41** If you just counted how many positive interactions, how many positive interactions did you have with your spouse yesterday? I said, "Pass me the salt," and they did, does that count? It's hardly that. But people know how often they fought, and they know how often they had sex, so they can report these. Anyway, it's the overall ratio of good to bad. If it's 5 to 1, as we'll get back to in a few moments, 3 or 4 to 1 is usually about the break-even point. So 5 to 1, you're safely above average and doing well.
- 2.42** Other relationships, just be good enough. Comparing it to perfection and so on sets you up for disappointment. But be happy with what you have, and then mainly, to make the relationship work, avoid the bad things, the negative things. Other findings, laboratories, studies, and so on show that bad things make you learn more, make you think more, and so on.
- 2.43** I noticed the quarterback of the, I think, the Philadelphia football team, after he lost the Super Bowl in a very close game a couple of years ago, he said, "You win or you learn." I don't know if he made that up, probably is a standard line, but it was a good approach that, yes, something bad happens, well, you learn. Sure enough, they came back and won it a couple years later. So generally, negative information stimulates the mind to do a lot more thinking, produces learning, and so on.
- 2.44** Indeed, this is always this question in education—should you focus on punishing wrong answers or rewarding right answers? People want to think, well, wouldn't it be nice? We can get rid of bad grades and bad marks and just concentrate on positive feedback. But it doesn't work as well. People learn more from criticism than from praise. And now, criticism, punishment, and so on will have side effects. So, like, in raising your kids, if you have strict discipline and treat them badly and so on, they may resent you and have other negative effects. But in terms of learning and shaping the character, that works better.
- 2.45** There's even a remarkable study looking at not rewarding the learners, but the teachers. I think this was done in the Chicago system some years ago, and they were trying to get the test scores of the school kids up to a certain amount. And so they told some of the teachers, "Well, if your class, if enough of them reach this criterion, you'll get a, I forget how much it was, a several-thousand-dollar raise, but a nice substantial a chunk of money." OK, that sounds good.

- 2.46** The other teachers were told, "Here, we're giving you the same amount of money, but if your students don't reach this level, you have to give it back." So it's the same financial incentive, the same criterion, everything the same, except for one that was positive, one was negative. One is that, "Oh, you could get this bonus later." Others are, "Well, you got this bonus now, but you have to give it back if they don't do it well."
- 2.47** Guess whose kids learned better? The ones whose teachers got the money up front and didn't want to have to give it back. The threat of the negative thing happening motivated them to do a better job of teaching, as compared to the ones who had the chance of something good happening later on.
- 2.48** It's even true in the brain. This is not in my field, but other people have studied the brain and will show various stimuli, and you can show them pictures of happy things or sad things or disgusting things or whatever. Anyway, the brain, they measure the amount of reaction in the brain, the electrical activity. And there's clearly more activity to—there's more reaction to negative than to positive things. And that's well replicated and well documented.
- 2.49** And negative reactions seem to stay in the brain. So there's interesting studies with fear, for example. You learn to be afraid of something that's bad, but you can get over it in a sense, but it's still there in your brain. Your brain never forgets the negative things, so somewhat permanent. Turning from the brain to a more simple perceptual thing—people see a crowd of faces. The gaze converges on—here's nine different faces. One of them has an angry expression on his face, and people tend to notice that faster. If there's one of them having, say, a happy face, that doesn't zero in on that nearly as much. So it looks like your attention automatically will zero in on the angry face in the crowd, and again, showing this hypersensitivity to negative stuff.
- 2.50** Emotions. If I say, "OK, write down a list of a bunch of emotions, whatever comes to mind," and then you look at them, people have two or three times as many negative emotions as positive. And this has been replicated in multiple different languages. It's not a peculiarity of English.
- 2.51** There just seem to be more words for bad emotions than for good ones. In terms of managing their emotional states, you could think, well, what do you use self-control for? Emotion regulation. You want to get out of bad moods and you want to get into good moods. Well, people have a lot more strategies and spend a lot more time getting rid of the bad moods than they do trying to cultivate the good ones. So the bad has higher impact.
- 2.52** And so, they really learn how to handle that and how to escape that. Feeling better is nice, too, but it's sort of a bonus. Experiences, people have more memory for emotionally bad experiences than good ones. This is somewhat complicated because people do have defenses and defense mechanisms and so on, but it's usually in there.

And if you look at people's emotional memories, that, again, have a higher number. A bad experience does really stick in the mind quite powerfully.

- 2.53** I talked about the forecasting effect in the previous lecture. So people overpredict their future emotions, but they predict more strongly, more severely overpredict, for the bad ones than for the good ones. So for example, a study where there was this really desirable dormitory at University of Virginia, and everybody wanted to get in, and so they had to have a lottery for who was going to be allowed to live there.
- 2.54** And so they asked people, "How happy would you be if you got in? How unhappy would you be if you didn't?" So they predicted, "Oh, if I didn't get in, I'd be really crushed," and so on. In fact, of course, they got in or they didn't, and they pretty quickly got over it. So there wasn't a lot of lasting things. But they tend to think, "It would take me a long time to get over it."
- 2.55** Things like, "What if you lost your job, or you found out your spouse is having an affair and they left you, or things like that?" They'd say, "Oh, it'd be years before I'd get over it." Well, they feel bad. I mean, they're correct in terms of what emotion they feel, but they tend to get over it a lot faster. Anyway, the overpredicting, again, stronger effect, indicating the greater sensitivity to the negative feelings than the first ones.
- 2.56** So, another thing is making friends. Now, this is an interesting chapter in the history of social psychology. When social psychologists started to say, "OK, what causes people to come to like each other?", well, a bunch of them managed to get connected with an entire dormitory. I think it was just newly built at Stanford. It was for married students. So everybody was moving in new. They said, "All right, let's sign everybody up and we can see who becomes friends. And we'll keep track."
- 2.57** And they gave them all kinds of questionnaires, and they had their theories. Well, is it going to be having, again, the same hobbies, same interests, same political views or whatever? All these fancy theories that turned out to be the strongest effect was who lived near each other. If you lived on the same floor, you made more friends with those people than somebody up or down another floor. Even the people who lived by the stairs, if I recall correctly, they had more friends because they bumped into people more often. There was no elevator. Being by the stairs, you saw more people.
- 2.58** OK, so propinquity, nearness, causes attraction. This was in all the textbooks. It was a classic study and so on. It was never replicated for about 20 years. Finally, 20 years later, somebody else said, "Well, we should do the same study again and just make sure it's really true." They had another opportunity, so they did; and they did replicate it. They got the same effect. But what they did differently that time is, they also looked at who became enemies. And becoming enemies also depended on who was close to there.
- 2.59** And the enemy effect was bigger than the friend effect. So your next-door neighbors play on their stereo late at night or whatever, cooking stuff that smells bad, I don't

know. So, yeah, people are more likely than average to become friends with their next-door neighbor than somebody upstairs, but they're even more likely to become enemies to their next-door neighbor. So that being close to each other, it just multiplies the effect. It doesn't push it in a positive direction, like, as I said, all the textbooks had said for 20 years.

- 2.60** Social support, we know, is good. People have talked about that, having people who care about you and give you encouragement and reward you and so on. Yes, that's good. But when they also ask for questions about, "Are people undermining you? Are there people in your social world who are working against you?", that has bigger effects. So, yeah, I mean, no denying that social support is a positive thing. Everybody wants that or needs that to some degree, goes with the need to belong. But having people in your circle who are working against you and so on, that has an even stronger effect on your well-being.
- 2.61** Parenting. Well, a lot of people wonder about what's the right thing to do? How can I be a good parent and so on, look for parental influences on intelligence? I remember when my baby was born, I asked the experts in intelligence research in the department, "What should I do?" I wanted them to tell me which Mozart sonatas I should play in her crib and everything. All they said is, "Don't drop her on her head," which was not very helpful. That was already my plan.
- 2.62** Do parents have some influence? Well, it turns out, parents do, but mainly in the negative direction. You can make your kids stupider. You can't really make them smarter. This is shown partly as well by looking at socioeconomic status. The behavioral genetics people are very sophisticated about how much genetic effects are. But what happens among the rich people and the good parents and all the other optimal things, the kid's intelligence ends up being whatever the genetic handshake allows the genes to shine through.
- 2.63** The bad parents can make the kid dumber. If they're abusive or neglectful or whatever, failing to stimulate the child, just plop them in front of the TV instead of reading to them, things like that—a bad parent can really damage it. And so, then the genetic prediction from parent to child is much weaker than it is for the more optimal or good parents.
- 2.64** Indeed, the great developmental psychologist Sandra Scarr remarked once on how mothers feel such competition to try to be the perfect mother, and they worry about, "Do I have to do everything right and manage everything," and so on. And Scarr said, if you look at the data, if you're in the top 95%, that's about all you can do. You just try to be a good-enough mother. The bottom 5%, again, abusive and neglectful or something, that's bad for your kid, so don't do that.
- 2.65** But to worry about, "How much should I spend on the birthday party? Or do they need to have me read this book, rather than that book, and stuff?", that kind of stuff doesn't

seem to make a lot of difference. Just getting off the topic a little bit, the evidence suggests that the peer group often has a big influence on how the kids turn out. The parents don't have that much, unless they're among the really worst ones. So I think the good-enough mother message is rather reassuring, and you can stop beating yourself up and worrying about doing everything perfectly.

- 2.66** Information processing. People think more about bad news and bad information than good ones. So when something bad happens— bad news, bad stories, bad information coming in—people look and try to understand it. Remember I mentioned the the angry face in the crowd and so on. They also think more about unexpected information. But people think most when something unexpected and bad happens. You almost cannot stop yourself from, "What caused that? Why did that happen? What is wrong?" And so on.
- 2.67** The mind automatically gets going on that. Something unexpected good happens, you don't really question it that much. You might some, but it's not nearly as thorough an automatic response there. Morality is another thing where you can see the greater power of bad. I mean, a liar isn't someone who lies all the time. You only have to lie now and then to be called a liar.
- 2.68** And morality researchers have done things like, how many good deeds are needed to offset a bad one? Suppose somebody killed someone. What would they have to do to rehabilitate themselves? So suppose they went out and saved one life; would that make up for killing someone else? Well, no. How about 10 lives and so on? But anyway, for some bad deeds, there's no answer.
- 2.69** And one of the hypothetical questions is, suppose Adolf Hitler came back and decided he wanted to be a good person. What would he have to do to get into our good graces? Well, that's a lost cause. But for lesser things, people can think of something, but again, it's a lot more good needed to offset just a few bad things. And again, you get labeled as a bad person morally rather easily, and it's much harder to shake off.
- 2.70** Stereotypes. There are good stereotypes and bad stereotypes. Research shows the bad ones form faster than the good ones and are harder to disconfirm. It takes more contrary evidence to overcome it. So, yeah, the stereotype formation, where you just come up with a general impression of some group and apply it to them, it can be either good or bad, and the mind does work that way. But again, it works faster and more strongly with negative things than positive things.
- 2.71** And people's reactions to feedback, some of the work we did in my lab, we looked at, well, do people try to make themselves look better, self-enhancement, or do they protect themselves against failure? And sometimes they're the same thing. You want to do a good job on the test, so you're protected against looking stupid and you're able to look smart. OK, those are together. But should you take a chance on going out, doing some public performance?

- 2.72** I remember walking with a colleague through Amsterdam, and they had one of those great big chess sets that's built into the ground, and the pieces are this high, and people are playing the game while others are watching. My colleague said, "Why would anyone do that?" I said, "Well, it's cool. You challenge yourself." And he said, "Yeah, but you could look stupid." So he's showing the self-protection response.
- 2.73** "I don't want to take a chance on something bad happening," where others, especially people with high self-esteem or whatever, will, "Oh, yeah, I will take the chance. I would do it." Anyway, self-protection is a stronger motive overall. Both of them are real. People want to look good. They want to not look bad. But it's more important to not look bad than to look good, in many cases.
- 2.74** As I said, so there's a lot. There's even more in the original paper, if you want to read them. But as I said, our idea was to look for exceptions because then we would see when this applies and when it doesn't. That would give us some idea of why it comes about—only we just couldn't find exceptions. It just appeared to be true everywhere. Over the years since we published this, I've had people come up to me and say, "I think I found a case where good is stronger than bad." I say, "Yeah, good. I would like to see one."
- 2.75** But somehow it never seems to peter out. There have been three or four of those, and somehow it never seems to pan out. I'm sorry. So it just appeared to be a general pattern, which is disappointing. It would have been a more interesting theory if it were true here and not true there. And we even had all our own pet theories about, maybe in the future, good is stronger than bad. No, in the future, bad is stronger than good there, too. But it's compensated. If it's so pervasive, it must be one of the basic features of the mind.
- 2.76** And that leads to a couple other questions. OK, first of all, how much stronger is bad, bad things than good things? In our book, we called it the rule of four, roughly four. There's no exact amount. There was, for a time, a couple researchers tried to say that there was a very precise mathematical ratio of bad to good, of how much stronger bad things were. But then the math on that fell apart when some other mathematicians looked at it, and it isn't there. So it's not an exact number.
- 2.77** With money, it looks like it's about twice. Would you flip a coin where, if it's heads, you get \$10, and tails, you lose \$10? People aren't really eager to do that. They start being willing when it's about, if it's heads, you get \$20. If you lose, you only lose \$10. So it's twice as much. That's about the break-even point for money things. But in social things, intangibles, then the number tends to be a little bit better.
- 2.78** And again, as I said, it ranges from two to five. The Gottman ratio in relationships, do five things. And that's, of course, something to keep in mind in your relationships. You don't go, "I did something to bother my wife or my husband yesterday, so I should do

something nice to make it up to them." No, do four or five nice things to make it up for them. That's what will make the relationship work better.

- 2.79** So why is this? Well, what we came around to is, life has to win every day. Death only has to win once. So, yeah, you have to be extra careful about any possibility of any negative thing. I mean, food poisoning, a negative reaction to food, is a hypothetical example. Suppose you'd never had fish, and then someday when you're, say, you're 18 or whatever, for a week, they cook you fish for four days in a row.
- 2.80** And two of these are delicious, wonderful, some of the best meals you had, and two are really awful, and they make you puke and violently ill. So it's two to two. Is that break-even? No, you're never going to eat fish again if that happens. It's much more negative. And of course, it's very important for the way we evolved, living out in the woods, in the forest, to be careful what you eat and avoid eating things that will poison or punish you, because that's much more dangerous.
- 2.81** If you miss out on a few good things, well, it's too bad, but it's not nearly as bad as making the other kind of mistake. So, my assumption at present is, this is such a fundamental principle. There's even some signs it's true in animals. It's harder to measure all these things in animals, but it looks like they will try harder and learn faster to avoid bad things than to get a good thing. So when, again, the conditions we evolved, one mistake is much more costly than any benefit from one positive thing.
- 2.82** All right, let me go a little bit farther on some things today, some of the applications and implications of all this. So media certainly focused by emphasizing negative things. Lots of films about violence destruction and so on, lots of movies and so on, showing people having problems and conflicts and so on. Happiness doesn't really make good entertainment.
- 2.83** I forget who is the literary critic who wrote it, but he said, "Nobody ever wrote a great novel about a happy marriage." There have been a few that tried, and I read a couple pages of those, and you're quickly just sick of it. But reading about problems people have, bad marriages, stresses, and so on, that seems to be endlessly fascinating to the human mind.
- 2.84** Politics, too. Politicians focus on the negative—a lot more what's wrong with the other opponent, the person I'm running against, than focusing on what's positive about themselves. And periodically, they say, "Oh, that negative campaigning is bad. I want to be different," but the thing is, the negative campaigning works; depicting how awful your opponent is really will sway votes, and even for other things— getting people scared about there's going to be a disaster or there's going to be a problem.
- 2.85** I've been seeing this, I mean, of course, I assume most of us want to take care of the environment and pass it on in good shape to future generations. But the environment movement has just predicted one disaster after another, most of which have not happened. So there's some credibility, but that's what they do to get people motivated

and convince them they need to act in positive ways. "Oh, if you don't, it'll be a disaster."

- 2.86** Indeed, Jonathan Haidt and others have recently written about safety addiction, that people are more and more concerned about safety, which is ironic because life is safer than it's ever been in the past. Parents today, a lot of them are worried about letting their kids play in their front yard without an adult supervising them, as if they'll die there. And the chances of something bad happening, being kidnapped or killed or anything, are minuscule.
- 2.87** But the media report single cases. And in a big country, you can usually find something bad happening somewhere. And then this whips up this fear and appeals to people's bias toward negative things. And so, ultimately, people miss out on opportunities because of excessive concern over these threats, which statistically are really negligible. I mentioned the environmental movement, somewhat sacrificing credibility by predicting one disaster after another that have mostly not happened.
- 2.88** On the other hand, on the positive side, life is really good, and it's mostly getting better. I spent much of my career wondering, is things really getting better or worse? There's more pollution and crowding and blah, blah, blah. Well, there's both. Things are getting better, some things are getting worse, but more are getting better than are getting worse. It's clear, a positive trend there, despite the journalists and the books and the podcasts and everything talking about negative things.
- 2.89** So yeah, but psychologically, bad things are stronger than good. But life is good because there's a lot of good things. It's a force of numbers. Yeah, one bad thing will have a huge impact, more than one good thing. But you can have a lot of good things, and eventually, they overcome the few bad ones. So what we need to do, practically in society and in psychology, is help people cultivate how to—help you learn how to cultivate these positive experiences and good outcomes and so on. Even if, individually, they're not as strong as the others, they'll do better in the long run.
- 2.90** And again, the three or four to one is the break-even point. Where you can use the power of bad? Well, for your relationships, as I said, give priority to avoiding the bad things. For getting along with others, whether it's in marriage or in a group or whatever, reducing negatives, reducing conflict and problems and so on, that will do more for the total health of the relationship than cultivating positive things. But positive things are good, too.
- 2.91** But as I said, you want to cultivate a lot of positive things, if you have to outweigh a few negative ones. Delivering criticism, I'll come back to that in a moment on the next slide. And watch out for the safety addiction, which has really gotten out of hand. People are much more concerned about tiny, negligible threats that mostly have no chance.
- 2.92** So the term "crisis crisis", as my friend Tierney came up with this, from his career in journalism, he just said, it's always one crisis after another. And so, don't get carried

away by too much about any particular, the latest crisis. It's just the latest in the series, and it's probably overhyped and exaggerated to sell more papers or get more clicks or whatever. So, crisis is a permanent state of affairs. It's just, which one it is will change, and it's probably not as bad as they're saying.

2.93 In other words, they exaggerate it. So, it may be bad, but it's probably not as bad— and how people respond to it sometimes will even make things worse. So that's something to keep in mind when following the media on all the negative stuff that goes on. Covid was a recent and powerful example of this, about which he's written a number of columns. The American news media were much more negative in talking about Covid than the media in other countries.

2.94 There's probably a good reason. I don't know that it's that Americans are demented or pessimistic or anything like that, but it's a more competitive market. It's a big market. And so, to compete, you emphasize the negative, and that's what the media do. There are businesses to attract people's interest and attention, to get them to pay attention there. So the American media were more negative about it.

2.95 For example, when schools started to reopen in some places, mostly it was fine. It turned out, the children were not at much risk and so on. But what the news media did was run around to find a single case of some kid who got sick or whatever, make a big story about that, rather than report that, "Oh, look, most of the kids are fine. It didn't make any, it didn't have any problem. And they're getting back to school and learning again," which is the other part, is that the lockdowns of the schools really did lasting damage to that generation of students.

2.96 Some are suggesting they might never fully recover from that. And certainly they lost—I mean, it wasn't a total loss, but it was a year or two of learning that was really severely damaged during this. So this is his point—how you respond to it caused more harm than the crisis or the problem itself.

2.97 In New York, for example, they worried about running out of hospital beds. So the governor said, "OK, well, old people get sick, just take them to old people's home." Well, old people were extra vulnerable to Covid. So putting sick people in their old folks' home with them ended up killing more people. So again, the cure was worse than the problem. The solution did more damage than if they just let the epidemic run its course.

2.98 All right, what you can do in terms of coping with it, it's part of the mind, so get used to it, it's there. Avoid perfectionism. You're not going to be perfect, and focus on being good enough; that'll do it. Try to minimize the bad things, so that things are under your control. The first thing to do is, avoid doing harm or causing negative things. Follow the rule of four. Try to look for four positive things to overcome a negative one.

2.99 Remember the big picture. Your mind will overreact when something bad happens. But remember, life is mostly good and mostly getting better. So you can keep a positive

outcome, even though, of course, bad things do happen at some point to just about all of us. When they do learn from them, that's a natural response. The mind automatically learns when negative things happen. So take advantage of that, and then cultivate lots of small good things. Look for positive pleasures and even little stuff that you can do.

- 2.100** So using it, learning, as I said, learning automatically happens more when something bad happens. In relationships, minimize the negatives. Don't be fooled by the worst-case scenarios because that's what gets carried away, the power of bad. "Oh, this will be a disaster." It goes back in the environmental movement. I always remember the first, it started, I think, 1962. There was a book, *Silent Spring*, and it said, because of the pollution in the environment, pretty soon there are not going to be any birds. And that was a prediction. Well, we still have birds. But the worst cases got a lot of attention.
- 2.101** And make a point of celebrating the good things as well. Don't just mourn the bad ones. So yeah, the mind automatically will react more strongly to the bad ones. But OK, make a fuss over the good ones. In business, bad reviews have more impact than good ones. So some of the most successful businesses put a lot of effort into dealing with those tiny few customers who cause trouble or negative reviews.
- 2.102** I have an example of that in a moment. But bad customers, some of the businesses call them terrorists, who complain and post a negative review that can really damage someone. In the book Tierney did on this, he had a number of cases where some people were driven out of business by one bad review that wasn't even really fair.
- 2.103** Bad members of the team. If you're putting together a business, a group of people, to work together, it's more important to prevent one bad person to get in. One bad person will undermine the whole group, whereas one good person is not going to save a bad group. Some bosses have what they call the no-asshole rule. So when they hire someone, the first thing is to try to determine, is this person an asshole? Because they might disrupt my whole operation and do way more damage.
- 2.104** I guess there's a Russian saying, "A spoonful of tar will ruin a barrel of honey, but a spoonful of honey won't do anything for a barrel of tar." So a small amount goes the bad way. In delivering feedback, too, if you're the boss and you have to tell people how they did in the performance reviews, that's part of it. It's awkward for both people and so on, and what happens is, you don't want to say the bad things, and the other person doesn't want to hear them, so it's hard to get this done.
- 2.105** And so people equivocate, or they start with, "Well, there's a bunch of good things you did," and so on. But the person is listening there, "Come on, come on, come on." They're waiting for the bad news. Most people say they want to hear the bad news first. If you're the boss, you've got to deliver the feedback. Start out right away with the negative things, and then you realize they're not perhaps as bad as they thought, if that's all it is.

- 2.106** "You're not going to be fired or whatever, but you got to improve in this." Then the person is all ready to hear the good news because, OK, they survived the bad, and they're ready to hear some positive things, too. So it's not the way, intuitively, people want to do it. They want to postpone the bad news. But probably the most effective thing is to lead with that and then get around to the good ones.
- 2.107** In terms of reviews, there's a study that was actually done using these things. So these are two parallel reviews of an apartment building, and they're basically the same amount of information, and it was either posted on Pat's blog or on ApartmentReviews.com. So, which one is more credible? Well, logically, if you look at the negative one, it's kind of dubious. I mean, how can all the people in the building be mean and unfriendly? And if the rent is not affordable, how come you're living there in the first place?
- 2.108** So I would be skeptical of the second one. The first one is more credible. But people found the second one more credible, paid more attention to it. On the good one, it mattered also whether it was just on somebody's blog or was on a website. That didn't matter in the negative one; wherever it was, the negative review had more information. So, again, if you're running a business, the bad information seems to have more impact and so on.
- 2.109** And then we came up with suggestions in terms of running your life to deal with the greater power of bad. We call it the low-bad diet. Here, a number of suggestions. As I said already, forget about being perfect. Focus on being good enough and eliminating the bad things. That's a much easier thing to obtain. And we'll do more good in total than trying to get to perfection. In fact, on a recent project I did, I was just surprised at how many mental illnesses are correlated with perfectionism. Trying to be perfect is really mentally unhealthy.
- 2.110** All right, that's one. When something goes bad, focus on the big picture, learn from it. OK, I got a setback, but go on. Like the quarterback said, "You win or you learn." So learn from it, take a lesson, move forward, get on it. The media is full of negative information, bad news, and so on. So after you read something bad, go ahead and read something positive, and so on. There are a number of websites with positive information or humor and so on. The Onion, the Babylon Bee, and so on are very funny, satirical things. So make a point of looking for something pleasant, or share positive things with others as well.
- 2.111** Nostalgia, thinking about how nice things were in the past. Psychologists were very skeptical of this for a long time. But this guy in Great Britain, that's a great research program showing, no, nostalgia is mostly positive. "Yeah, that's gone, but I had a lovely time back there, and things were good there." It's one of the reasons older people are happier than younger people, is they don't have to worry as much about the future. There isn't as much of the future. They're more focused on, "Yeah, I had some good experiences in the past, and I can enjoy them and hark back to them."

- 2.112** And remind yourself, life is mostly getting better everywhere, all over the world. People are getting healthier and wealthier. More comforts, less violence, total violence in the world. It seems like it's high because of what we hear on the news, but it's much lower than it was in bygone eras as well. Wisdom and information are increasing as well.
- 2.113** So, to conclude, yeah, bad things have stronger impact than good things. But good can win, and life can be good, because there's a lot more good, even if each individual thing has a lesser impact—and that generally is what happens. Thank you.

Lecture 3: Evolved for Culture

At a glance — Evolved for culture.

- 3.1 All right. With this lecture, I'm going to explain what I came around to, gradually became the foundation of my thinking, my quest to understand the human mind. What is it? Why is it the way it is? And the term I use is, we are cultural animals, and I will unpack that in a couple minutes. But I want to explain the concept to you and why it's interesting and important.
- 3.2 As an example of issues to discuss, if we're trying to come up with a big-picture understanding, here are a couple standard findings in social psychology. Conformity—you all know the classic Asch studies, where you can see on the screen that A is the correct answer, but everybody else says it's B. Often you go along and say B, also. Why do people do that?
- 3.3 Or helping—people study how much money people donate to somebody in need. Well, they donate a lot more if they're identified than anonymously, if somebody's looking, if somebody's paying attention, than confidential or private donations. Or prejudice—people readily, quickly form stereotypes of outgroups. But today, people are trying to overcome those and not try to act on those. So why is that natural to form these stereotypes? And why are people today trying to do the opposite?
- 3.4 So the question is, how do we understand human nature? What is the human mind all about? Why do people think, feel, and act the way they do? So the idea is to come up with a big integrative picture. Now, sometimes I start by putting up a picture of something like this and having students try to guess what it is. And you won't be able to guess it. I used to do windsurfing, and this was a device for increasing the downhaul, so that the mast would sail properly.
- 3.5 And the reason I bring this up is that, when you try to understand what something is, you try to figure out, what was it designed to do? So this has the crank there, and you tie the rope there. But its function is a key to what it is. And so we take the same approach to the human mind. Let's try to understand, what was it designed to do? Now, some of the big-picture thinkers in psychology have used the term the social animal. I just like to say thinkers from A to A.
- 3.6 Aristotle said we were social animals way back a millennia ago. And more recently, the social psychologist, Aronson, had a series of books with the same title as social animal. So yes, that's correct; we are social animals. But does that really get at the essence of what makes us human? There are a lot of social animals. To say we're social animals, which is true, just calls attention to how we are similar to all these others. But it doesn't get at what sets us apart, what's special about human beings.

- 3.7** So that's why I moved to the term the cultural animal, which goes a step further. And I think that captures more uniquely what is special about human beings. Now, in social science, when you try to understand the big picture, people look either to nature or culture. And that's just a standard way of thinking, so they're often looked at as opposing forces. And so it's either, they're born that way or it was nurture, how they were brought up and how they were taught and what they learned, one or the other.
- 3.8** So how much of each? I mean, even if both are there. So this question is applied to all kinds of categories of people—geniuses, criminals, athletes, homosexuals, leaders. You see the same kinds of argument. Are they born that way, or did they learn to become that way? Or is there something about the socialization and culture that made them that way? Example of explaining people with culture. We focus on how you were brought up or your parents and your peers.
- 3.9** Culture is learned behavior, so it's not inborn. So somehow along the way, you had to learn it. The influences that teach you, people emphasize parents. I said there's been some reaction against that, that a lot of kids are more influenced by their peer group, but others as well, teachers, media, personal experiences, and so on, all that. It's what happened after you were born, essentially.
- 3.10** And there's some arguments that people can be shaped in widely different ways by culture, which is certainly true to some extent, but it doesn't explain very well what's similar about people everywhere. If it's culture, then different cultures should produce different results, unless there's some reason the cultures are all the same.
- 3.11** Now, conversely, let's turn around and look at nature. Nature explanations—you're born a certain way, you've got genetic traits— they typically emphasize evolution, that we evolved from other animals and from prior humans. Our patterns of thinking and feeling and wanting and so on were shaped by our genetic inheritance, from what worked in natural selection, as in producing more offspring.
- 3.12** Survival and reproduction are the main criteria. The purists say it's just reproduction, but survival is important, too. You can't reproduce very well if you don't survive. A lot of the basic desires for food and safety and so on seem to be rather similar everywhere. So that makes sense that they are somehow innate. If there were people who didn't desire food and never ate anything, they would probably not succeed all that well in terms of leaving a lot of offspring.
- 3.13** This works. It doesn't easily explain what is different about humans versus other animals. If you're looking, evolution looks for continuities, as well as about what's different about humans in different cultural settings. So the essence of the message I want to argue today is that the human mind was made by nature for culture. So the basics of the human mind were shaped by evolution, but they were shaped in a way that enabled us to participate in culture.

- 3.14** Instead of looking at nature and culture as two competing explanations, we're looking at them kind of working together and saying nature created us to take advantage of the benefits of culture. In terms of what produces the human mind, nature and culture influenced each other. It's not just that nature came first and then culture on top of that, but culture, as we became more capable of culture, it made changes to our biological hardware. So it was bidirectional.
- 3.15** So, in a sense, culture is our biological strategy. It's how we solve the problems of survival and reproduction. All species have to solve those or else they go extinct. We solve it in a very unusual way by organizing our social life with shared information and all the rest of what culture is about. Not too many other animals do that. Some of them have a little bit of culture, but nowhere near to the extent that we do.
- 3.16** And we point out, it's a very successful strategy, too. Our population has been rising and rising, even while most of the other mammals on the planet, their populations are declining. So life is temporary. Nature has not been able to make any living thing that will live forever. So the only way that species can continue is to live as long as you can and then reproduce and create new life that will carry on the species. Again, all species have to solve this, and we do it with culture.
- 3.17** Now, for context, it's interesting to say, psychology, there's hardly any psychology of trees. Why not? Why don't we study trees? Well, there are several obvious reasons. They're pretty simple in terms of psychological reactions. They don't do any behavior that we can talk of. There's a little discussion of plant behavior, like turning toward the sun now and then or something, but basically, no.
- 3.18** Inner psychological processes. I mean, trees don't have brains. They don't have emotions, as far as we can tell. Again, that's not much for psychologists to do. Trees basically survive. They take things from the environment. They take water, food, sunlight, and turn it into their own life, into thriving. The point is, what's there inside the tree is to help it get what it needs from the environment. That's got to be true of humans as well; what's there inside us is to enable us to get what we need from the environment, so we can survive and reproduce and continue life and prevent our species from going extinct.
- 3.19** All animals need things from the environment. No life is self-sufficient, and so they're built to be able to get these things. They need food, for one thing. So they have ways to find food with their eyes and their nose and so on. And hunger motivates them to go look for food and eat it when they can get it. They have mechanisms for extracting the nutrients from the stuff in the environment, the teeth, which chew things up, and then they swallow, and digestion.
- 3.20** Turns out that one of the beginnings of the brain's central nervous system was to help with digestion, from taking in nutrients and converting them into calories that you

could use. You want to eat, you also want to avoid being eaten. So there's fear and sensitivity, nose to smell dangerous animals coming around, too.

- 3.21 And then, as I said, no life lives forever, so you have to create new life. So animals have to figure out a way to reproduce, to make babies or whatever they call them. And there's the desire for that, the motivation to engage in whatever sexual activity will do it. And the organs, of course, that produce the gametes and so on and make that possible.
- 3.22 So, OK, now animals do have brains. So you could ask, well, why do they have brains? It's a big difference between animals and plants, of course. Plants don't really seem to do much thinking. But why did animals develop this organ, the brain? And then why does it evolve gradually to become more and more intelligent? And then why didn't they become even more intelligent? Why aren't there any squirrels running around with Einstein brains?
- 3.23 It's a simple way of thinking of evolution, that if two animals are competing for some food, then the smart one will have an advantage and it'll get it. And so it'll be more likely to live and pass on its genes. So all species should become more and more important, but that's obviously not what happens. So why is that? So think a little bit about the brain. It has, obviously, costs and benefits.
- 3.24 One of the costs is, it's extremely metabolically expensive. The usual statistic, they say the human brain is about 2% of your body mass, but it consumes 20 or 25% of the calories. So, way disproportionately a lot of what you eat goes to making energy for the brain. That's one of the reasons that we don't see the squirrels with Einstein brains, because he wouldn't be able to find enough more nuts or acorns to support that giant brain and would starve to death.
- 3.25 So the brain, to get bigger, it can only do it if it pays for itself by somehow bringing in more calories. Now, Robin Dunbar, whose work I'm relying on here, he had a classic line of research where he compared all different species, not even including humans, just comparing the others, and looking at the size of their brain. It's a proxy for how smart they are, and you always correct for body size. So it's the size of the brain as a proportion of the body.
- 3.26 And so what kind of animals have the biggest brains? And there are all kinds of interesting theories. Certain kinds of food might be more difficult. It's called the fruit-seeking brain theory because fruit ripens and then spoils, and so you have to keep in mind and keep updating in your environment. The bananas on this tree aren't ripe yet, but this one is getting ripe, and this one was ripe but some other monkeys came and ate it the other day. So you have to keep updating it. No, that didn't work.
- 3.27 Territory. If you move around a lot more, you have to remember where you were going and so on. A little bit of difference in the motor cortex, but basically no. Dangerous. You got to outsmart the predators who want to kill and eat you. If it's a more dangerous

environment, you need a bigger brain. No, that didn't work either. What he found was what he called the social brain theory. The animals with the bigger brains, proportional to body weight, had the bigger and more complex social networks.

- 3.28** So it's not for dealing with the physical environment, the trees, the dangers, the food, and so on. It's for dealing with each other that the brain evolves. So, and to the giant human brain, by extension there, it's there for advanced social interaction. It didn't evolve so that we could do more science individually or figure our way around the environment or anything like that. It evolved so we could deal with each other, develop more complex social systems, and in that way, produce more resources and be more successful.
- 3.29** One of the slogans in my career is "inner processors serve interpersonal functions." What's inside your mind is there to help us relate positively to each other. So let's look at animals in the social environment for a moment. Being social improves the animal's chances. Most animals are not social, it turns out. Most animal species are solitary, especially if you include all the bugs and so on. But at some point, animals learn to be social, to cooperate and hang around together. And there are a variety of advantages to that.
- 3.30** If they hunt in packs, they can get more food than if they hunt alone. I mean, a lone wolf can kill a rabbit and eat it, but it can't kill a large animal like a moose or something. But a bunch of wolves working together can kill the moose. So there's more food in their same environment if they're social, if they work together; but to be social, you need more brain. You need more social for the wolves, to coordinate, to work together to kill the moose and get more food for them. They need more than the solitary animals might require.
- 3.31** So think of some of the requirements that are needed just for the brain to be social. You have to recognize the others, the friends; you have to have some desire to be with them. Maybe this is the beginnings and evolution of the need to belong. The herd instinct, to be a cruder form of that. Herds just hang around together. There's already safety in numbers. But to coordinate, say, in a group hunt, you need that much more than that.
- 3.32** So you have to want to do things that others are doing. You have to have some, at least, hardwired form of cooperation, so you will work together and do somewhat of the same things. Obviously, in social life, there are always disputes. Who gets the best piece when the wolves have all killed the moose? Who gets the best chunk of it? So there has to be some way that they can resolve these disputes. Nature has sort of come up with aggression, that the bigger, stronger one gets the better piece and leaves the leftovers for the weaker ones.
- 3.33** Hierarchies often form. Many groups have a sort of a dominance hierarchy; and so, to get to the top, you have to have the ambition, the drive, to want to fight your way at

some point to tackle the number one, usually the alpha male, and to feed them. So it has to be some interest and desire in doing that. Also, maybe some degree of self-control to follow the rules of the group. If you're not the alpha male, well, of course, it's natural. You see food and you're hungry to go and eat it. But after alpha male beats you up a couple of times, you learn, "OK, I'll wait my turn, even though I'm hungry."

- 3.34** That's maybe the earliest evolutionary forerunner of self-control, is just refraining that to avoid getting beaten up. More complex social environments will have groups of rivals, alliances; a couple will work together, things like that. You see that in a number of species. But again, you're needing more brain, more psychology, to make a friend or have a partner. "I can't beat up the dangerous one, but if you and I work together, we could take them down." That sort of thing, but again, more complex.
- 3.35** And over time, more learning phase, the adolescence, the childhood where your behavior is highly amenable to change, that's going to require more care from adults. You got it for a while, so you have a longer period of learning after you're born. OK, as I said, humans are social animals, too. We have a lot of the same motivations, abilities, as these other animals. As I've said, we have the drive to connect with others and the need to belong, to form and maintain relationships.
- 3.36** Humans use aggression to resolve disputes. In the simpler past, we used it a lot more than we do now. Might makes right. Culturally, developed other ways of resolving disputes, but people still fall back on aggression sometimes, even in the modern civilization. We have more self-control than other animals, in terms of how much we can adjust our behavior to rules and things like that. Rivalries, friendships, alliances.
- 3.37** So we inherited these traits from the social animals. Maybe we carried some of them farther. But as I said, we are social animals. Now, the core is that, to be social, that's our strategy for solving the problems of survival and reproduction. And forming social groups already helped with that, but human evolution took this a step further. Culture is basically a better way of being social. It's an improvement, and it adds a lot of benefits to us, even in pure biological terms of surviving reproduction.
- 3.38** As I said, our species is flourishing, even while most of the mammal populations are declining. So, yeah, it's correct. Humans are social animals, but we've taken that to another level, and being cultural animals, too. And it works really well for us. So what do I mean by culture? Well, there are a lot of different definitions and so on, but I'm going to talk about a new kind of social system.
- 3.39** Ants have social systems and, as I said, wolves are social animals. They have groups, too, but there's usually one powerful one, lording it over the others and not a whole lot more. But we have much more advanced, complicated social systems, which are therefore more powerful, more flexible, can get a lot more done. Culture is based on ideas, on meaning, on information. Culture has a lot of shared understanding.

- 3.40** So it's not just that I know something and you know something too, but I know that you know it, and you know that I know that you know it. And so we can communicate and interact on the basis of shared understanding. That's a big advance there. Shared beliefs, also shared understandings of how to. This is one of the arguments I gather in anthropology. What makes a culture? Is it the shared beliefs that people all have, or is it how to get things done?
- 3.41** And I say, and the praxis people say, "Well, Americans don't all have the same belief. They argue constantly about this and that and have elections and all that." But praxis, you move to a new city and how do you get a job and how do you get an apartment? Well, there are certain standard ways of doing it. You apply for jobs with the system, and you rent an apartment by looking in these places, and you get your news this way, and if you're injured, you go to the hospital. So learning how things work in a culture, that gets things done. That's a part of the culture, too.
- 3.42** The purpose of the culture is to enable people to live together and satisfy their needs. There's this theory Marvin Harris came up with in the 1970s. I think his definitive book was in the '90s. He called it Cultural Materialism. I didn't like this theory when I first heard it. I'm kind of romantic. I don't think culture is all kinds of fancy things, and the fancy things are there, but he's right. The basic things all cultures have to do is enable the people to live together in some way that they don't all die.
- 3.43** They have to have water, for example. They have to set up where they have water and food and they can get it and share it through the group. And they have to be able to provide for the young, to produce children and so on, otherwise the society would be gone. Through most of history, there's a lot of conflict with enemies, with other groups. So the culture has to be able to defend and protect itself, one way or the other, against others who might want to come and steal all their food or kill them.
- 3.44** So we evolved the brain to do culture. Culture makes the brain a lot more powerful itself, the individual brain. I mean, just think of anything; think of mathematics. If you started out not knowing anything about mathematics and never talked to another human being, you wouldn't get far. You might figure out how to count or something, but that's not there. But now we learn fairly sophisticated mathematics that humans, over the centuries, have built up this common knowledge.
- 3.45** And so we can learn it in school, and we can do long division and calculus, differentiate equations and things like that. So the brain is able to do a lot more by being part of a group with all this shared information. So the brain evolved, as I said, to do culture, to enable us not just to figure out the physical world for ourselves, that's a small part of it, but rather to participate in the social world and make use of all the information that we can get from other people.
- 3.46** I put up here the analogy to computers. I'm old enough to remember when personal computers first came out, and they were just by themselves, and they were handy to

have. You could work on it, but there was no connection to other computers. There was no Internet. There was no communicating from one to another. You had something, you'd put it on a disk and take it over to another computer and put it in there.

- 3.47** But as the Internet appeared and started being there, computers started being better and better at connecting, and the early ones had all kinds of complicated things. We had to dial up something on a phone and put it in a little deal, whatever. But by the turn of the century, the computers were where you could bring the computer home and open it up and turn it on, and it would already be looking for the Internet connection.
- 3.48** So the way computers evolved to be better and better at connecting with the world of shared information, that's kind of an analogy for what the human brain did, that the amount of information you can get by yourself is so much more limited than the amount you can get by sharing and talking with other people.
- 3.49** All right. Why culture? As I said, it's a better way of doing social life. And with that, the brain can do things that it can't do by itself; and likewise, the social system can do things that it couldn't do with simpler systems. Ultimately, all leads back to the biological criteria of survival and reproduction that enabled us to succeed better at that. And as I said, our species has been very successful at that.
- 3.50** In 1800, I understand there were about one billion humans. Here we are, two and a quarter centuries later, and there are eight billion of us. No other mammals can match that. Maybe the dogs and cats, I'm not sure, but they're riding our coattails. So human mind evolved to take advantage of these things. So here are a few advantages I just want to point out that enables culture to contribute better to our survival and reproduction.
- 3.51** First would be language. You can't have a language by yourself. Language is something that is shared, so you need a group for it, and usually a fairly large group. Obviously, with language, you can communicate better, so you can share information. Some animals communicate a bit by barks and squawks and things like that, but no very complicated ideas can get expressed that way; whereas with language, all kinds of remarkable things can be done.
- 3.52** So improves the sharing of information; also improves the storing of information. If you have language, you can organize it in your own mind better, store it there, and retrieve it better. Thinking, you can manipulate information, logical reasoning, things like that, much better with language than without. And also, you can transcend the immediate present.
- 3.53** With language, you can talk about things that are happening far away on the other side of the world or the other side of the forest or in very distant places, or you can think about things that will happen in the future or things that happened in the distant past.

Most animals, we forget this, they basically live in the here and now, with very limited — they learn lessons from the past, that they remember the lesson. It's not clear they even remember the experience that taught them that lesson. But we can reconstruct things that happened long ago and simulate things that may happen in the distant future.

- 3.54 Second advantage of culture is the accumulation of knowledge. When a chimpanzee, that's our closest biological relative, when a chimpanzee dies, it has in its brain all the information that it learned from its own experience during its life, and a little bit from observing the few around it, but then that's gone. That information is gone.
- 3.55 With culture, the lessons stay in the group, not in the individual mind, so they outlive the individual. Only one person has to invent a product, and then it's in the culture, and then everybody has it. So you don't have to discover everything or invent everything for yourselves. And crucially, each generation can start where the previous one left off, so progress is facilitated by culture.
- 3.56 Wolves, as I said, are highly social animals. If you go up to Canada somewhere and find a pack of wolves living out in the wild, they probably live pretty much the same way a pack of wolves did 10,000 years ago. They haven't discovered democracy. They haven't started a free market of exchange. They haven't reinvented the roles of female wolves. They haven't done all these things. It's a very stable system. But with humans, as we accumulate social innovations, knowledge innovations, and so on, they're stored in the group, and then the next group, the next generation, can build on that. That's a huge, huge advantage.
- 3.57 Third, cultures have division of labor. That's a basic economic point, but it's indisputable it works. Different people do different tasks. So what that means is that everybody can become an expert at what they do. I've been a professor. I've spent my life at universities. I'm not sure I ever got my own dinner from nature. Maybe when I was eight years old, I caught a fish, my grandmother cooked it, but that would be the only time. If I'd had to feed myself from nature, I would have eaten a lot less. I'd probably be dead.
- 3.58 Same with building my own housing. But instead, experts take care of the food, they produce it, and other experts build the houses and so on. So everything gets done better. The quality of life thus improves by division of labor. Multiple evidence that that's true. It's axiomatic in economics. It's one of the foundational points, and Adam Smith wrote, this is a classic book, *The Wealth of Nations*, that started the field of economics. He looked at a pin factory, and it works better if different people do different jobs, rather than having one person do the whole thing.
- 3.59 There's a more recent study with flutes, which is more complicated than pins. In one factory, each person made the whole flute himself, and the other, they just divided it up on a kind of assembly line deal, and the division of labor produced a lot more flutes

at a lot cheaper price. Anyway, it's just better. Everything gets done more efficiently, more effectively, with division of labor, and life improves that way.

- 3.60** And last is trade, that they have a network of exchange. Matt Ridley in his recent book said, "Trade is 10 times older than farming," which is kind of astonishing. If you're farming, it's something like 10 or 12,000 years old. If that means trade goes back over 100,000 years, it's something our species keep on very early. No other species, as far as I can tell, does anything that really constitutes as trade. They don't have a marketplace, but it really works. It makes things better off.
- 3.61** It enables people to interact with others, even strangers, in ways that are beneficial to both. It took me a long time to realize this, but a marketplace is really an exercise in cooperation. We can think of the antagonistic things, because you're tempted to cheat somebody or the buyer wants a lower price and so on. But basically, the buyer wants to buy, the seller wants to sell. And so if they can find a price that's acceptable to both of them, they're both better off. And that's the definition of cooperation, an interaction that improves the outcomes for both parties.
- 3.62** And as I read the evolutionary literature, cooperation and communication are the two things that humans stand out as far beyond all the other great apes, and indeed all the other animal species in the world. So marketplaces, we learn to cooperate in this way by trading very early in our species history. There are even some arguments that that may be why we prevailed over the Neanderthals, because they didn't trade as much.
- 3.63** It's hard to reconstruct that, but you look at burial sites, and there's the stuff buried there. Is it of local origin, as it was from Neanderthals, which means there wasn't much of a trade network, whereas our Cro-Magnon ancestors would be buried with stuff that was from hundreds of miles away, indicating a bigger trade network. Anyway, economic history has also always confirmed that trade increases wealth. You look back through history, there are countries that engaged in a lot of trade, got rich and prosperous; the ones that didn't languished and went backward.
- 3.64** There's some, like that famous city Bruges in Belgium that was a big trading hub for a while. And then I guess the river got silted up and didn't function anymore for trade, and then its progress stopped. It stopped becoming wealthy. So, anyway, trade increases wealth, makes life better. And so, overall improvements there. Those are four big advantages to culture—a good reason why our species, once we evolved the brainpower to make a system like that, would produce the positive benefits in terms of just even survival and reproduction, of increasing the population.
- 3.65** To understand, that's not what I'm going to go in the next part, is apply this idea for explaining a variety of phenomena. Again, we're assuming that the driving force in human evolution, specifically us, where we departed from the other great apes, was capacities, traits, in order to do culture. In fact, at this point, I would go so far as to say, most and possibly all the distinctively human traits, the essence of human, what makes

us human, are the result of biological adaptations to enable us to do culture, to participate in these advanced social systems. These are the things that set us apart from others.

- 3.66** We can think, what traits would these evolve? Look at some of the psychological traits. Why would they be particularly suited for culture? Let's just go through a few of these here. Intelligence. Obviously, I already mentioned the social brain theory, that the bigger brain, presumably the more intelligent brain, among other animals, is there to enable it to have a more complex social life.
- 3.67** Well, with culture, there's a whole lot more. There's not just more social alliances and contingencies and relationships to keep track of, but there's a lot more information to understand. My assumption indeed is that we name ourselves *Homo sapiens*, which is Latin for clever dudes. And so why did the intelligence there—the early theories about why intelligence evolved was that, well, first we evolved this big brain, and then we started walking upright, so we wouldn't bonk our fancy brain into the trees and things.
- 3.68** And once we started walking upright, that freed up our hands. And so we started making tools, and pretty soon we had the iPhone and all the other stuff. But then they discovered some of the early humans walked upright, even though they still had tiny little pea brains. So that raised the question, what's going on there? Obviously, the intelligence did not come first. That wasn't the driving force. But they stood up. There are probably multiple reasons for this, but one is to free the hands to communicate. So communication may have come first.
- 3.69** Plenty of evidence. For example, the other apes can be taught to do sign language. We can't teach them to talk. They don't have enough, the vocal system isn't good enough for that, but they can learn primitive language to share in sign language. And so there's a pretty good case that earliest human communication was sign language, was through hand signals and so on. Once that was there, that created the selection advantage for the bigger brain, because we're sharing information now; there's more for the brain to do.
- 3.70** Remember, the brain has to pay for itself because of its huge needs for a lot of calories. So, once there's information being shared, there's more value in having a big brain. You're not just learning from your own direct experiences, you're learning from others and communicating and so on. Anyway, culture is based on information, as well as rules and norms and things like that, so you need a brain to learn all of those. So there's more payoff for being intelligent in a cultural social system than, say, if you lived alone.
- 3.71** Talking. All cultures have language, and this is obviously a distinctive human trait. Anthropologists love to find a culture that's an exception to everything. That's what makes their career. Nobody's found one that doesn't have language. And no animal society has language as far as they can tell. There is some communication, certainly, but in terms of what linguists would say, this is a real language, no. So it is one of the

essentially defining human traits, is language, but language requires a whole lot more capabilities.

- 3.72** To talk, you need the vocal apparatus that can make a lot of different sounds. They had to move the larynx down or something, which is actually bad. It increases your risk of choking to death. So it'll only happen if it's worth it because you can communicate better. And the advantage from communicating outweighs the greater danger, which again fits the idea that people were first communicating with hand signals. And then you just use vocal sounds to call people's attention and stuff like that.
- 3.73** And so we gradually switched, because a lot of people still use their hands when they're talking. Obviously, the hands are the extra aspect, and the talking is the main thing, but that required a big change. You need ears that can hear very small differences. If you have a dog, you say different words, "eats, east, please, sees, ears," the dog can't really tell the difference to those. You had to change the ear to be able to detect these subtle differences in sound. A variety of other changes go with talking.
- 3.74** So, yeah, there's, the grammatical brain, too, is more complicated in that you had to build in the infrastructure to understand how different sequences of the same words have different meanings. And so, we do that well. Even the smartest other animals, that seems to be beyond them, that different sequences of the same words have different meanings. So talking, language, again, a big adaptation, essential to being humans.
- 3.75** Theory of mind is a bit awkward term, but it means understanding that there are other people with mental states that might be different from my own. So theory of mind, which even the smartest great apes, they'll pass some simple tests of it. They'll infer what somebody else's intention is, if it's competitive, but not if it's cooperative. Somebody explained this to me once.
- 3.76** An ape goes into a room, and there are two buckets, and the ape knows that one of them has food in it, but it doesn't know which one. So it sees you come in from the other end. If you start running to this one, the ape figures out, "OK, that's the one with the food. I will run over there and beat you with it." But if you come in and you're not hungry and you say, "Oh, it's that one. Go eat that one. That's the one," that's completely beyond what the ape can understand. They don't know why you would be telling it where the food is.
- 3.77** So, limited understanding of the minds of others in other creatures, but very extensive one in ours, and we're constantly thinking, "I wonder what she thought about this, or why did he say that, and does she know that he knows?" And so on. So that's an important advance, and again, important for culture. We developed the capacities to infer what others are thinking and feeling, often becoming quite skilled at that.
- 3.78** We assume other people have inner states somewhat similar to our own, and we can tell similar and different ones. We can relate to those pretty well. And we also assume

we can communicate, and that I can say something and you'll know what I mean. And when you say something, I know what you mean. That, too, really helps us connect and work together in culture. All very useful for not just language and talking, but for cooperating, for doing things together, making deals, things like that.

- 3.79 The self. We'll talk about that in another lecture that we're on, but the human self, much more complex than anything like it in nature. Obviously, there's beginnings of selfhood in a fair number of animal species. But the human self, the way it identifies and understand itself, much more complicated and advanced. But again, that's because it's participating in a much more complicated social system, so it needs these advanced capabilities.
- 3.80 It's highly sensitive to how it's viewed by others. People are, it's called public self-consciousness. You worry about what other people look at you, and are you dressed properly, are you speaking, saying the right things, and so on? It's not just for minor stuff, but for forming relationships and getting on with others. For cooperation, you need a lot of trust. So people are very concerned with what can they do to get other people to trust them, which facilitates working together and so on.
- 3.81 The self invokes the roles you play in society. A lot of names come from jobs that presumably an ancestor did. And so each person, in division of labor, you seek a unique role in the group. You do your job, and other people do other jobs, and you identify with that. If we all do our jobs, then the system works, and people survive and reproduce, and the population thrives.
- 3.82 We maintain concepts of ourselves, extensive information, somewhat accurate, somewhat distorted sometimes. But people do maintain their understanding of what they are like, what are their strengths and weaknesses, where do they belong, what can they do, how do they fit in society, how good are they at these different things. And then, of course, in a complex culture, you change roles sometimes. You get promoted or you move up or down.
- 3.83 And so, the self is flexible enough to still be the same person, but changing roles, even home and work and other things like that, show again the power of the human self to operate in a very advanced social system. Basically, what the self is about is connecting the physical body, the animal creature that we all are, to the social system, to the group, the self. The purpose of the self is to connect the individual to the group. If you were a solitary being with no group, you wouldn't really need much of a self.
- 3.84 OK, one or two more. The mind, different terms for it, but basically, the two chambers, the conscious and the unconscious. This goes back at least to Freud, if not earlier. And so the mind definitely has both processes. Our thinking about it has changed so much since Freud, who looked at the unconscious as a dungeon where naughty thoughts were banished, so you didn't have to be aware of them. Now it's more like your support system working behind the scenes to make sure everything runs smoothly.

- 3.85** But in any case, we have the two systems. Kahneman called them System 1 and System 2, which is an interesting dodge of not trying to come up with a precise term to explain them, but probably correct. Most of the animals have what we call the automatic or unconscious system. Their consciousness is much more limited. So the human one is more advanced. What does it do? What are its functions? Well, people show, you can explore novel situations better, think through new things.
- 3.86** The automatic reactions are fine for familiar environments and for simple learning. But if you want more complicated things, you need to use these consciousness discussions, ideas. You're explaining why you did something. I mentioned the importance of getting people to trust you. So morality is a fundamental in most societies. Well, you need to learn the rules and follow them so that other people will trust you.
- 3.87** And if you did something that some people said, "Hey, you shouldn't be doing that," you have to be able to explain why you did it, why it was all right, consistent with the rules of the group. The conscious system is much more flexible than the automatic system. So for novel, unfamiliar kind of situations, people do a lot more conscious thinking then, whereas you can react out of habit and automatic response to a familiar situation.
- 3.88** Logic depends on consciousness, too, and it contributes to learning. When you learn tennis, you first pay a lot of conscious attention of, where do you put your feet and how do you hold the racket, and all those things? After a while, all those become automatic. So there's this big transfer from conscious to the automatic, so that's another advantage of having a conscious system to improve learning.
- 3.89** Self-control overrides one response to allow another. That's really essential in a culture. Cultures have lots of rules about what you should or should not do. You can't just do everything that you want, and so you use self-control to hold up some of your impulses that you shouldn't act on, that are against the law, against the rule, against the norms, and so on.
- 3.90** As overwhelming evidence shows, people with better self-control do a whole lot better in life in our society. It leads to more success interpersonally and professionally and financially, in health and everything else. So it's a powerful cultural trait, which is why I think it's much more advanced in humans than in others. Free will, I will have another lecture on down the road, but we're talking here about a more advanced form of controlling your actions than what other animals have.
- 3.91** People argue about the term, but I think everybody agrees that the human mental system for controlling action is way more advanced and sophisticated and really qualitatively different from what the other animals have. Capacity for rational choice, for thinking about reasons and exactly what is the best thing to do in multiple steps. Morality, free will strongly tied to morality in all the analyses, morality, again essential

for living in culture, to know the rules so that people will trust you and cooperate with you and so on.

3.92 Self-control is an important part of free will as well. If we want to go to trade, economic decision-making, do you want to make a deal? Is it worth it? How many fish are worth trading for, how many beaver pelts or whatever the exchange is? You need advanced capability for that, too. As I said, no other animals do trade or have marketplaces or anything like that. The ability to mentally think out when there are alternatives, to think out all the different ones and analyze them and pick the best one, again, that's much more advanced in humans than even our closest relatives.

3.93 All right, last few points. I've emphasized nature and culture together. They're to explain human psychology instead of saying, oh, it's one or the other, that they're sort of competing forces. Culture basically builds on nature, and indeed, nature created the basis so we could do culture. Things like, as I said, the ability to speak, which required changing how the mouth and throat are organized in some risky ways. That was effect of culture changing nature, that our biological makeup is different, so that we could be capable of language.

3.94 So nature prepared the traits useful for culture, and culture is natural for humans. And indeed, as far as we can tell back in history, people lived together in groups that share information and have language and understand some rules about property and sharing and so forth. To be sure, there is some conflict between them, nature and culture. Probably the biggest, I think, is selfishness, that it's just natural for every organism to be selfish, to look out for itself.

3.95 Whereas culture wants, sometimes you need to make some degrees of sacrifices, at least not taking what you want all the time. This can be as simple as something like waiting your turn, rather than going first to grab what you want, or going into battle and risking your life to defend your group against enemies, whether it's wild animals or other competing tribes or groups. So a lot of the conflicts in nature and culture revolve around selfishness versus doing what's best for the group.

3.96 I started this with the three social psychology phenomena and said why we can explain them. Let me just quickly revisit those, and then we'll be done. Conformity. Why do people give a wrong answer when everybody else has given it, even though they can plainly see that it's a wrong answer? And the helping and the prejudice ones. Well, for conformity, yeah, they'll say it because part of our way of getting information is from other people.

3.97 Most animals just rely on the evidence of their own senses to learn about the world. They don't really share a lot deliberately, but we do. Often, if everybody says one thing and you think something else, maybe they're right. It's worth that. And then also maybe you want to go along with the group and not stand out and maybe make somebody mad at you if you disagree with it. Anyway, we're naturally predisposed to

look to each other for information, and we rely on it in some ways, in ways that are sometimes excessive or paradoxical.

- 3.98** Second, the helping one. Well, in culture, it depends on how well other people view you, whether they view you positively or not. So you're going to an altruistic good deed. You're going to donate money for someone. You want to get credit for it. It's better for you. I mean, it's nice to do, and you might do it in private or anonymously anyway. But if other people are looking, you want to look like a generous person that others can count on, that they will share with you, because they think you will share with them. That makes you more appealing as a partner.
- 3.99** And so again, people are more selfish when they're in private and more generous and altruistic and helpful in public. And last, prejudice. People quickly form negative impressions of out-groups, but today in modern society, people are trying to not think in terms of stereotypes. On evolution, other groups were dangerous. So it made sense to be dangerous. They might be physically dangerous. They might have diseases that you wouldn't have a resistance to. They might be behaviorally dangerous. They might come to kill you and steal your spouses and your food and your resources and so on.
- 3.100** So we evolved, rather, in a context of very much ongoing competition among groups. So it made sense to have negative views of the others. But now, in a multicultural society, we want to all get along, and we need to overcome this natural tendency, so that we can look at others and be friends and treat them as equals and participate together, so that a society of much more mixed categories of people will flourish and function well.
- 3.101** And again, you need self-control to alter your reactions, which is one thing they showed, is that the non-prejudiced people, they know the same stereotypes. They just use self-control so that they don't dwell on them and think them, and then they replace them with egalitarian thoughts. So, point of the lecture was to provide a context for social psychology. What is the human mind designed for? What is its basic purpose?
- 3.102** I think the answer is to participate in this advanced social system, with multiple roles and interlocking sets of that and lots of shared information and norms for how things are done and all the right. So that's what separates us from other animals and explains a lot about human nature. And in the lectures to come, I'll be referring back to this as we look at one thing or another.
- 3.103** This is the starting point, is that the essentially human traits are the result of biological adaptations to enable us to benefit from culture. And again, it's worked extremely well. Our species is thriving while most of the other mammals are not. So in pure biological terms, we've hit on the best solution. Thank you. We have a few minutes for questions.
- 3.104** So what did humans get right evolutionarily that our social cultures have progressed, whereas wolves have not progressed in 10,000 years or more?

- 3.105** OK. Now, we can't say wolves are doing it wrong. They are still surviving. Their species has not gone extinct. But there is a sense of, just look at the population. I mean, biology measures success by the population. Ours is thriving, and theirs is probably going down to some degree. So we learned ways to work together in social systems that will enable life to increase. Some of the basic things. You have to be able to produce more resources than you could get individually.
- 3.106** Now, the wolves already have that to some degree by hunting in a group. Like I said, one wolf can't kill a moose, but a pack of wolves working together can do it. So they started in that direction, but we are much more advanced than that. Farming by itself is an enormous innovation, so try to imagine the human population without farming. We'd have to all be finding our food and killing it every day. So managing the environment to grow food produces more resources.
- 3.107** And we also share it pretty effectively. We have more advanced ways of sharing it than other animals. I mean, the wolves all dive in, and I guess the biggest one gets to eat first, and then the others get their share. But we have all fancy ways of—the government will take money from the productive ones and transfer it to the poor ones to help them. And the churches will take care of people, too. So we have better ways of spreading the resources around. But the first thing is, is producing a lot more resources.
- 3.108** We communicate better. We store information. As I said, they don't have changes that much. Human society is constantly changing. Now, multiple things go into that. We can imagine a different world, probably too much. We keep imagining how everything should be perfect and could be changed, and then you fix one thing, and another problem arises. But we can imagine it being different.
- 3.109** And again, we can accumulate information over time so we can store what worked and what is successful, so each generation can start where the previous one left off, and then we have progress. The wolves don't have technology. Technology doesn't come overnight. It's a long process of slow innovations, and each person making some small improvement that then the group uses, and then future ones build on that. Those are a couple of the key things we got right. Yes?
- 3.110** I was curious about culture and economics a little bit. You had said some phrases like culture being exchange, cooperation, progress. I think you said the line, you invent a product and it's part of the group. I'm curious about things like patents or exclusion, paywalls or whatever. Are those just forms of aggression that regulate the culture, or are those anticultural?
- 3.111** Well, I think those are designed to regulate between, to foster innovation, but encourage it. So the idea of patents, this is outside my—I'm not an expert on the law or anything, but the idea of a patent's, if you invent something, you should get the profits from it. If you, say, you're a genius and you come up with some great product, and as soon as you show it to the world, somebody else says, "Oh, I can make those a lot

cheaper and faster," and so they take it over, and you don't get anything from it, so then what's the incentive for you to come up with another invention? Somebody else is just going to steal it.

3.112 So the patent says, if you file a patent, then this is your innovation. It belongs to you for a couple years. I think there's a limited time, like with the drugs, they let the drug company— because it has to, it costs a lot of money to develop a new drug, and a lot of things don't work. So they do the research, they come up with something, they can't let it just be stolen right away. They have to recoup their investment.

3.113 So the government says, "Well, you can get the selective profits on the thing you invented for a couple years, and then that expires, and then it's in the public domain, and then other people can make it cheaper and furnish their version of it." So it's trying to keep the incentive to reward the innovator, which is why people put in the work and effort to develop things in the first place, but then also to ultimately to share it with others.

3.114 The markets do work by incentives, so there has to be some reward. In my view, it's why the communist kind of systems don't produce as much innovation, because you're not supposed to be allowed to benefit from the work, so why then do it?

Lecture 4: The Function of Emotion

At a glance — The function of emotion.

- 4.1 All right. Today, we're going to start by addressing one of the basic psychological questions—what is the function of emotions? Why do we have them? They've been shown, in another lecture I'll talk about self-destructive behavior, and a lot of emotions seem to increase people's tendency to do things that are bad for them. Well, that doesn't make any sense. Evolution would select against that.
- 4.2 So if that were the main thing, evolution would have gradually phased emotions out, but they must be doing something really positive and useful and good and adaptive for them to survive. Well, what are the functions? One thing is, they direct attention. Things that have emotional impact grab people's attention more. So the mind can pay attention to all kinds of different things. Emotion is one useful way to guide it.
- 4.3 Emotions do improve learning. There's plenty of evidence that people in emotional states, partly because they pay more attention, are more likely to learn, learn lessons, and store things for future. Liking and disliking really helps make decisions, that people who lack emotions have trouble making up their minds. Some people, because of brain damage or whatever, that have a lack of emotional reactions, and they can think through all the pluses and minuses of the different options, but to pick one is really difficult for them.
- 4.4 Emotions improve interpersonal relations. Negative emotions, a lot of them come when anything happens to damage a relationship. You think of anger, grief, or sadness, jealousy, and so on. Meanwhile, starting a new relationship or strengthening a relationship is often an occasion for a very positive emotion, so they serve that as well. But what about directly causing behavior? I think that's the most widespread assumption. The idea is anger makes you fight, and fear makes you run away, and all those things are somehow useful.
- 4.5 And so, I started out with that assumption, too, that emotions are there to directly cause behavior— only, I looked closely, that gradually fell apart, and I started to have to look for an alternative theory. So it's fair to ask, does emotion really cause behavior, especially in an adaptive way? As I said, if it makes you do stupid things or things you'll regret, that can't be the evolved purpose because evolution would select against it.
- 4.6 So does emotion cause behavior? Does it go the other way around? Does behavior cause emotion? Maybe then behavior starts to pursue emotion. You do things looking to get a good result, or possibly something else. Now, one of the classic debates about emotion is how much it's tied to cognition, thinking. There's an epic moment in, I think, back in 1980, when two leading psychologists published articles back-to-back in the field's top journal, *American Psychologist*, arguing opposite sides of this.

- 4.7 So one fellow, Richard Lazarus, he said, yes, of course, emotions are heavily based on cognitions. You can't have emotions without thoughts. It's how you understand things, how you evaluate things, what you think of is going on. That's essential to it. The other position was held by Robert Zajonc, who said, no, and his article was called, "Preferences don't need inferences." He certainly conceded that lots of emotions have cognitions in them, but he said there are some that aren't.
- 4.8 And particularly, he was looking at people's liking and disliking reactions. He would show people, and this is his research, show people Chinese characters, if they didn't speak Chinese, just sort of a shape, and ask how much you liked it or disliked it. And there are things like, if you'd seen it before, you liked it more, even though you didn't recognize it; you didn't know you'd seen it before. He said there's not really much or any thinking going on there. It's just, you look at it and you have a positive or negative reaction.
- 4.9 Anyway, the two opposing positions were stated. About 25 years later, I was tasked with heading a summer institute on emotions and decision-making with the economist, George Loewenstein. And George got a grant so we could bring in experts on emotion and come and talk to us, the two of us as senior people and about a dozen younger scholars and researchers. So we were talking about emotion in various aspects. And so all these people would come for a day, give a talk in the morning, and hang around with us.
- 4.10 I got to talk to them at length, and I reminded them of this debate between Zajonc and Lazarus. And they all agreed that this had been a productive debate, that this had been a good thing. And I asked them, was it still going on? They all thought that the debate had been resolved. So I asked them, well, who won? And they were about evenly split. So they all thought the matter had been settled at one side or the other, but they didn't agree with which one.
- 4.11 So what that made me realize is that there are two different phenomena that go under the name of emotion, and they were talking about different things, which is often the reason that any kind of scholarly argument will go on forever. We'll cover this in another talk about free will. When people are using different definitions, well, then the evidence only pertains to one side and not to how the other side is thinking about it.
- 4.12 So for one, what we understand is the full-blown conscious emotion. When you're really angry about something or you're really happy about something, you have a strong feeling and you know why it is. And of course, those are saturated with thoughts, because you know what went on to provoke the reaction; and you're thinking about the environment, understanding things, and so on.
- 4.13 What science was talking about, though, is more of an automatic reaction. Yeah, you see something and you immediately have a liking or disliking of it, a fraction of a second before there's time to do any serious thinking. So preferences— I like this, I don't like

this—you can have right away, or go to an art exhibit and look at modern art, and you can just look at things, and you don't really know what it is, but you like it or dislike it. You often have an emotional reaction right away.

- 4.14 So these are both real phenomena; and so, the different scholars, depending on which one they were focusing on, thought, yeah, the evidence had solidified the support for their argument, and hadn't really acknowledged the evidence for the other side. I spoke in a lecture yesterday about rejection and people's reaction to that. Remember, the conscious emotion was absent, but there were unconscious emotions.
- 4.15 People were not reactive to it, but again, the mind started making positive connections unconsciously, automatically, without the person realizing what was going on. So something was happening in the unconscious emotional system. And intuitively, people didn't know. When we asked them to predict the results of the experiment, they got it wrong. They thought people would be really upset after the overt rejection, and the unconscious one, they had no inkling that that was happening, which means to be unconscious.
- 4.16 So that was one sign to me that we need to understand both of these different phenomena and perhaps how they work together, if we're going to understand the function of emotion. Anyway, as I said, the standard theory in emotion is that it's there to cause behavior, at least to prepare for action. You'll see this in lots of books about emotion. They pretty much all use the same example of fear, and that fear makes you run away.
- 4.17 And that's useful if our ancestor didn't have fear, and then the sabre-tooth tiger came up, and then you'd say, "Oh, nice kitty," and pet him, and then you'd be eaten and dead. And so, the ones who had enough fear to run away and hide were the ones who survived and reproduced and passed their genes down to us. It turns out, fear is not a good representative of other emotions, and that's not even really how fear works in the wild.
- 4.18 Explained this to me, Jeffrey Gray, who's a researcher who'd studied animals and studied emotions in them. He said, OK, is this really plausible? The mouse sees the cat. The mouse thinks, "Oh, that's a cat. That's not good." And so then the body starts generating a fear reaction. The heart beats faster, and all the things that go with an emotional arousal, and then the brain sees the fear in the body and says, "OK, I guess I'm afraid. I'd better turn and run away." That's way too late.
- 4.19 If you go jogging in the woods or something, I do it with a dog, and the birds take off the minute they hear us coming. They don't wait around for an emotion to generate. They're sitting there. The fleeing has to happen instantaneously. It might contribute to it or reinforce it or something, but it's not the starting cause of the behavior.
- 4.20 I remember too talking about this with one of my friends, who's another scholar, and he said he was in Germany and he'd traveled to stay with his wife, who lived in another

city, and somehow the traffic lights made a different signal in the south and in the north, and so he thought he should go. He started going, and suddenly he realized the cars are going on both sides. And he said, he'd got to steer this way and that way. And he got to the other side and pulled off. And then he was terrified.

- 4.21** He said, "During the crisis, I didn't feel any fear. I was just totally focused on not having a crash." But the emotion came too late to do anything. And then, again, there are not very many other examples to fit this. I'll go with some of the criticisms that really made me break with the theory. But let me give you a foretaste of what the second alternative theory is. Emotion is a feedback system. It helps you react to what's happening now or, as I said, contributes to learning, so that you will store the message better and learn better in future opportunities.
- 4.22** So the slow, the full-blown emotion takes a while to develop, and that makes you think about things. It's even, in this age of artificial intelligence and robots coming and so on, and think of how a robot features the world. Well, as soon as the situation is done, it moves on to the next one. But emotion might keep us dwelling on what happened, so we continue to think about it, and are more likely to learn from it, to extract the lesson.
- 4.23** So in that sense, the purpose of the full-blown emotion is to say, "OK, this was good or this was bad, and in this particular way," and keep you dwelling on it, and so you end up learning more about it. And it leaves a little trace then in your psyche, so that the next time a similar situation arises, you have this immediate twinge of, "Oh, don't do that. That didn't work out so well." And you don't even have to redredge up the whole memory of what went wrong the last time. You just have an automatic, "Oh, that's good, or that's bad."
- 4.24** Ultimately, though, you do learn what actions produce what emotions, and you learn to anticipate the emotions. So in terms of driving behavior, it might be more, the emotions you expect to feel or not to feel, based on what you do, rather than the emotion directly causing the behavior. Indeed, we're going to talk about self-regulation fairly soon, but one of the questions is, a researcher who studied self-regulation, self-control for many years, why can't we directly regulate our emotions?
- 4.25** We can directly regulate what we do. We can directly regulate our thoughts to some degree. But emotions, it's really hard to say, "I'm going to stop feeling guilty." The answer is that the emotions are there to guide us. If you could just, by active will, erase feeling guilty, guilt would lose all its power to influence us. So you have to control emotions indirectly by not doing things that are going to end up making you feel guilty.
- 4.26** So, if fear was a good example of the first theory, of emotion directly causing behavior, guilt is a good example of the second one. You do something, then you feel guilty afterwards, and so you reflect on that and, "Oh, I shouldn't have done that." You learn your lesson, and the next time you're in that situation, the guilt makes you—the twinge of guilt, the anticipation—makes you not do that.

- 4.27 To criticize the direct causation theory, there are some reasons to say no, emotion is not really well suited for directly causing behavior. One, of course, it's slow. That was the point of the Jeffrey Gray example of the mouse and the cat. It takes a while for the emotion to develop, and action sometimes has to be done rapidly. It's also not that specific. As a number of researchers have pointed out, there are lots of behaviors, and there are not that many emotions.
- 4.28 The idea that the behavior is somehow contained in the emotion, that's not really going to make sense. They use the example of fear even. OK, maybe fear makes people run away, but in other cases, fear might make people sell their stocks, or listen to the radio, or do any number of other things. The behavior has to be somewhere in between the individual and the situation.
- 4.29 There could be maybe a broad approach and avoidance. So good emotions, this is one of the basic categories psychologists use to sort behaviors—so maybe positive emotions generally promote approach, and the negative ones promote avoidance. There might be something like that. But as I said, there's not much evidence for the idea that emotions directly cause behavior. Schwartz and Kohler, these researchers have been studying emotion their whole careers, said the immediate effects of emotion are more mental than behavioral.
- 4.30 Emotion has lots of effects on what you think, but in the psychological research literature, a lot fewer effects on behavior. I think I counted in one of their reviews, something like 20 pages of summarizing one experiment after another, showing emotion affecting thinking; but emotion affecting behavior, they had about half a page of that. As I said, just fear is really the cause of fleeing? Probably not. It's much too slow. The mouse has to run away as soon as it sees the cat, if it's going to have a chance to survive, because cats are faster and so forth.
- 4.31 Now, as I said, looking at the research literature, lots of studies find effects on thinking, not very many on behavior. The links to behavior are inconsistent, mostly generalized things; and sometimes they're destructive. There's no question that sometimes emotional states cause people to do things that are costly to themselves, even like getting into a fight with somebody who's going to hurt you.
- 4.32 Or the wisdom that, if you're upset, you can go ahead and write the email, but don't hit send until you stop being upset. So it might be useful to you to articulate your feelings, but don't act on it. Once you send it, it's out in the world, and it's better to decide that when you're not in the midst of some emotional state. So the destructive thing again means that that's probably not the evolved function.
- 4.33 And even when there is evidence that emotion leads to behavior, so you manipulate an experiment, say, some emotional state, and then measure the behavior afterwards, maybe it's not that the emotion caused the behavior, but they're thinking that the behavior will help them change the emotional state. It's a reaction to it. For example,

one of the classic findings was that sad people help more than other people. If you put people into a sad mood, they are more helpful than people who are in a neutral or happy mood.

- 4.34 Well, is it because sadness somehow contains helpful behavior inside it? Or is it because maybe they know that if they do a good deed and help someone, that will make them feel better? And a bunch of researchers here in Arizona developed this clever procedure to test that. What they do is, give everyone a pill and say, "This is a pill we're doing research on." It actually had no effect. But they told people, "We're testing this to make sure it'll be OK, medicine or vitamin or something like that."
- 4.35 But they said, "One of the side effects we found is that whatever emotional state you're in, it will freeze it and lock it in." They said, "You know how drugs can affect emotions?" And they used the example of marijuana, cannabis, which increases your emotional state. It intensifies whatever you're feeling. They said, "Well, this doesn't do that. It doesn't intensify it. But whatever state you're in, you're going to be in about the same emotional state for an hour, and then it'll go away, and there are no lasting effects."
- 4.36 OK, so people are willing to take it. But the beauty of that procedure is that then there's no point in doing something to try to feel better or feel different, right? Your emotion state is basically locked in. They tried this with the sadness thing. They put people into a sad mood. The ones who had taken the pill, though, didn't help, unlike the others. They replicated the usual finding in the control condition. Sad moods make people more helpful. But if people believe, "I can't really stop being sad for the next hour," then they don't help.
- 4.37 We were inspired by that. We copied that procedure. A lot of the classic emotion-to-behavior links disappear with that. My friend Bushman had studies showing that frustration or anger leads to more aggression. OK, that's one of the most replicated findings in society, but you can get rid of that if people think their mood is frozen. So, angry people engage in aggression because they think it'll make them feel better. "I think after I punch them out or shock them or whatever, that'll discharge my anger."
- 4.38 Now, whether it works or not is unclear, but sometimes does the opposite. But that's the belief, and that's what does it, and you take away that expectation. We looked at people who are more depressed or upset. They would eat more sweets, binge-eating stuff. When people are sad, they do that. That, again, disappeared with the mood-freeze manipulation. Or people who are more upset seek more immediate gratification, rather than delayed gratification.
- 4.39 That's one of the classic psychological tensions, between do what's best for yourself now or best down the road. Well, upset people go more to the immediate present. But even that one disappeared with the mood freeze. So my point is, lots of the evidence that even looks like emotion causes behavior isn't really that. It's really engaging in

behavior, in order to change your mood. So again, the idea that emotion is a direct cause of behavior is rather weak.

- 4.40 We said, OK, let's take this on the road and do a big major test of it. We did a giant meta-analysis, which statistically combines the results of a lot of different studies. Now, usually, you're taking all studies on the same topic. You want to do a men analysis. Are men taller than women? So you do a lot of studies and you find all the studies in the literature that have measured the height of the men and the women, and you combine them statistically, and you can calculate how big the difference is on average and so forth.
- 4.41 For this, we just took all studies on anything, anything that studied emotion as mediator variable. So the basic approach there is, if the emotion mediates, then the emotion is the direct cause of behavior. I spoke about this yesterday with the rejection, the theory that rejection would cause people to be upset, and the emotion, therefore, would cause the behavior. It didn't work in that one, but we looked at lots and lots of studies.
- 4.42 I forget how many, but it was over about a 30-year period in the top journal in the field, looking for any example of mediation by emotion. So for example, being insulted does make people angry and it does make them more aggressive, and you can test that statistically. So if you take those analyses and control for the anger, then being insulted no longer causes aggression. You got it? So we just look for any mediation by any emotion.
- 4.43 And of course, even some of the ones that survived that, some of those were probably, as the mood-freeze would show, that that wasn't really the emotion-causing behavior, but that would be the best evidence. So 4,000 articles went into this meta-analysis, 200 of them were tested for emotion-mediating behavior, 143 emotion-mediating judgment. The 4,000 was the total published over that period, so about 22 years.
- 4.44 Anyway, emotion did mediate about 17%, and that's at the 5% level of significance. So it's just barely above chance. And in terms of mediating judgment, about the same, 18% were significant at the 5% level. So yeah, there's a little sign of emotion-causing behavior. And remember, if you take out the ones that the mood-freeze would have eliminated, that the emotion is causing the behavior because people are expecting the behavior to change their emotional state, then it might get down to nothing. It might really not show much at all.
- 4.45 The general idea is no. I won't say there's nothing. Sometimes emotion does probably directly cause behavior, but pretty small. The few studies that looked at anticipated emotion, which is more relevant to the second theory, and those mediated very well. 80 and 90% of those were significant, at the 5% level. People think, "Well, if I do this, I'll regret it." OK, they expect to regret it, then they don't do it, so that does work.

- 4.46 And so that's really interesting evidence for the second, for the feedback theory. The anticipated emotion is powerful. Of course, lots, lots more studies measure how are you feeling right now, than how do you expect to feel if you did this or that? So, call for emotion researchers to shift gears and pay more attention to anticipated emotion.
- 4.47 This is interesting, too, because space in a top journal is very precious. People really compete over it. To have the journal filled up with all these null results, hundreds of null results, is surprising. The only reason is that everybody assumes that that's what's really happening, is that emotion is mediating behavior. And so, even if that's not your theory, you'd have to show that that's not it. Otherwise, the journal reviewers will say, "Well, what about emotion? Maybe they felt this way or that way, and that's why they did what they did."
- 4.48 So it is a widespread assumption that emotion causes behavior, informing lots of studies, but the data don't really back it up very well at all. OK, so on to the second theory, the emotion as a feedback system. The emotion helps the brain know whether events were good or bad. And so, generally, you want to feel good, so you learn which actions will produce feeling good and feeling bad. And of course, it's probably more important to avoid the bad than to pursue the good, but people will do both.
- 4.49 And emotion, therefore, guides behavior. My colleagues came up with a nice example of this. Suppose you're a young man out shopping with your girlfriend, and so she comes out, and it's kind of dull for you, but you got to help and shop and so on. So she comes out and says, "Well, do I look fat in this dress?" And now, she's told you honest communication is really important, and a good relationship means you always have to tell the truth with each other. And you know there's money riding on this.
- 4.50 So you say, "Well, not any more than your other dresses," or something like that. Yes, to some extent. So let's assume that doesn't go well. That's a big negative reaction. So the next time you're shopping with your new girlfriend, because you might not have that one anymore, and then she says, "Do I look fat in this?" You say, "No, no, no, not at all." So in that kind of way, the emotion reacts, leaves a trace; and so, the next time there's an immediate automatic twinge of, "Oh, don't say she looks fat."
- 4.51 OK. So that's how that theory works. Let's turn now to some of the issues in terms of learning to support this. This is one of the key theories for the second—the key points from the second theory is that emotion improves learning. As I said, there's plenty of evidence that emotion changes how people think. If you put people in an emotional state, their thought processes change much more reliably than their behavior changes. Emotion makes them reflect on multiple causes, in terms of what could have produced this or that result.
- 4.52 One thing people do a lot is think back to something that happened, especially something that didn't go well, and replay it in their mind, only counterfactually. "Oh, if only I hadn't said that," or "if only I had done something differently." And emotion is a

big driver of that. In fact, Neal Roeser was one of the pioneers of studying this thing, said negative emotion is the biggest cause of people engaging in this counterfactual thinking, of imagining how things could have turned out differently, how they could have acted differently, and so on.

- 4.53** It's a very powerful learning technique. I mean, an animal goes through an experience and learns from it, reward or punishment, one time, and that's it. Each event leaves one memory trace. But a person can think back over something, like 30 or 40 times. And so you can learn much more powerfully from the same incident. And you're not only replaying exactly what happened, you're playing it, "Suppose I had done this, or suppose I had said that, or suppose I hadn't gone there in the first place."
- 4.54** All these different possibilities enable you then to learn much more effectively, assuming you simulate correctly. But again, you have to learn to do that. There's evidence, too, that studies—Pennebaker started doing this decades ago—have people write about some really unpleasant experience that they had had. It had lasting positive effects on them because they would think about it and replay it. It had been upsetting at the time, but to go through and write and describe what happened enabled them to confront it.
- 4.55** It showed in some of his studies that people were actually healthier afterwards, such as counting how many times they went to the student health clinic the rest of the year. So, somehow reliving the experience, going back through the emotions and experiences, and reliving it and reconsidering it, had positive beneficial effects.
- 4.56** In a very different context, researchers have studied habits. When you're doing something new, you often have a lot of reaction. But habits generally happen without emotion. Habits are emotionless. So again, that suggests that emotion is part of the learning process; that when you're acquiring a habit, you're learning, "This stimulus, this situation, I do this." Once that's settled, the learning is more or less done, and then it happens without a lot of emotional reaction.
- 4.57** As I said, emotion strengthens the memory trace. That's been clear. The psychologists who study memory, yeah, there's a bunch of things they're learning, and a couple of them have big emotional cues in there. Those are getting remembered much better. And also emotion somehow narrows your focus, so that whatever you're going through, you zero in on the key, the important facts, about it.
- 4.58** Moral reasoning is an interesting one. The study of morality in psychology for decades was dominated by that Lawrence Kohlberg approach, which would ask people, give them moral dilemmas, and they'd have to engage in moral reasoning, and then they'd code, how did the person resolve it, to suggest what stage of moral reasoning they were. A man is sick, or his wife is sick; he can't afford the medicine that would save her. Should he steal the medicine if he has a chance? And it wasn't that there's a right or wrong answer, but they looked at the level of sophistication in the moral arguments.

- 4.59 Then there came this scandal that had started to appear, that when people face a moral dilemma themselves, they don't engage in moral reasoning. For a while, people were saying, "Oh, moral reasoning is nothing." Since then, they've come around that, well, in the crisis of the moment, they do what feels right. But if it turns out badly or whatever, or if they have to explain it to others, then moral reasoning is important. It might change how they react in the future.
- 4.60 Again, consistent with our emotion theory here, if you do something that feels right to you, and it blows up in your face, then you'll reflect on that and think about it, and then you'll engage in moral reasoning. And so, that the next time you're in a similar situation, you'll have this immediate twinge of, "Oh, don't do the same thing that led to a bad result last time." So the moral reasoning is used; it's just often not there in the moment of decision, in the moment of choice. When the person has to choose, then they often follow their moral intuitions. But the intuitions are residues of emotional learning from the past.
- 4.61 This even raises the possibility that all human behavior is guided by wanting to feel good or not bad. We call this ubiquitous emotion regulation. It's carrying the argument probably a bit farther. Usually, things that try to explain everything end up being overstated.
- 4.62 But contrary to the previous assumption that emotion is generally the cause of behavior, this says, well, at least a lot of times, behavior is done aiming to produce things that will make you feel good and not feel bad. And that's why guilt works. It's a productive, very socially constructive emotion. It's good for relationships and things, but you don't really have to feel guilty very much. You just have to avoid doing things that you'll feel guilty about later on.
- 4.63 If you think of nature designing the human mind, well, that's a pretty good way. If you do things that will make you feel good and not make you feel bad, that's a reasonable way to get through life and solve most decisions in a fairly constructive fashion. And last, as I said, anticipated emotion then becomes crucial to that. You have to correctly understand that, if I do this, then I will be sorry, and you change your behavior on that basis.
- 4.64 As I said, from the meta-analysis, anticipated emotion does mediate behavior much more effectively and adaptively than currently felt emotional state. It promotes good and beneficial behavior when you anticipate emotion. As with guilt, "I won't do this because I'll feel guilty tomorrow if I do this." That's a good adaptive reaction. Anticipated regret, people studying financial decisions and so on, people avoid things that they think they'll regret in the future.
- 4.65 Now, it's not perfectly rational. It does bias people towards safety and the status quo. If you're avoiding something that might make you feel bad or guilty or regretful or whatever, you may end up not taking a few chances that might have been logical or

rational to do. But still, you can't go too wrong sticking with safety and the status quo, as long as the status quo is pretty good.

- 4.66 I've spoken, I think already a bit, about this research on people predicting their emotion states. They call it affective forecasting, forecasting our emotional weather. And remember, the general finding is that people predict the emotion will be stronger and last longer than it actually does, in the event that it comes through. So, say a young professor working to get tenure at the university, "Oh, if I don't get it, it will be so awful. I'll be miserable for years." So that motivates you to work harder to do it.
- 4.67 I mean, the people who don't get tenure, they get over it surprisingly fast. So the actual emotion is less than the anticipated emotion, but that fits this theory perfectly well. The operative part of something should be the big part. It would be really a problem for the feedback theory if people expected, "Oh, I only feel a little bit bad," and then the thing happened, and they feel terrible about it. That's not very useful. So you have to expect big, strong emotional reactions.
- 4.68 Are you pursuing a relationship? "If I do well, I'll be so happy. This person will marry me and stay with me, and we'll have a wonderful life together forever. And if I screw it up, I'll be broken-hearted, and I'll never have another chance, and I'll be miserable for the rest of my life. So it's useful to think that way." It makes you try harder to make the relationship succeed. There's no need for you to actually be miserable for the rest of your life if it doesn't work out. But thinking that you will be is a useful motivational aid.
- 4.69 So, as I've mentioned in the meta-analysis, the anticipated emotion predicted behavior, mediated behavior, much better than actual emotion, although there was a lot less data, so I wouldn't draw strong conclusions from that much. Anyway, so what we said thus far, the main function of emotion isn't for causing behavior directly. Rather, it's for helping you learn from your experience. You evaluate what you just did, and how it turned out is good or bad.
- 4.70 The full conscious emotion helps you dwell on the experience, analyze it, think about it, analyze it counterfactually, so that you learn things. And then it leaves these residues of a little automatic twinge of positive or negative feeling, so the next time you're in that same situation, or the same stimulus or opportunity or whatever arises, you have an immediate good or bad feeling about it, and that helps guide your behavior in that case.
- 4.71 All right. In that connection, I want to talk a little bit more about guilt, since it's a good example of the second example of the feedback theory. You do something bad, you feel guilty, you reflect on it, and then you learn not to do that again. Now, guilt has a bad reputation. A lot of people share the view here that guilt is this mechanism we use to control people. It's an illusion. It's a kind of social control mechanism and very unhealthy. It does terrible things to our bodies. There are much better ways to control our behavior than that, rather than extraordinary use of guilt.

- 4.72 All right, people have this idea we should get rid of guilt and so on, but consider the source. This was spoken by Ted Bundy, a notorious serial killer who murdered a couple dozen young women. If he'd had maybe a little more guilt, he wouldn't have done all those terrible things. So we're going to give guilt a more positive reconsideration. Guilt is distinguished from shame in the research literature. Shame is more global.
- 4.73 So guilt is, "I did a bad thing." Shame is, "I'm a bad person." And there's plenty of research on that. Shame is much more destructive, guilt is positive. So if you're raising your children or whatever, yes, you want them to have a healthy sense of guilt when they do something bad; you don't want to shame them because being a bad person, there's nothing you can do about that. People react to shame either by withdrawing, just hiding out from it, or by lashing out and aggressing. Neither of those is very constructive.
- 4.74 But guilt makes people learn lessons and change their behavior in a positive fashion, so guilt is an important foundation. You really don't want to have someone who has no sense of guilt to be your boss or your spouse or any important person in your life. You want people to be willing to have it. Anyway, guilt is good for relationships. Plenty of evidence for that. It depends on concern for each other. You're much more guilty, say, about not returning a call to your mother than not returning a call to your insurance agent. Or something like that.
- 4.75 The psychological basis, quite plausibly, is that it's tied to the anxiety about being alone, separation anxiety and the fear of that, the need to belong being very fundamental. So feeling bad when you're ruining your chances to connect is one logical source of guilt. It makes people avoid hurting others, also avoid ignoring others, makes them pay attention to other people who are important to them. Guilt makes people apologize, try to make up what they did wrong—all these are very constructive for repairing a relationship if something bad has happened.
- 4.76 Expresses caring for it, too, for the other person, even if you can't resolve it. So you come home late, and your spouse has made a nice dinner, and it's now all cold and spoiled. Well, if you feel guilty, it isn't going to fix the dinner. So it can't make up for it. But it shows that you care enough and that you're really sorry that they did this for you, and you weren't able to appreciate it and share it with them. So just feeling guilty is already a sign of positive caring. And that, too, is good for relationships.
- 4.77 And guilt has also always been known as one of the weapons of the weak. So we associate it, even in the traditional family, we associate guilt induction more with the mother than the father. The father had more power. And sure enough, if guilt works, it can reduce inequity and power. It can give the less powerful person in any relationship more of a chance to get their way, because of the other's concern for it, and hence guilt.

- 4.78 Now, people have long argued that guilt is a private emotion, something that goes on in the privacy of one's mind, but more and more, I think that's wrong, that it is a fundamentally interpersonal emotion. So we did an experiment where we had people write stories about different emotions that they had had, tell an experience from your own life when you felt sad and so on.
- 4.79 And then if there was another person in there, we coded who was the other person, what was your relationship to the other person, on the scale from zero being alone, and one being a stranger that you're never going to see again, all the way up to the close relationship partner, who is someone who is really close to you. And as you can see from this, guilt is by far the most relationship-oriented of the emotions. I mean, being afraid, a lot of times people are afraid of strangers. That'll happen, in a stranger or someone you hardly know, and so on, and sadness, anxiety, and so on.
- 4.80 A lot of those happen outside of close relationships. But guilt, you see, the average is almost four out of five, and the scale was, that's at least an acquaintance, someone you have an ongoing relationship with, and then their close relationship partner. I'm sure there were plenty of those as well. So we do occasionally feel guilty about toward a stranger, but it's much more common in an important relationship.
- 4.81 And again, that supports the idea, it supports good relationships through this feedback mechanism. You did something, you often didn't realize what you were doing, that it would have this negative effect on the other person. And so, "Oh, I didn't mean to make you feel bad. I didn't mean to do that." So they feel guilty, and then they apologize, make up, show that they care, and ideally learn not to do it again.
- 4.82 And experience sampling study. We had people carry around one of those beepers, or nowadays, you just do it with a phone. And so whenever it goes off, you report on what you're doing and what you're feeling. 13% of the responses indicated some guilt. If you extend this to how much of waking life it is, it suggests that the average person, this was in the Netherlands, would feel about two hours of guilt a day. Now, mostly pretty mild guilt. "Oh, I wish I hadn't done that, or I cut somebody off in traffic or whatever."
- 4.83 It occurs with all different kinds of activities, associated more guilt on certain things. But it was highest when you're thinking about yourself in relation to other people. Again, the argument being that guilt is very constructive for strengthening and benefiting relationship. Now, people do feel guilty when they're alone, but they're usually thinking about something they did with someone else. So it's the thinking back on, "Oh, what I said today really hurt someone's feelings or offended someone," and so on.
- 4.84 In another, we had people tell stories about when they made somebody angry and felt guilty, or they made somebody angry and didn't feel guilty. So it's the same thing. You did something that somebody else thought was wrong. But we could code then, what are the differences between the ones where you have guilt and where you don't?

Here's some of the answers. Learned a lesson. Almost half of the ones where they felt guilty said they learned a lesson.

- 4.85 And this is something they spontaneously put in the story. It wasn't our asking them the questions; did they mention that they learned a lesson? Almost half. But when they didn't feel guilty, no, they didn't learn any lesson for that. Change the behavior. Again, lots of them said that when they felt guilty, not otherwise. Apologizing, guilt makes people apologize. As I say, it's a good relationship repair step. Confessed what they did wrong, were they selfish? Selfishness seems to be a big factor in making people feel guilty.
- 4.86 And is the other person someone who mattered to you, the person you angered? Was it someone who mattered to you? Much more common in the guilt story. So yes, when we do something to make somebody we care about mad, that's often a cause to feel guilty. Doing the same thing and make somebody we don't care about feel mad, you do feel sometimes guilty, but not nearly as much.
- 4.87 In terms of memories of inducing guilt, these are almost all in close relationships and making other people feel guilty. People do that as well. They often felt better after making the other person feel guilty. The ones who did it also, we call it meta-guilt, which is guilt over inducing guilt. I made so-and-so feel guilty, and I felt guilty about doing that. I had a couple of stories I brought along that we could read there, so these are interesting to illustrate it.
- 4.88 She says, "I always try to make my boyfriend feel guilty when he doesn't write me or phone me, as he does not live nearby. Usually after one week, I call him up to make sure he's alive. Then I joke around to feel out the situation as to why he hasn't written or phoned. I usually bring in the fact that I write and phone, and ask him point-blank, why can't he treat me the way I treat him?" So inducing guilt to get the boyfriend to pay more attention. That was a common theme.
- 4.89 There's one, "When I was about 18 and 19 years old, I was still living at home with my parents, and I stayed out all night at a party, got home 8 or 9 AM the next day. I didn't call. It didn't even cross my mind to call home. I did get home, kind of hungover, tired. My mother was waiting for me with the biggest guilt trip known to man. She started that, 'You don't love me. You make me worry so much. I thought you were dead' routine. She told me, worst of all, that she was disappointed in me.
- 4.90 This is my mother, who fawned over every little achievement I had from kindergarten to getting my driver's license. The way she made me feel stuck with me. I made up, of course, after lots of crying and explaining, but to this day, I can still picture the look of disappointment on her face and the tone of her voice. I hope to God I never make her feel like that again." So obviously, very constructive change in her behavior, based on feeling guilty in terms of how she treated her mother.

- 4.91 "Last summer, I worked two full-time jobs so that I could move out this year, whilst my two younger siblings, both in university too, had an all-expenses-paid trip to Europe for three months. So every time I needed money, I mentioned to my mother that I didn't have a vacation last summer. She feels guilty and gives me money, which I usually don't need.
- 4.92 However, I'm fully cognizant of the fact that she feels bad about it and won't be relieved of her anxiety until she pays for me to go on a trip, too. But I won't let her. Thus, this guilt trip bit should work fine for at least five or six months, then it will be business as usual." So, using guilt to get money from the parent, and a very blunt and calculating in terms of how well it works.
- 4.93 "I broke up with my girlfriend that I had been seeing since the beginning of the year. She made me feel guilty by asking me what was wrong with her that had made me want to break it off with her. She asked if she was ugly, fat, unattractive, boring, etc. She followed these questions with bouts of tears and self-pity. Of course, this made me feel really guilty. I tried to explain to her that I still cared for her, but I just didn't want to see her anymore.
- 4.94 Truly, however, my real reason for guilt was the fact that I found her extremely dull and boring, and I had just gone out with her for such a long time, probably just for sex." So there's feeling guilt. We don't see that constructive result from it, but perhaps he will look for other things rather than just sex in future partners.
- 4.95 "My closest female friend induced guilt in me to prevent me from making other plans on her birthday, which was also the weekend closest to my boyfriend's birthday. She had mentioned several weeks earlier that she would like to get a group of people together to go to Yuk Yuk's, a nightclub, to celebrate her birthday. So this would take up both nights of the weekend closest to my boyfriend's birthday. I told her I would think about it.
- 4.96 Several weeks went by, and she never mentioned it again. The Wednesday before that weekend, my boyfriend and I were invited to go to a movie with a different group of friends. He was more interested in doing this than accepting my friend's invitation, but I felt I should check with her first, since I had made a tentative commitment. I phoned her to explain the situation, and she complained that it seemed like no one had time for her on her birthday. She wasn't sure anyone would go out with her, so she might end up just sitting home alone on her birthday.
- 4.97 I resented this obvious attempt to manipulate me, since she had entirely forgotten my birthday, and since we were already intending to go out with dates for dinner and dancing near her birthday. Nonetheless, she seemed so upset that I agreed to go along with her." So there's inducing guilt, and it does get her way. There's certainly some resentment, which is, all right, you at least get what you want. I got one or two more.

- 4.98 "At home, my husband and I share most responsibilities. Once I felt I was doing too much, he wasn't home, and I prepared his favorite food, claimed all I could, and did all the things I could to make him feel guilty. When he came home, I pretended I was sick and disappointed in him. It worked. He felt guilty. What a mean wife," she comments on it. So there's the meta-guilt, guilt over inducing guilt. She does get her message across, and we hope he did improve from that, but she did feel guilty about inducing guilt.
- 4.99 All right, I like this one. "When I was younger, I beat up on my sisters. This always angered my mother, who would punish me whenever I did. However, when my sisters would beat up on me, I would be punished for doing something to cause them to hit me. Obviously, this was unfair. I tried to tell my mother that this was happening, but she didn't see it that way. One day, I tormented my sister to the point that she hit me. My mother saw the incident and sent me to my room for tormenting my sister, yet did not punish her at all.
- 4.100 I said, 'Wait a minute,' and pointed out her inconsistency. I paralleled the incident with others where my sisters would torment me, causing me to hit them, and I would get in trouble, and they wouldn't. I made my case and waited. She could not argue back. She suddenly realized that she was wrong and felt very guilty. She started to apologize, but I was in control now and decided to make her suffer.
- 4.101 I said, 'Do you think an apology will fix the years that I have taken this punishment unfairly?' She had no reply, and I could see that she felt very guilty—so guilty, in fact, that from that point on, she was hesitant about punishing me for anything. Oh, the benefits of guilt." There's someone using guilt. Obviously, it took a while to get there, but it did have an intended behavioral effect.
- 4.102 Anyway, I'm running out of time. I want to move on. I got one more thing to cover. To understand guilt, again, as an example of the feedback theory of emotion, mainly occurs in close relationships, strengthens the relationships in various ways, and can improve behavior. Even if you hardly ever feel guilty, if you're careful to do things to avoid feeling guilty, that's really the best possible outcome. You don't need to feel guilty all that much, as long as you avoid it.
- 4.103 All right, the last part is a little research project I did coming out of the feedback theory of emotion. It started with the assumption, OK, let me say that. The basic idea is that, because emotion is associated with learning, when people have emotions, they will think they learned something. And so it will create this illusion of learning. People don't always know exactly. It's hard to assess how much you've learned, but emotion can be used as a cue for it.
- 4.104 We got interested hearing people say, "Oh, I learned a lot on this or that experience." Often, it was an emotional one. You press them, really; what exactly did they learn? Sometimes it's rather vacuous. We were doing studies on rejection, which I spoke

about yesterday. My wife's a social scientist, too, and so we watch these reality shows that are all about rejection, like *The Bachelor*.

- 4.105** And we were watching them late in one, and the guy was talking about it, and you know how the show works. There's 15 different women or something, and they're all competing for his favors, and one by one, he rejects them. And he's saying, "Oh, it's been such an intense experience. I learned so much about myself." And my wife turns to me and says, "Really? What did he learn? He likes having 15 beautiful women in love with him. He likes riding around in limousines and drinking champagne. Big insight."
- 4.106** But undoubtedly, it was a very emotional thing to reject these people, one after another. People go on these group retreats, sometimes too for business groups to get to know each other, and they do these trust falls and things. So they get a lot of emotion. My sister did one. "Oh, we learned so much about each other." But what did you really learn? If this is true, it's a way to scam teacher ratings for professors who are concerned about how their students rate them.
- 4.107** It doesn't really—giving them a lot of information isn't the important thing. It's creating an emotional state. I read an account somewhere, somebody hired some actors to give lectures at a university. Of course, they didn't really know what they were talking about, but they're very good at creating emotions. The students gave them really high ratings, even though, presumably, they didn't learn that much.
- 4.108** So, once again, the core idea, people estimate their learning from their emotional reactions. Calculating how much you learn, how can you do that? Even at the end of the semester, can you really remember what your mind was like before you took the semester, and how much information has come into it since then? But you can remember the emotion.
- 4.109** And since they're often linked, it would be a reasonable cue. But that means, again, people could make mistakes on it. It fits with, again, the feedback system of emotion. Emotion stimulates learning. So if there's emotion, you can infer there might have been learning. All right. Let me skip ahead to a couple of the experiments that we did. Here's the hypothesis.
- 4.110** People learn when they're emotional. If we create emotions, arbitrary emotions, irrelevant to what they're learning, they may think they learn more. So they'll mistake the emotion for learning. So, first experiment, and my group came up with these ideas for the experiment. I really intuitively couldn't imagine that these would work; they were so obvious. So first they read a biography of George Bernard Shaw, the writer. And then we had people get into emotional states by remembering one time you felt excited, one time when you felt angry.
- 4.111** Different people were assigned to different emotions. We had good and bad ones, high and low arousal amounts. And then they rated, how much did they learn about George Bernard Shaw? And then we also gave them a quiz to see how much they did learn,

based on what they'd read. Those were the four emotions, and then there was a control where they just wrote about, what did you do today, this morning, before you came to the experiment? Presumably, not very emotional.

- 4.112 We measured how much they thought they learned, how much information did you learn, how much more do you know now, and so on. Anyway, the results, we all had our pet theories about which emotion would promote more learning. But no, the big effect was just all the emotions, which had nothing to do with George Bernard Shaw, but they all made people think they learned more. I was surprised by this, but a clear effect.
- 4.113 So, all four of the emotion conditions rated that they learned more than the control. Again, we had just first had them read, and then put them in an emotional state based on remembering. The emotion had obviously nothing to do with what they'd read. There was no improvement in actual learning. We tested them. They didn't really learn any more. And the effect of emotion was there, even after we controlled for what they did actually learn.
- 4.114 No difference among the emotional states. Statistically, they were all about the same. And we measured, did you have personal interest in George Bernard Shaw? Whether they did or not, that didn't seem to make any difference. So this randomly created emotion, irrelevant emotion, made people think that they had actually learned more, although they didn't.
- 4.115 Second study, we said, well, how about if we manipulate the emotion first? Well, we'll create the emotional state first, and then see. So this time, instead of George Bernard Shaw, they read something on biology of jellyfish. Rated their learning, same effect. See the white bar over on the left? That's the control condition. They learned something about jellyfish. All the different emotional states thought that they learned more about jellyfish.
- 4.116 Third one, we wanted to test some alternative explanations. Maybe emotion just makes you rate everything in a more extreme fashion, not just learning. Or maybe you somehow changed what you thought you knew beforehand. So it gave you a bigger boost, even though you ended up at the same place. But no, all five emotional states, again, rated their learning higher than the neutral control.
- 4.117 But the learning was increased by the emotion. It did not affect things like, how much do you like dark chocolates? How much do you like winter? So it wasn't that emotion increases all ratings, just specifically ratings of learning. And as usual, no difference in actual learning. So, conclusions from this, emotional states increase your impression of how much you learn, even though the emotion is obviously irrelevant to what you're learning for.
- 4.118 We found no differences in actual learning. Certainly, if they read the thing before the emotional state, there couldn't possibly be any. We ruled out the alternate

explanations. There is this interesting illusion of learning aspect. All right, so just to conclude, the function of emotion is not to cause behavior directly. It will do it, I think, sometimes, although again sometimes the results when that happens are not really what you would want. Rather, the proper adaptive function of emotion is to stimulate your mental processing, so that it improves learning, hence it works as a cue, hence the illusions of learning.

4.119 Anticipating the emotion as you learn, "If I do this, I'll feel happy; if I do this, I'll be sorry," you learn to do that. That helps guide behavior. That's very constructive. It means often behavior is pursuing emotion. You're trying to do things that will make you feel good and not feel guilty or bad in other ways. The automatic traces of emotion influence current behavior, often based on past feelings. All right, that's it. Thank you.

Lecture 5: Self-Control

At a glance — Self-control.

- 5.1 All right. For this lecture, I'm going to focus on one of the important functions of the mind and the self, namely how the self changes itself. This is something I've been studying since about 1990. When I was in graduate school, I decided I would try to understand the self. In fact, that was still the end of the hippie days, and people were looking for self-knowledge and all that, and the hippies did it by taking drugs.
- 5.2 But the scholars were interested in using research to understand it, too. I wasn't really so much interested in myself, but the self in general. And so I started off studying self-esteem and some other things. But then some of the leading people in the field started saying, "How the self regulates itself, that's the key to the whole system." So I thought, "I'd better learn about that and inform myself." So I set out reading the literature on that and did some experiments.
- 5.3 Anyway, it's been a long theme of my career, so I'm going to describe today how my thinking developed and what I learned gradually as the theories were refined and so on. So what is self-control? I use self-control and self-regulation interchangeably. Some people use them with a slight difference, that self-control is just the conscious forms, and self-regulation might include how the body keeps a stable temperature, but that's not something I pay much attention to.
- 5.4 What is it? Well, essentially, it's changing yourself, changing your responses, overriding one response to substitute for another. You use self-control to regulate your thinking processes, your emotions. Impulse control is a very familiar one, not yielding to temptation, not doing things you shouldn't do, and so on. And last, task performance. There may be others, too, but those are the four big ones.
- 5.5 Overriding responses. I like the term regulate because it means not just change, as in any change, but change based on an idea, a standard of how things should be. So when the government tries to regulate how buildings are built, it doesn't just say, "Do it differently." It says, "No, there are certain standards. The toilets have to flush, the doors have to open, there have to be windows, heating and cooling systems," and all those. And they inspect to make sure that you do it. So it's changing based on some idea of how things should or should not be.
- 5.6 Vital for human social life. As I said, we evolved to live in culture. Culture has lots of rules. People need to change themselves, override their responses, resist their temptations that are illegal or antisocial and so on. And it's not perfect, but if most of the people follow most of the rules most of the time, then the system works pretty well.

- 5.7 One of the basic ideas in psychology, going back at least to Freud, is that people have motivational conflict, that you want to do this, but you know you shouldn't, or you want to do this, but you want to do something else that's incompatible. And so, self-regulation is a way of saying, "I should give in to this motivation, not to that one. I should pursue this and not that." So it's useful when you have a motivational conflict.
- 5.8 As I said, it's one of the keys to understanding the self. Certainly, morality is mostly rules about things you should not do that are morally wrong. And morality is a set of rules that enables society to function, enables people to live well together. So self-control, morality requires self-control. We sometimes call it the moral muscle, which you'll see in a little while why.
- 5.9 But it's essential for being, well, I feel like doing this, but that would be morally wrong, so I have to resist that impulse, make myself do something else. Indeed, it's one of the bases for free will. The philosophical discussions of free will often invoke issues of self-control.
- 5.10 Now, what are the benefits from it? Well, I should say, I mentioned already, I started my career studying things like self-esteem. There was a big fascination in society with self-esteem. I noticed that there are a lot of positive correlations. People with high self-esteem are doing better in a lot of spheres of life than others. And so they got this idea, well, if we just raise everybody's self-esteem, then everybody will do better and society will work better.
- 5.11 Only, that's the correlation-causation mistake. When they started tracking people over time, it started to emerge that self-esteem is more a result than a cause. For example, if they measured students' grades they got one year and their self-esteem, and then the same thing next year, well, their self-esteem the first year did not predict their grades the second year, but their grades the first year did predict their self-esteem the second year.
- 5.12 So the grades come first, the self-esteem comes after. And so the idea that, well, if we just raise all the kids' self-esteem, then they'll get better grades, it doesn't really work that way. So I got disappointed with self-esteem. But as we'll see, self-control is the real deal. It really does contribute. Self-control comes first and predicts all kinds of benefits later on. So studies even measuring kids' self-control when they're 10 years old, sometimes, say, rated by their teachers or whatever, will predict whether they're unemployed as adults and other things like that.
- 5.13 Anyway, good self-control over time as a personality trait predicts greater success in school and in work later, including earning higher salaries and more money, predicts better relationships. The partners of people with good self-control are happier than others. It makes oneself happy, reduces stress. People—we think, self-control, just some sort of puritanical self-discipline, making yourself do the right thing. No, no.

They're really happier in multiple ways of measuring happiness, and they all show their benefits to it.

- 5.14 A better adjustment, better mental health. High self-control actually does even lead to higher self-esteem in the long run, by virtue of helping you achieve more and be more successful in life. Fewer mental health problems, physical health also. You take care of yourself. People with good self-control do everything from exercise better to flossing better to not smoking, things like that.
- 5.15 Fewer money problems. They are less likely to be heavily in debt. Self-control produces all sorts of positive behavioral things. People with good self-control are less likely to be arrested, less likely to show prejudice against others, engage in abuse or violence or anything like that. And at the far end of life, they live longer. Psychologists looked for a long time for, are there any traits that help people live longer? And there was a slight boost for intelligence. Smart people live longer.
- 5.16 But self-control, which is also known as conscientiousness, conscientiousness is a big trait that includes a couple other things, but I think it's mainly self-control. And yes, those people actually live longer because they take better care of themselves—again, along the way, better health, fewer behavioral problems, and so on. And conversely, if we look at the problems people are having in life, various things that go wrong for them, there's usually some element of self-control failure involved in those.
- 5.17 All right, so how does it work then? There are three or four basic components to self-control. One is, you have to have the standards. You have to know what it is that you want to do. What is your ideal weight? How much do you want to work each day? What do you want to get done? Things like that. Monitoring is the second, keeping track. That is the argument that self-awareness evolved precisely to enable us to regulate ourselves better.
- 5.18 So you can be aware of yourself, keep track of yourself, attend to yourself, and know, well, I'm doing this, but I should be doing something else in order to be acceptable to the group, in order to get ahead in life. So monitoring of the behavior. It's very hard to regulate something without keeping track of it. So I gave the example of the government regulating how buildings are made. Well, they don't just make the rules. They have to send out inspectors to go and look in the room and see, does the heating and cooling system actually work and so forth?
- 5.19 And then there's the capacity to change, which is usually some energy aspect, willpower. That's going to be the particular focus of my work and what I'm going to talk about today. Some people will list a fourth one. You have to want to be good. You have the standards, but if you don't care about the standards, if you don't care about being a good person or losing weight or whatever, then you're not likely to do it.
- 5.20 So some motivational basis, either that's implicit in the others, or you can list it as a separate factor. But those are the three or four things that we need to do to

understand how self-control works. All right. So my work has been focused mostly on the limited willpower aspect. And when I first started this, I wrote a book without doing any studies. I just read all the published studies that had been done that I could find. And I started noticing there were patterns, that it looked like the capacity for self-control was limited, as if some energy got used up.

- 5.21 So for example, people, when they're quitting smoking, they can resist for a while, but especially if there are other demands on their self-control, that's when they relapse, that's when they give in. A variety of things show this pattern. It looked like there's a limited amount of willpower. At the time, this was a fairly radical theory. Evidence, too, that even on the same task, that your ability to regulate yourself, to make yourself do things, deteriorates over time.
- 5.22 In perception psychology, they study vigilance, like you have to watch a screen just in case to see some pattern shows up or something happens. It's a little bit akin to the sonar operators during the Second World War and the ships that had to be watching for submarines. And even there, their ability to concentrate and to spot anything did deteriorate over time. They did lose it. And there's no loss of motivation. That's a life-or-death thing. If it was a submarine, it could sink your ship and kill you and everybody else on it. So the motivation is high, but still they couldn't sustain it indefinitely.
- 5.23 More recent studies showing when people try to do two or three things at once, that they do badly at both. People who are trying to diet and quit smoking, well, they both take willpower. And if you put your willpower into one, that leaves less for the other, and so you end up doing a bad job at both. Also, those people, the study found that they were unpleasant to live with, which means they're not regulating their emotions very well. They're getting grumpy and irritable because they're both trying to lose weight and quit smoking.
- 5.24 So that's what [led] to this theory, that there's a limited amount of energy that's somehow gets depleted. So if you use up your willpower, then your performance will gradually go downhill, which seemed to be the pattern implicit in a variety of findings, although most people didn't talk about it very much back then. It seems to be limited to some aspect of the body. I'll come back to that.
- 5.25 This is less conclusive, but glucose is a chemical in your bloodstream that carries the energy from your stomach to the muscles and to the brain. So maybe when that isn't adequate, then the energy isn't going to be there anymore. And down the road, we started extending this, and it applies to other things, too, the limited amount of energy. So making decisions, thinking carefully, logical reasoning, and so on, that that resource might also be used for that.
- 5.26 Let me describe some of the experiments we did to test this idea that your willpower, your capacity for self-control, is limited. I should say, back in the '90s when I started doing this, this was such a bizarre notion. Psychology then was all dominated,

computers were spreading through society. People were getting personal computers, and so all the analogies were "the brain is like a little computer." All the theories about how we process information.

- 5.27 And those are very valuable. I mean, there was a lot of stuff there, but there was no room to talk about energy in those. I used to joke, when I would give early talks, that energy models were so out of fashion that we weren't even against them anymore. In terms of the self, we used Freud's term ego for the self, because I couldn't find any other theorist who talked about the self who had invoked energy since then.
- 5.28 I wasn't really buying all of Freud's theory, but the idea is that the self is made of limited energy, which Freud had said, and he was kind of vague about how it worked and where it came from. But at least he gave it a role of energy in the past half-century. Hardly anybody had talked about energy as an aspect of self, but they also weren't able to explain about how the self exerted control over the world or made decisions or things like that. I think those were linked problems.
- 5.29 So, one of our early experiments, we can't tell people when they come to the lab for an experiment, you can't tell them, "We're going to test your self-control," because then they're all nervous about that and self-conscious, and then they'll try to do a good job. So you have to give them a cover story. We told people we're studying how well they could remember taste sensations. That's plausible. And we said, "Because of that, we need you to not eat for three hours before the experiment."
- 5.30 OK, so they come to the experiment and they've skipped lunch or whatever, and they come down. And I guess this part then was a little mean. We set up a microwave oven in the lab and baked chocolate chip cookies there. You know how it blows out that aroma? So the lab smelled really good. We knew it was good because people with offices across the hall were saying, "I'm in my office trying to do my statistics and smelling your darn cookies all day. Really, really hard." So it was very tempting.
- 5.31 Anyway, the person comes, is hungry, missed lunch, comes in and smells really good, and he sits down and there's a plate of these cookies. Next to them and also some chocolates in case they weren't into cookies. Very, very tempting, all that. And also there was a bowl of radishes. And in one condition, the experimenter says, "Well, you're going to be in the radish condition. So we're going to need you to eat a bunch of these radishes and don't touch the cookies. They're for other people in the experiment."
- 5.32 So they're really tempted, but they've got to eat the radishes. Others, we told, "Don't touch the radishes. So we need you to eat the cookies." That was fine with them. And then we had a no-food control group. But the ones we were interested in, who had to sit there seeing those cookies and smelling those cookies and wanting those cookies, but instead had to resist that temptation and make themselves eat those stupid old radishes—we left them alone for a few minutes to maximize the temptation.

- 5.33** Of course, we didn't trust them. So we secretly observed through a little control window, and there were longing glances at the cookies. Some people picked them up and smelled them and put them back. Some people dropped them on the floor and put them back, so we learned not to eat them ourselves. But anyway, nobody bit into the forbidden food, and everybody made themselves eat at least part of a radish. And then, as I said, in the other two conditions, one had, it was no food, the others ate the cookies. That was fine; they weren't longing for the radishes.
- 5.34** So then we took them to a different room where there's no food, anything, and borrowed this procedure from stress research, which is how long do you keep trying before you give up? So we had unsolvable problems. They thought they were solvable. But of course, if they solved them, it would mess up the experiment. So we wanted to see how long do they keep going before they quit?
- 5.35** So we said, work on these problems. When you're done them, ring the bell, or when you give up, if you can't do them, just ring the bell. Since nobody could get them, it was just a matter of, when do they give up? Well, as you see, there's a huge difference. The five minutes of resisting the temptation to eat the chocolate chip cookies took 10 minutes off of their perseverance in the frustrating puzzle despite failure, so a rather strong and a big effect. And the point is, resisting that temptation, using self-control, it took something out of them that they no longer had to help themselves keep on trying and keep on going and keep working at these difficult puzzles.
- 5.36** All right. Lots of laboratory studies have shown this with lots of different procedures because there's no one perfect procedure for this. At this point, there are hundreds of replications of this. But one thing we did wonder about is, is this really just some kind of a lab phenomenon? Social psychologists occasionally do that. They'll find something that, in a lab situation, you can do it, but it isn't really there in real life. So I was kind of pleased, and we happened to do a study on something else. We hadn't even really thought about it as self-control.
- 5.37** It was a study of desires, of motivations, because motivation is one of the fundamental categories of psychology, and yet we don't even know how often do people feel motivations. We had people, we beeped them as they went around their daily life and say, "Are you having a desire? Did you have a desire in the last 15 minutes? If so, what are you desiring? Are you acting on it? Are you resisting it?" And so forth.
- 5.38** Anyway, we did this and we had the first article already written up. And then the lead researcher said, "We have a way of testing your theory of ego depletion with these data, because we know if they've resisted other desires earlier in the day. We can see, how often did they resist other desires and did they resist successfully and how long ago was it? And then we could see, does that gradually sap their ability to resist further desires?"

- 5.39 So if they already resisted one desire, and then they have another one that they have to resist, are they going to end up giving in and doing it or not? So we're able to track whether they resisted the desire or not, and whether they gave in or not, because they might resist it and then give in. And then at the bottom, we built this composite score of, how much have they reported resisting other desires earlier in the day?
- 5.40 Of course, we missed a lot. We just beeped them six times during the day, so we didn't capture everything. But still, it could produce at least some version of the effect, as it indeed did. If you look at the top line, whether they did not resist the later desire, then were they likely to go ahead and do what they desired to do? And that was quite steady regardless of depletion, right? About 70%.
- 5.41 You might think it's 100%, but sometimes people want to do something, and they can't. Or it's raining outside, they can't play tennis or whatever they feel like doing. But if you look at the bottom line, the red line, sure, as the more they had resisted and the more recently they had resisted other desires and thus depleted some of their willpower, the more likely they were to give in, if they were trying to resist the subsequent desire. So again, having used up your willpower makes you more prone to give in to desires later on, even though you're trying to resist something.
- 5.42 I should mention here some publicity about this research. It has been a number of times, a New York Times bestseller book and so on. But at one point, a big group of people tried to replicate the effect and didn't find it. And so there's a lot of publicity, where, "Oh, ego depletion doesn't work." But that's really wrong, and since then it's been very strongly rehabilitated, and I think has a good case to be the best replicated finding of social psychology.
- 5.43 I always ask people, when I say this, if you can think of another one, if you disagree, tell me another one that has a better record. And so far, I've only gotten one nomination. That was a pretty good case, too, but still, it's very strong. So already seven or eight years ago, there were 800 published significant replications, none in the opposite direction. Many different labs, many different methods.
- 5.44 Pre-registration is a new thing that we've been doing last 10 years in psychology, where you have to specify in advance how big, how much data you're going to collect, what analyses you're going to use, all that. There are a number of people who've done that and got this effect to support it. There are lots of real-world findings, too. Things like physicians, as the day wears on, they're less likely to wash their hands between patients, more likely to give in if the patients ask them for antibiotics. You're not supposed to prescribe too many antibiotics.
- 5.45 But anyway, a whole bunch of things like that, accountants and others. So there's lots of real-world findings, too. And then these multi-lab studies. Actually the original one that was published that didn't work, when they finally made the data public, somebody else went back in and analyzed the data and said, "Well, you didn't really test the

hypothesis. You didn't deplete people." If you look at, to the extent that you did, the theory was actually supported.

- 5.46** And then there was one other that kind of didn't get it, depending on what kind of analysis you did. And then there are two others, more recent ones, that are very strongly supportive of it. Anyway, if you've heard that this stuff failed, there was a big brouhaha in the news for a while about that. But as usual, they don't follow up on the news and say, "Oh, never mind. It really does work." But I think any reasonable scientist would say this is totally a slam dunk.
- 5.47** All right. So the initial idea was that somehow you'd used up your willpower, that you had a very limited supply. And when it was gone, then you couldn't self-regulate anymore. It was simple-minded, but it seemed to be something like that. But then some findings changed that. They said, well, even though people are depleted, it looks like they can self-regulate if you give them a reason to. So you offer them a financial incentive or others or something. And yes, well, they can do it.
- 5.48** It turns out, if you offer them an incentive, they can perform well on the second task, but then you give them a third task, then they're really depleted. So it does seem like there's a limited amount of willpower, but people can find more if they need it. So anyway, the theory revised, it is really these effects are conservation effects. You're conserving your energy. It's not that it's all gone.
- 5.49** I frequently use the analogy to physical muscles. It turns out, physical muscles are much the same way. You don't run full speed until you're exhausted and then face-plant down on the track or in the snow. You may run really fast at first, but then you start pacing yourself and conserving your energy. It turns out, as I said, muscle tiredness is that, too. There is a point at which a muscle cannot function anymore, but you start feeling tired long before that.
- 5.50** And what the exercise researchers find is, they have somebody in the lab and they'll be doing physical exercises, and then they're tired, and they say, "OK, that's enough." But they'll say, "I will give you \$10 if you can press as hard as you did on the first trial when you first came in." And they can. So you can still get your muscles to do that, as long as you haven't gotten to the extreme place where you just can't do it.
- 5.51** I talked about windsurfing the other day. I used to notice this in windsurfing. It was so much fun, I would do it till I was exhausted. At some point in the middle of the late afternoon, I would reach—I'd be in the water, reaching for the sail, and the muscle really could hardly do it anymore. Then I knew it was time to get to shore and get off the water. But being moderately tired, you really can overcome that.
- 5.52** Anyway, it's the same thing. The mind sends out signals to conserve energy long before you're running out of energy, and that seems to be what depletion is. So the analogy of self-control to a muscle has served me very well over the decades, and whenever I get puzzled about something. So the depletion effect, yes, after you exert it, it gets tired,

but it's conserving energy. It's not that the energy is all gone, that you don't have any anymore.

- 5.53** And that led us to wonder about extension. Muscles get stronger when you exercise them. Right after exercise, of course, it's tired. But if you exercise regularly, it will actually improve and get stronger. And that's been shown with self-control, too, that if you have people do self-control exercises, even for just a couple weeks, it seems to be enough that then they show improvements on lab tests that have nothing to do with whatever exercise they did.
- 5.54** So, for example, in some of the studies for exercise, we borrow things from meditative traditions. For example, if you're right-handed, use your left hand for a bunch of things like opening doors, drinking from a cup, brushing your teeth, a variety of things like that. So the point is, you walk up to a door, you're right-handed, and you automatically reach with your right-hand. You have to override that and instead use your left hand. So that uses self-control.
- 5.55** So we have people do that for a week, and then for another week, maybe have them do verbal exercises, speak in complete sentences, say yes and no, rather than yeah or no or other things. Any variety of things. Don't use acronyms or abbreviations, say the full name of whatever it is. So again, just people have to watch themselves, override their habit, and make themselves do something else. Two weeks of that, and then we test them on something completely different, like holding your hand in ice water, and they do significantly better at it. So it's a general domain thing. It's one muscle used for all these different things, and that's a key point.
- 5.56** Some people say, "I have good self-control for this and not so good for that." Well, yeah, because it's a limited supply of energy. So you channel it into one thing rather than the other. You channel it to get your work done on time or to keep your desk neat, keep up with haircuts and things like that. You can put it into different areas.
- 5.57** Anyway, the four major ones, as I said, controlling your thoughts, which is things like making yourself concentrate, avoiding distraction, shutting that annoying thought out of your mind of a song you don't like or maybe ex-lover or something like that. So controlling thoughts is important.
- 5.58** Controlling feelings is a little more difficult. It has to be done indirectly. But basically, people are usually trying to get themselves into a better mood, to either get out of a bad one or get into a good one. But sometimes you want to get into a worse mood, if you're feeling all great and you have to go to a funeral; you can't go there with a big grin on your face. "Isn't life great?"
- 5.59** Impulse control is the traditional one. Not just, say, resisting temptations to drink or smoke or whatever, but it could be aggressive impulses or sexual desires to sleep with the wrong person. Self-control is important for those. Even saying something you shouldn't say. So impulse control is the important one.

- 5.60** And last, task performance is another application of self-control, making yourself continue working when you feel like quitting, when you're tired. Maybe trading off speed versus accuracy. Do I want to speed up and get more done? Or do I want to slow down and make sure I don't make any mistakes? All those sorts of things are there, too. So I'm not saying those are all of them, but it's a pretty thorough list of most of the applications of self-control.
- 5.61** Now, this has been applied to a whole lot of things. Let me just run through a few here as some examples. For one thing, intelligent thought, reasoning. Yes, when people are depleted, their scores are lower. This is a pet issue for me. I had colleagues who were intelligence researchers, and intelligence is like the big brother. It's been studied in psychology for now about a century and a half. There's lots of work on it and so on. And really, in the big picture, psychology's found two things that are pretty much always good for you.
- 5.62** One is that intelligence, I mean, intelligent people do better at almost every job you put them in, from nuclear physicist to janitor or waitress, and produces a variety of other benefits, too. Intelligence is good, has very little downside. Being less intelligent is really not helpful in just about any setting. And the same for self-control. So self-control had come along later.
- 5.63** So when I started finding these things, intelligence researchers said, "Well, you're probably really just studying self-control." They said, "You're probably really just studying intelligence." So we were able to show, well, maybe you, with all your stuff on intelligence, maybe that's really about self-control. Because look at this, when people's self-control is down, they do worse on an IQ test. It was a fairly sizable drop, too. But making yourself think carefully, discipline your thinking process, consider all the evidence—that's how you get problems right on one of these intelligence tests.
- 5.64** And again, it mainly applies to reasoning. It doesn't apply to things like vocabulary or other automatic responses that are also part of intelligence testing. But when your self-control is down, where you have to actually do something actively with your mind, take some information and reason to something else, that's where things deteriorate.
- 5.65** Lots of interpersonal effects of self-control. For one, obviously, when people are depleted, they become more aggressive to others. There are a lot of aggressive impulses, and people restrain them most of the time, too. As they know, that's not appropriate. A variety of things are found with couples. People have self-serving attributions that they think, they take more credit for success, and they deny blame for failure.
- 5.66** Well, when you have a couple, a romantic couple, working on a problem together, and you tell them, "Well, you guys, you failed. You did one of the worst jobs of anything we've seen in the experiment," mostly they have partner-serving attributions. So the man will say, "Oh, it was my fault. I insisted on doing it my way." And the woman says,

"No, no, no, it was my fault. I made a dumb suggestion that got us off on the wrong track."

- 5.67** But when they're depleted, then they start blaming each other. And so that's a risk area for marriage and for relationships, especially in the modern world, where people work hard and come home depleted at the end of the day, and they're not going to be as careful to do the right things and say the right things to preserve the relationship.
- 5.68** There's even a study that we did. It was building on an earlier one, which got couples who were happily dating and said, "Well, we need you to rate some stimuli for other experiments coming." And so they sat down, and the stimuli were pictures of attractive members of the opposite sex in bathing suits and things like that. And they didn't care about the rating so much as how long did the people spend looking at these attractive pictures.
- 5.69** And that predicted whether they broke up eight months later. The researchers contacted them at the end of the school year and saw if they were still dating or not. And the ones who'd spend a lot of time lingering over, "Oh, she looks really nice," they were presumably less protecting against other temptation. And so they were more likely to have broken up during the course of the school year.
- 5.70** Anyway, what we did as a follow-up was deplete people and give them the same measure. We didn't need the follow-up. But sure enough, the people who were happy in relationships just went right through. "That's a six." Went right through them. But the people who were depleted spent a lot more time in, "Oh, oh." So again, there's something natural when you're in a relationship that protects you by making you not really pay attention to attractive alternatives. But when your willpower is down, that doesn't function as well, and you don't self-regulate as effectively.
- 5.71** All right. Impulsive spending. I wasn't involved in this one, but a couple of very smart people looked at that—the idea of that when people are depleted, they spend more impulsively. So they had people come to an experiment and do some tasks which depleted them or not. And then they were paid \$10 doing the experiment for money.
- 5.72** And as they left, the experimenter said, "Oh, by the way, the bookstore is having a promotion, is offering all these things. Do you want to buy any of these products, stickers with the university logo on them, or pens and coffee cups and things like that?" And the people who were not depleted just said no. I think the average was 50 cents out of the \$10. But the depleted people spent over half of their money, over \$6, out of the \$10 that they'd earned. They hadn't come to shop. They didn't need this stuff. But the impulse just got to them because their willpower was reduced.
- 5.73** And so, instead of going home with the money, they went home with a coffee cup and a pencil or something like that. They also, in one of the follow-up studies, showed people are more willing to pay a higher price for the same set of items, the same good. What's the maximum? In marketing research, what's the maximum price you'd be

willing to pay for this product? Well, it's a higher product if your willpower is depleted. So again, don't shop when you're depleted. You'll come home having paid a lot more and bought a lot of junk that you don't really need.

- 5.74** So what happens when you're depleted, the automatic system is still working. The automatic responses are there, and they may gain relative influence because the self-regulatory capacity that's supposed to override them doesn't do it. These German people did a study, for example, that they measured how much do you like candy, in both your automatic and your conscious. So consciously, people say, "Oh, it's not good for me." Unconsciously, "I like chocolate." And they depleted some and looked at how much they ate.
- 5.75** The non-depleted people, what they thought about candy, their conscious reaction dictated how much they ate. But for the depleted people, it was their unconscious attitude that dictated. The conscious thing wasn't guiding it anymore. So again, that sort of shows how this system works. I have values that I believe in and so on, but I also have unconscious feelings and urges and so on. And as the conscious structure gets weakened by depleting your willpower, the unconscious one takes control and can guide behavior and do things that you might end up being sorry about.
- 5.76** Lots of studies also show, sure enough, consistent with the moral muscle idea, that when people's willpower is depleted, they're more likely to do immoral things. Studies of cheating and lying and so on, a number of clever procedures. We have people take a test and score their own test, and then bring it in and say how many they got right and get paid based on that. And the depleted people somehow claim substantially more correct answers. And as far as I can tell, they didn't really get any more correct answers. But they're more willing to claim they did, because they get more money when there's no way to follow up.
- 5.77** We usually study this ego depletion as a temporary state. You come home at the end of a hard day and, "I don't want to make any decisions. And I don't want to be tempted or anything like that." But can it become chronic? Can it become a state that's permanent? This is an interesting idea that's just being explored now. It might apply to things like burnout, where people just feel they can't fight it anymore.
- 5.78** We thought of several possible pathways for it, and there's a bit of evidence for each of them. The first one is that people have the usual amount of self-control, but they're just in a job or whatever that always overtaxes them. So these jobs are frustrating, you don't have enough resources, you have too many clients needing your services or whatever. It's the same if it's not a job and taking care of people, a sick partner, family member, something like that. People put a lot of self-discipline into doing that, and so that can make them chronically low in willpower.
- 5.79** The second pathway into it is, you get depleted, but you don't recover that well. Vacations were invented in the 1800s and sold to management because the workers

would actually perform better the rest of the year if they could get a break. It used to be, I think US Steel, they said, even at the turn of the 20th century, the standard was six days a week, a 12-hour shift. So there was a lot of work, but vacations, having time to recover and feel, then people can come back to work with a new positive attitude and so on. But people don't recover if they're on vacation, they get sick or they don't enjoy the vacation or whatever. Also, not getting enough food or sleep might prevent people from recovering because that's how many people do it.

- 5.80** There's a third possibility that some people just have lower willpower or lower energy to begin with. There's a little bit of evidence of that. That, I think, is more tentative. Anyway, the idea that depletion could be chronic state, that's at present mostly a speculative idea at present.
- 5.81** All right, I've talked about self-control, but does it more, is this energy, this willpower energy, used for anything else? Well, yes, there were a couple other things. One is making decisions. The term decision fatigue has been taken up; it came out of our research, that maybe making decisions uses up willpower. And so, after you make a lot of decisions, then your willpower is lower, and the decisions will get worse, self-control will get worse.
- 5.82** One of our studies, we had people make a series of choices among products, or thought about all the products and rated whether they'd used them or something, but didn't make any decisions. So you wanted to deplete people by having them make a lot of choices. And then a classic test of self-control is how long can you hold your hand in ice water. You maintain it at 1 degree Celsius, 33-34 degrees Fahrenheit. And so it's cold, and your hand wants to pull out. So how long can you make yourself keep it there?
- 5.83** And you see, the people who had made choices gave up in half the time, less than half, as the people who hadn't. So again, making those decisions—and we told them they would get to keep one of the products—making those decisions took something out of them that wasn't there anymore, available to help them keep their hand in the unpleasant cold water.
- 5.84** Just to get moving a little faster now, so we look, when your willpower is depleted, how does your decision-making change? The study I just showed is, the effect of making decisions depletes you, and the effect is on self-control. So now let's use self-control to deplete them and see how their decision-making changes. So one is that people just don't want to make a decision.
- 5.85** I knew a guy with—his wife had a very stressful job as a therapist. She comes home the other day, she says, "I'll do anything you want. I just don't want to make any decisions. We can go out to dinner, tell me what to cook. I'll cook it." He says, "Well, we got to plan our summer vacation." She says, "No, no." So, and sure enough, in lab studies too, people prefer not to decide.

- 5.86** I went to a bunch of the decision-making researcher conferences, and they say, "Well, which would you pick out of these two computers or two phones or two shoes or whatever?" They have different features and they look at how people do it, but they never give them the choice of just saying, "Neither. I'm not going to decide. I'll keep looking." So we just repeated the studies, including that option. And sure enough, the depleted people were more likely to say, "I can't decide. I'll keep looking. I don't want to pick either one."
- 5.87** Compromise. People are naturally compromisers. And even with things like price and quality, the price goes up, the quality goes up. You don't want to have a high price, but you do want to get quality. So people will look for the sweet spot where you get the most quality for the price you're paying and so on. Depleted people are more like, "Just give me the cheapest." They pick one dimension and maximize on it. Sometimes they'll say, "Give me the highest quality one," if they don't care about money.
- 5.88** But instead of compromising between the two, they'll just make an easier decision. And the decision-making researchers, I think Uri Simonsohn is one of the early leaders, said, well, a compromise is very mentally complex as a kind of decision, because you're integrating different perspectives or different criteria and putting them together in some way. So it's no wonder then, when people are depleted, they want to have a simple decision-making. So it's just, "Pick one dimension, give me the cheapest. That's fine."
- 5.89** Sometimes there's a default option, and people become more likely to choose that when they're depleted. A lot of options and systems and so on. They say, "Well, you can have anything you want, but the standard one is this." There's a clever study done with Audi dealers. Some marketing researchers got in touch with Audi dealers. Apparently, when you buy an Audi, a car, you make a whole series of decisions, and they all have a standard option. So there's a standard sequence they go through.
- 5.90** But the researchers got the dealers to just scramble the sequence at which they asked people these questions. And sure enough, as they went along and their willpower got depleted and decision fatigue set in, they're more likely to say whatever standard. I'll just take whatever standard. It turned out, the Audi dealers already had an inkling of this. So they had come around to set things up, so that the first decision you make is like, what interior fabric do you want?
- 5.91** And there's so many different options, and you've got to look at the different colors and types of fabric and all these things. You use up basically all your willpower, deciding which shade of taupe would be the right thing for the interior of your car. And so later on, when it comes to things like rust-proofing, "Yeah, whatever, that's fine." And so it drives up the price of the car. So anyway, keep that in mind.
- 5.92** The idea for this first came from one of our postdocs who was getting married, and she did the bridal registry thing, where you go to a store and you pick all the things you

want people to give you for your wedding. And it's a nice thing to do, but she said, "Wow, when I was done with that, you could have talked me into anything. I was so wiped out, just making all those choices. What design do you want on the spoon and what shape should the gravy boat be?" All those things you have to make the decisions. Anyway, the point is, people become more prone to the default option.

- 5.93** Also just leaving things as they are, rather than changing. It's like an investment thing where there was a pretty good—you had some money in the bank, hypothetically. You could leave it there or you could invest it in this thing, which had a moderate risk but high return, so it's a pretty good thing. Then you have to take more risk. But the more depleted they were, the more they were just, "I'll leave the money in the bank. I'm not going to deal with it or make any changes." So again, it shows this general passivity, not wanting to decide, not wanting to make changes.
- 5.94** Impulsive behavior increases more. One of the early studies I liked, it wasn't perfect, but it was well done. They did it at Yale, and they got students going to the library and coming home from the library after several hours of study. And they offered them—this was back when you rented videos—and so they offered them a certificate to rent a video from a local store. And they had to choose what movie to watch.
- 5.95** And there were some high-brow intellectual artistic movies, and then there were some low-brow exploding cars, naked women, that kind of thing. So the students going to the the library, you know, Yale is an elite school, and they would tend to pick the intellectual, artistic, stimulating movie, going to the library. Coming home from the library, they wanted the exploding cars, the low-brow stuff. So again, the lower impulses tend to take over as you get depleted.
- 5.96** One more finding, irrational bias. Usually, when you have to make a decision, you have a lot of information, and only some of it is relevant. And do you screen it out? Actually, this study is complicated. So let me just say, irrelevant information creeps in and distorts the decision more when you're depleted than when you're not. Sometimes people are good at that.
- 5.97** Anyway, that's the set of things. So the general pattern is a shift toward low-effort ways of thinking—not thinking things through, just give me the cheapest, whatever. Not compromise, irrational bias, and so on. Also, looking what's best right now or leaving things as they are, which is kind of the safe option. That's essentially what's happened. When your willpower is down, decision fatigue happens.
- 5.98** I should mention, there's our former president looking very dapper in his gray suit. We were doing, much of this research was getting publicity during his presidency. And I don't think he read our research reports, but somebody on his staff read the New York Times featured article on it. Anyway, he was interviewed. He said, well, he wears only gray and blue suits because he doesn't want to make a lot of decisions. He learned that there's a limited amount of energy for making decisions. Of course, the President only

gets the tough decisions. The easy ones get made lower down the chain of command. Only the difficult ones come up to the top guy, so it's a lot of decision demands.

- 5.99** And I've heard other people, the Facebook guy, Zuckerberg, was doing the same thing, wearing the same clothes every day, so as not to waste your willpower making decisions in the morning. And he mentioned the research that making decisions degrades your ability to make further decisions. And that's what he was wanting to conserve for the important things for our country. I don't know about the more recent presidents, whether they follow the same system or not.
- 5.100** So just a couple more things that apply. So making choices, making decisions, that depletes willpower. Another was active or passive— if you become more passive when you're depleted, just go along with things, whatever, rather than taking initiative. One of the early experiments I liked, we depleted people or not, and then said, OK, for the second part of the experiment, the experimenter took you into a room and says, "The computer will give you instructions. Just follow the prompts on the computer."
- 5.101** And the experimenter leaves, and the computer just goes to blue screen, and nothing happens. And so the measure was, how long do they sit there staring at the blank screen? We'd taken their phones away so they couldn't entertain themselves. But clearly, something is wrong. This is another thing we borrowed from stress research. How long do you just sit there before you go and say, "Hey, something's happened"?
- 5.102** And as you see, the depleted people sat there twice as long, doing nothing. It was harder for them to summon up the initiative to go get the experimenter, say, "Hey, you said the computer would give me instructions, and it's not working, not doing anything." Let's see. So again, it looks not just for initiative and something's wrong, but other behaviors. Again, there'd be more passive, more likely to do nothing. So what can happen when you're depleted, the top-down mental control isn't there, so the automatic mind takes over and does whatever is simpler.
- 5.103** We had an experiment we borrowed from a procedure back in the '70s, where you give people some tasks to do on the computer, or in the '70s, it was on paper. It said, "Oh, by the way, help yourself to these nuts." And they put out a bowl of nuts. They're either in the shells or not, so peanuts in shells or not. And the difference is, of course, it's a lot more work to eat peanuts in the shells because you have to break open the shell and take them out. And so if they were depleted, and the nuts were easy, were there, were already out of the shells, all you had to do is grab a handful and throw it in your mouth, then the depleted people ate more. But if they had the shells on, then they ate significantly less. And it only worked for people who were hungry. I know it was complicated, but still, it showed effects in the opposite direction.
- 5.104** We had to do this to satisfy the reviewers who said, "Well, sometimes depleted people do more things, like they're more aggressive. So it isn't just a general passivity," and that's correct. But this, we showed, they basically do whatever is easier. They feel like

eating, they'll eat more. If they're depleted, if it's easy, but if they have to work to get them out of the shells, no, it's not worth it.

- 5.105** And the fourth and final one was planning, that planning takes willpower. People with higher self-control do more planning. After people are depleted, they don't want to engage in planning as much. They show a reluctance to do it and so forth. Having plans also will help you function when you are depleted. If you know exactly what you're supposed to do, that helps you get it done when your willpower is down.
- 5.106** I should mention there's another line of work that we've been doing more recently. Being in an uncertain situations seems to—we start to think it doesn't really use up your willpower, but it mimics it, because uncertainty is a cue to conserve energy. So depletion effects are conserving effects. So yeah, you go to conserve your energy more.
- 5.107** All right, one or two more things to cover. I've talked about self-control basically is a state. Thus far that your willpower is high, you can regulate yourself well. When your willpower is depleted, then it's not so well. But we also know there are differences among people, personality differences. I mean, just think about your friends. Some of them have better self-control than others.
- 5.108** And so there's also a trait scale that we developed that we can use to test people for their level of self-control, based on self-report. And it works very well; thousands of studies have now used it. One of the interesting questions was, well, who would be more depleted? Who would have the ego depletion effect more strongly? Is it people with high self-control, because they have farther to fall? Or people with low self-control, because they're already on the borderline and they might fall earlier?
- 5.109** And we looked for this in many studies, and basically, the finding is, there's no difference, which is somewhat of a surprise. High and low self-control people get depleted at the same rate and show the same size effects. The one exception to this was one where we had those studies where we follow people around in their daily life and beep them and had them report on feeling depleted, which is a little more difficult, and see what else was going on.
- 5.110** And there, in daily life outside the lab, people with low trait self-control had much more frequent depletion. So what is this? Is there something wrong with the lab, or is it something weird about daily life? Well, in the lab, everybody does the same task. So everybody gets depleted to roughly the same degree. In daily life, people with low self-control shoot themselves in the foot a lot, and they spend a lot of time digging themselves out of holes that they put themselves in. So daily life is much more difficult for people with poor self-control.
- 5.111** I mean, you don't pay your bills, and then the electricity is shut off, or you say something you shouldn't say to your partner, and then you're sleeping on the couch, and all kinds of things like that. People with low self-control have more interpersonal conflicts than other people. And so, their lives from moment to moment are much

more difficult, and that's why they have a higher rate of depletion outside the laboratory. So again, if you're raising children or whatever, try to give them good self-control. It will make their life better in just about every way you can measure.

- 5.112** So high self-control people, it's not that they have more willpower, they just use it more effectively. This is a big change in our thinking, too, that they form good habits and they break bad habits, so that daily life goes habitually, and automatic pilot can run through, and it runs pretty smoothly. People with poor self-control don't do that as much, and so they expose themselves to more temptation, or they have habits that bring them difficulties and trouble and so forth.
- 5.113** I mentioned that study earlier on people's desires. When we went and looked, we had a trait self-control measure there. So we said, well, who is resisting desires more? I assumed it would be high self-control, since part of that is for resisting desire. People with high self-control should resist them more. But no, it was the opposite, that people with high self-control were less likely to report resisting desires over the course of the week that we tracked them. Well, that was a bit of a puzzle.
- 5.114** But what we realized is that they're not getting themselves into situations where they have to resist desires. So if you're trying to quit drinking, you don't go into a bar with your friends and say, "I'm just going to have a Diet Coke." But then after you sit there for a while, think, "Oh, that beer smells good." And so they get tempted. But people with good self-control just don't go into the bar; avoid the problem, and so they don't have to resist the desire.
- 5.115** So it was really less resistance, not because they indulge themselves all the time, but because they avoid situations where they have the problem. They felt less guilty, they had less stress, and so in our book, we called it playing offense. You use self-control to avoid problems, to prevent them, rather than to deal with them at the last minute.
- 5.116** I think we're in the final minute here, so let me go to one last point—the question of, what is it that gets depleted? Is willpower, we'd used it as kind of a metaphor, but is there something real in terms of the body energy? One of my students came up with the idea that, well, if resisting temptation weakens your self-control, would indulging temptation strengthen it? So I said, I didn't think it worked like that, but it would be really popular if it works. So go ahead and try it.
- 5.117** We were calling it the Mardi Gras effect, because Mardi Gras, people go out and have a big party and so on just before Lent, presumably to strengthen them or make them ready, able to tolerate whatever they're giving up for Lent. And so I said, "Go ahead and try it." They depleted people. They gave some of them ice cream and had some of them do a boring task, instead read some magazines. And then there was a third condition, just in case there was something special about eating, had them eat something that didn't taste very good. It was like ice cream without sugar, with some kind of quinine or some awful flavor in there. Yuck.

- 5.118** So anyway, I forgot about it because we were working on all the things. And he came in a month later and he said, "I ran the study. It didn't work." He was looking at his shoes the way the students do when the study failed. And I said, "OK, so the ice cream didn't overcome the depletion effect." He said, "Oh, no, it actually did. But so did the yucky dairy blech, blah, blah, thing. So it's obviously not yielding to temptation." I said, "Well, wait a minute, though. You still did get rid of the effect. And maybe it wasn't the pleasure from eating the ice cream, but maybe it was just getting the calories." He said, "Oh, that's interesting."
- 5.119** And it was great. He went over to the library for a month or two and read all these nutrition studies. And sure enough, nutritionists had found these things, too, that when the level of glucose in your blood is low, then you're prone to do all sorts of antisocial or problematic things. They didn't quite have a theory about it; they just showed it in different respects.
- 5.120** Somebody showed that juvenile delinquents who were arrested had really low blood sugar, which, OK, could be that being arrested is bad for your blood sugar. But it could be, if your blood glucose was low, that's why you did the things that got you arrested. You didn't have self-control enough to obey the law.
- 5.121** Or my favorite study, they have these ways of manipulating glucose in your bloodstream. They manipulate it all, and they had people work on a computer with a really frustrating computer problem. And the measure was, did they actually hit the computer out of frustration? And sure enough, the people with low blood glucose were more likely to perform violent acts against the computer.
- 5.122** Anyway, any time you're looking to merge mind and body things, you're asking for trouble. It's a difficult topic. Some people find these effects. A lot of people have, and some people don't. I'm not totally sure about it there, but my guess is that glucose is an aspect of it. Glucose is, again, the fuel for brain processes and so on. All thinking uses glucose. And so that may be what somehow gets used up or depleted when people are engaging in self-control and getting into this depleted willpower state.
- 5.123** We had a series of lab studies that we did. The first one was that we measured glucose before and after acts of self-control, and it went down. But that hasn't replicated very well, so I don't really believe that anymore. That might have been a fluke, the first one. But the other two are very solid. When the glucose is low, your self-control is poor. And as I said, the nutrition researchers had already established that a lot.
- 5.124** Diabetes researchers find the same thing; when people, diabetics, are prone to have low blood sugar, and then they have trouble making up their minds or resisting temptation or anything like that. And then also giving people glucose in the middle of an experiment replenishes and restores their performance. And we found that over and over in lots of things.

- 5.125** All right, we're at the end of our hour here. The new view, I think glucose is part of it, but I can't swear to it. It's complicated. The body doesn't know how much energy it has. The brain doesn't know. But it might estimate by how much you've used recently. I call it counting the ashes. They're byproducts of processing glucose in the brain. When those build up, that could be a cue to the brain to back off and not do that anymore. That could be what's mediating these effects.
- 5.126** Let me just sum up here. First of all, certainly a lot of psychological processes operate as if there's a limited energy supply that's needed for them. We resisted the folk term willpower for a long time, but I think that's what it is, and it is a fair term. Sure enough, it gets depleted. It's not that it's gone, but rather people get in a state where they think they need to conserve what's left, that the brain automatically conserves energy.
- 5.127** We conserve more than we need to. Remember, we evolved under situations where the food supply was much more erratic than it is today, and so it's adaptive to conserve energy wherever possible. We conserve more than we need to, which means you can exert self-control more than you think you can, even when you feel tired. Building character, exercising self-control, does make it stronger.
- 5.128** There's a big meta-analysis, actually there are two of them, done by one European and one Australian researcher recently, compiling the results of many studies that, yes, it does seem to work. Having regular exercises in self-control does indeed improve self-control down the road. So you can, again—the Victorian term was building character; it really appears to be true.
- 5.129** To understand self-control, we need to understand both the state fluctuations in depleting willpower and the trait that some people are chronically better at it than others. As I said, they do things like using their self-control to manage habits, to break bad ones and form good ones. The limited resource of willpower also is used in making decisions and taking initiative, as in being active rather than passive. And planning is also, should have been on there.
- 5.130** And last, as I said, I think that glucose is one of the biochemical mediators of the process, and there's a fair amount of evidence for it. Some people dispute it, and I'm not sure they're knowledgeable as well. But that's what it looks like to me. In any case, it's one of the most important traits to help people get through life successfully. So, worth cultivating in one's own life and one's loved ones, and also worthy for psychology to continue to study. Thank you.
- 5.131** I guess my question is, is there a measurable threshold from the, I guess, cognitively, consciously acting state and the depleted state?
- 5.132** What kind of threshold?

- 5.133 A threshold that determines the separation between, I guess, with the car example, from going with the decision you actually want to make, versus just taking the standard?
- 5.134 Well, I would look at it as conscious and unconscious are working together and sort of competing. And in some cases, the unconscious impulse will be resisted, but it takes conscious effort to do that. And as your consciousness gets tired or more oriented toward conserving energy, then you're more willing to give in. But it's not a firm bright red line or anything like that. It's just probably a gradual, greater tendency or greater vulnerability.

Lecture 6: Sexuality

At a glance — Sexuality.

- 6.1 All right. For this lecture, we'll go back to the nature and culture theme, particularly in relation to sex. I know in these times, the meanings of words keep getting debated and changed and so on. So I like to keep things simple. For me, sex refers to doing it, and gender is all things male and female.
- 6.2 Why would social psychologists be interested in sex, other than the fact that everybody's interested in sex? Well, I taught human sexuality for a while in one of my universities. Other fields cover important things like sexual health, problems, abnormal patterns, or weird desires, and so on. The physiology of how the sex organs work and stuff like that—that's not a social psychology thing very much. How to do it—they give advice for good best practices and stuff. That's not really our job either.
- 6.3 What psychology instead is that this is a way that people interact with each other. It's important for relationships, a certain type of relationships, at least. It's a highly interpersonal thing, except when you're doing it by yourself. We're trying to understand what are the motives, how is the self involved, what other aspects? Indeed, I have a textbook that I wrote for undergraduate social psychology, and we took upon the challenge.
- 6.4 One of my friends was a scholar of eating, and he said, "Eating is the perfect social psych variable because it relates to everything. You can have concerns with self and attitudes and aggression and helping and all sorts of other things." I said, "Well, the same thing could be true for sex." So we had one of those little boxes in every chapter applying, you know, prejudice. Are people prejudiced about who they have sex with? Again, attitudes and self-control and self-concept, things like that as well.
- 6.5 It's an important thing for social psychologists to chime in and contribute what they have to say as well. Now, in terms of understanding sex or understanding just about everything, when social scientists go to ultimate causes, they go back to nature and/or culture, with different people strongly favoring one or the other. So, in terms of sexual behaviors in various aspects, is this a nature thing, as in genetically predisposed or whatever, or is it a product of culture, of experiences, and socialization, and so forth?
- 6.6 So, for example, nature obviously designed us to make babies by having sex with members of the other gender. Why do some people want to have sex with members of their own gender? And so, highly debated for a long time. Is there some gene that predisposes people to be genetic? I mean, over the decades, I've seen exciting announcements in the news that they found the gay gene, and then after a while, it doesn't replicate. But the argument that there's some genetic basis still remains quite plausible.

- 6.7 Or is it something about personal decisions people make? Is it experiences that they have? And so, I mean, homosexuality is a particularly interesting one because, through much of history, both nature and culture were against it. It doesn't reproduce. So any genetic tendency to have homosexuality, you have to ask, why does that gene keep showing up since it doesn't make babies? And yet culture has often been against it as well. And yet it continues to survive.
- 6.8 And it's not just found in humans, which suggests there's some natural aspect to it. Plenty of animals show such behavior, but it's still an ongoing interesting puzzle. So most experts in sex will certainly say that there's some of both. It's not all nature or all culture, but they differ rather severely as to what mixture it is. How much of nature, how much culture, how much are you born a certain way, and how much is it a result of the experiences that you have?
- 6.9 Of course, humans have much more culture than other species. So people who study animal sexuality, there isn't nearly as much room for culture to have an influence, but there could potentially be enormous influence on human sexuality. I think there is. If you look at historical and cultural variations in sexual behavior, they are quite large, so there's certainly a role for each.
- 6.10 So, looking at the cultural theories, these call it sex as a social construction. So, yeah, those people know there's some role of hormones and biological, the sex organs and things like that. But still, I think mainly, who wants to do what with whom, how many times, in what positions and so on, this is a product of your culture, of your socialization, of what you learn, what other people are doing, what the norms are, influences, and so on.
- 6.11 Big emphasis in culture on the meaning. Culture operates by virtue of meaning. So what is signified by it, it's not just an animal activity, but rather it's sort of an expression of certain values or ideas or whatever that becomes highly important in the cultural approach. The feminists, particularly, very strongly on board. They generally oppose biological explanations. And so, and in terms of the researchers doing this, a lot of their cultural constructionist theorists are closely aligned with feminist views and values.
- 6.12 And one thing that really gave a boost to this is the sexual revolution. Occurred, some say it really was a 1970s thing, but it probably had its roots in the '60s and maybe even the late '50s, starting to be at it. The point is, there were so massive changes in sexual behavior over 20 or 30 years that there couldn't have been a genetic change. So that really demonstrated the power of culture, the sexual revolution, that people could change that much in such a short time.
- 6.13 It's not possible to attribute to that to something in the drinking water or genetic changes or anything like that. That made a strong boost to the cultural theories. Meanwhile, the nature theories, most of them take an evolutionary approach and emphasize how continuity with animals, that there are certainly biological factors that

contribute to sex. The idea is that today's desires come from those ancestors who produce the most offspring. Natural selection works by reproduction.

- 6.14 So in bygone eras, the feelings and attitudes and impulses that led to having the most children going forward were the ones that prevailed, and then came to spread through the gene pool, and then are more widespread today than the ones that they didn't lead to nearly as much. So I talk about reproductive contingencies; the males' and the females' involvement in reproduction is rather different, right from the start. We know which one gives birth and so on.
- 6.15 A number of researchers have pointed out that even among different animal species, although the roles sometimes differ between males and females across species, the species that has the bigger investment in making the baby in the first place is usually much more choosy about the other, because it's a much bigger job for it and a much bigger obligation.
- 6.16 So among humans, I mean, the man, what it takes to make the baby is about five minutes of fairly pleasurable activity, and nothing else is really needed. Whereas for the woman, it's the minimum of the nine-month pregnancy. She does have the five minutes of, hopefully, pleasant activity, and then has the baby, and usually it goes on after that. She has to take care of it and so on.
- 6.17 And one of the contingencies the evolutionary people point out is that, if a man sleeps with 100 different women in a year, he can make quite a few babies. Whereas if a woman sleeps with 100 different men in a year, she still can only have about one baby. So quantity is a strategy that works for men in terms of producing a lot of offspring, not so much for women. Women instead want to focus on quality. So human women are much more selective about what they want in a partner in order to consent to have sex with him.
- 6.18 And men are much more indifferent. The idea is, if the man has sex with a low-quality partner, which might produce a low-quality offspring, well, and then a more attractive one comes along, a higher-quality mate, well, no problem. In 20 minutes, he's ready to go again. Whereas for the woman, if she makes the same mistake, so to speak, well, then her reproductive system is tied up for the better part of a year. And so it makes more sense to wait and find the superior quality. So anyway, that's the nature thinking about sex.
- 6.19 Likewise, age of partner that people seek. And it's true just about all over the world, that people mate with the man being a bit older than the woman. And as David Buss has been a pioneer in this research, as he pointed out, well, imagine in the past that half the men desired old women and half the men desired young women, and so there was no difference across the population. And then everybody made it and got what they wanted.

- 6.20** Well, we'd still be descended from the men who made it with young women, because old women past a certain age can't have babies. I may have found the wrinkled 60-year-old woman really hot, but she didn't get pregnant, no matter how much they had sex. So that would be it. And meanwhile, women looking for quality, especially in terms of social status and resources, the older men usually have accumulated a bit more.
- 6.21** So it almost inevitably would evolve that it's fine for women to mate with the older men than the younger men, probably less mature, less able to provide, and thus less valuable as a partner. So the nature of people would also work out, yes, they certainly know the sexual revolution happened. But many aspects of sex are similar across different cultures and historical periods.
- 6.22** Anyway, I got into looking into this. I had just finished writing a book on why is there evil? I'd read all these terrible things people do to each other. And so I wanted to switch and next study something nicer. I remembered the hippie days, "Make love, not war." I said, "I know. Why don't I read the sex literature?" So I went to the library, got volume one, issue one, of the Journal of Sex Research, and read all the abstracts and some of the articles, and then just went through the next—it was about 30 years' worth, doing that without much idea of taking notes, looking for patterns.
- 6.23** And one thing I noticed was that the field—people were even talking about this, too—was divided into two theoretical camps, basically the nature and the culture camp, because the social constructionists and the feminists said sex is mainly a product of social and cultural influences, and they really emphasize that in their writing and their thinking. Genes, nah, they didn't think that that made much difference, maybe a little bit.
- 6.24** And on the other side, the evolutionary psychologists who'd come out later as a bit of a reaction against it, said that sexual desires are shaped by evolution based on that. And these weren't just two camps who had a friendly disagreement or different emphasis. They seemed to despise each other. They said it made it difficult for young scholars coming in the field, because you didn't know who was going to review your manuscript.
- 6.25** So, what they started doing is, not developing any theory about their findings. They would just report what they found. They took a survey and this happened and they found that, but being careful not to align. It was a very difficult situation for these two groups, being that hostile toward each other. So anyway, as I said, most experts recognize there's some degree of both, but the question of how much of one and how much of the other.
- 6.26** What I came around to thinking was that the answer to that differs by gender, that male sexuality is more driven by nature and genes and stuff. Female sexuality is more about culture or socialization, learning, meaning, and that sort of thing. I don't say, at the start, people get sensitive. There's no value judgment as to one better than the other; as far as I can tell, they're both in the reasonable range. I suppose there's a slight

adaptive advantage to having been more adaptive and more flexible in the female pattern to adjust to situations. But it's not a big difference.

- 6.27** So what this evolved into, this idea of erotic plasticity, which basically is how much is the sex drive shaped by social and cultural situational factors. The plasticity term comes from biology, which goes to flexibility in response to the environment. So some things are just hardwired biologically, and that's what you're going to do, no matter what. For a lot of animals, the reproduction is like that. And for others, it's much more flexible, and change is based on the situation. So the argument is, erotic plasticity is higher among women than among men.
- 6.28** OK, so to state this more systematically, the balance between nature and culture differs between men and women—female sexuality being more cultural, male more natural. Again, both have both; it's just the relative contribution of both may be different. There's no value judgment. Neither of them is inherently better than the other. And it can define plasticity as the degree to which social, cultural, and situational factors influence sexuality.
- 6.29** And some people get hung up if they don't know plasticity in the biological sense. They think that means fake or anything like that. No, that's not that at all. All right, so evidence for plasticity. Well, one thing that struck me was the sexual revolution. OK, it changed people's behavior, but it changed women much more fundamentally than it changed men.
- 6.30** Men, after the sexual revolution versus before it, they still wanted the same things, had the same feelings. They had a lot more chances to do what they wanted to do after the revolution. But they weren't that fundamentally different. Whereas women, at a much more fundamental level, changed how they related to their bodies, how they understood sexuality, the kinds of things they wanted, feelings and desires, and so on.
- 6.31** And just about everyone who wrote about the sexual revolution came to that same conclusion, that behavior changed in both. Obviously, they're both doing it, but it was much more sweeping and fundamental changes in women. Now, OK, sexual revolution, but psychologists aren't going to put much credence in that. It's a single historical event. How much can you draw and infer based on one historical event? So I tried to think, what sort of evidence could we find?
- 6.32** How could we test this hypothesis, that erotic plasticity is higher in the female than in the male? So one thing you could look at is, do specific people change over time, over the course of their adult life? If there's higher plasticity, you should show more variation moving from one decade of life, say, from your 30s to your 40s. There should be bigger changes there. That would be a sign of plasticity.
- 6.33** Second, specific cultural causes should have bigger impact on women than on men. If women's sexuality responds more to cultural factors, then we should be able to look at things like education and see, does getting a higher versus lower education have a

bigger effect on changing the woman's than the man's sexuality? And last, plasticity means responding to the situation. This is something social psychologists have studied for a long time.

- 6.34** Do attitudes predict behavior? Do people do what they say their attitudes support? And if it changes with the situation, this should be lower among women than among men. And that's a good, elegant prediction because you predict the social-cultural causes will have bigger effects. The consistency, attitude, behavior, consistency should be smaller among women. They could look for effects in both directions. Let's look at some evidence relevant to that.
- 6.35** So, the first one was change over time. Do individual women change more over time in the course of their adult lives than individual men, in terms of their sexual activities and feelings and so on? Well, the Kinsey Reports, which were the beginning of scientific research, not completely the beginning, but the huge boost to scientific study of sex, they already reported one difference.
- 6.36** You may know the history of the Kinsey Reports. First, they did lots of interviews and things. They published the first study on men, a kind of obscure thing, and it made a huge sensation. But the second book is very interesting, which is the one on women, because they had done the men already. They could say, well, look, the women are different in this way and that way.
- 6.37** So one of the surprises was that we interviewed women about the patterns of sexual activity over time, and there would be peaks and valleys, which they didn't really see in the men. One of the things Kinsey counted was total output, which she counted as the total number of orgasms you had from all sources, all activities, within a given period of time. So a woman might be having an active sex life with a man that she liked, then they break up, and then she'd have no sex at all for 6 to 8 months, and not miss it, apparently, and then find a new partner and start having enthusiastic sex again.
- 6.38** That kind of thing just didn't show up with a man. I mean, if a man was having sex and broke up with his partner, masturbation would start, if not that same afternoon, certainly by the weekend, so that the men kept their total output more at the same level of what their desire was, not showing the variation in the women. So that was one early sign.
- 6.39** Another thing is, do people take up new sexual practices in middle age? You can look at people when they start having sex in late adolescence, early young adulthood. But say, between the 30s and 40s, do they pick up new practices? Well, with men, it turns out, not so much. What the man desires to do at 25 is the same things he wants to do when he's 55, maybe a little bit less as he gets older. There's a decline, but very little adding new stuff.
- 6.40** Whereas women would show, would take up new things, might start masturbating for the first time in her life in her 40s, or have a lesbian affair, or want to try new things. So

in general, the pattern as you get older is less sex. But the exceptions of taking up, especially new practices and stuff, was almost exclusively found among women. So again, score one for the women.

- 6.41 In adjusting to marriage, both the husbands and wives agreed that the wife had adjusted more in terms of sexuality, adjusting to her husband, than the husband had adjusted to her. So that was one sign. We'll come back to that because it's an interesting twist, but that's one. One good measure of plasticity is looking at people who are homosexuals and say, "Have you ever had heterosexual sex?" In other words, if your preference is your own gender, have you ever gone to bed with someone of the other gender?
- 6.42 It happens in both, but much more among women. Far more of the lesbians reported having had sex with men than gay men had reported having sex with women. Also, there's more just even switching back and forth. One of the researchers, Lisa Diamond, had built up this big sample of lesbians that she wanted to track across time to learn about female homosexuality. She said, "Oh, what a headache," because she has this woman in her study, she's collecting these data, and then the woman breaks up with her girlfriend, and she's starting to date a guy. She says, "You're messing up my data. How can I include you in my study?"
- 6.43 But sure enough, there's much more switching back and forth. Men seem to be one or the other. Women will do both. There was a fair amount of research in the Journal of Sex Research on swinging for a while. This was late in the sexual revolution in the '70s, mostly in early '80s, which was consensual married couples would go to a motel or go to one of their houses and each have sex with the other's partner. And once the diseases started coming out, then that put the kibosh on some of that.
- 6.44 I guess there are certain ways it's still in practice, but this was a new thing being studied at the time. And one thing that was reported was, OK, so obviously, these people are open-minded. They're bringing their spouse along and trading off and having sex with somebody else's spouse. But in the course of doing this, do they ever have same-sex activity? One of the estimates was, like, three-quarters of the women at some point would try having sex with another woman, but it was something only like 4% of the men.
- 6.45 Being in an open-minded situation sexually inspired the women to say, "Oh, maybe try having a lesbian fling." But it didn't seem to have the men do it anyway. All kinds of pretty strong evidence. Another thing was in prison. It seemed to be, in prison, sexual partners are limited. You're only in a prison of your same gender. And so, women seemed to switch in and out of homosexuality quite easily.
- 6.46 They had no problem during, in prison, they would become a lesbian for a while, have sex with other women. And then when they got out, they would go right back to having sex with men. The male prisoners didn't show this nearly as much. Certainly, there's

some, but it was more of a problem for them and less of an easy adjustment. So that's changes within person over time. Again, multiple lines showing greater changes among women than among men.

- 6.47** So let's move on to the second way of testing now. Let's look at specific social-cultural causes and see if they have bigger effects on women, as, for example, the sexual revolution did. So one thing is education. Well, the first two, education and religion, were featured in the best data we have on sex, where this national survey conducted in the 1990s, because they really did get a representative sample. It's hard to get good data on sex. We know a lot less about sex than many other phenomena because we don't have people have sex in the lab very much.
- 6.48** And so we rely on self-report. And well, first of all, are self-reports accurate? Sometimes people exaggerate or whatever, and also some people just don't want to talk about sex. So what they find is that, if you rely on a voluntary sample and say, "We have a questionnaire about sexuality. Would you please fill out a questionnaire?", the people who say, "Sure, I'll fill out your questionnaire," they're more pro-sex people who are having more sex. A lot of the data about sex just overestimate how much sex is going on.
- 6.49** This study was really systematic. It got a representative national sample. It got over 80% of the people they approached to participate in it. Often, you only get 5 or 10%. As a full range, there's not nearly as much sex as reported in various other surveys. Anyway, one thing they looked at in chapter after chapter was education and religion. I remember reading their book saying, "They don't see the pattern? It's there in all their tables of supporting the plasticity argument."
- 6.50** Basically, highly educated women and low-educated women have very different sex lives. Highly religious women and very religious women, not religious women, have also very different sex lives. The differences for men are much smaller, almost disappointingly small. I mean, a lot of the religious men talk about sex being bad and so on, but you see what they're actually doing. It's often not all that different.
- 6.51** And from a social science perspective, trying to make scientific evaluation of the evidence is especially valuable because education tends to push for more sex, more experimentation, more partners, more different practices, and religion for less, the highly religious people. So we have two forces in society, education and religion pushing in different directions in sex, and both of them have bigger effects on female than on male sexuality.
- 6.52** I'll just point out, being scientifically responsible, these are mostly correlational data, so there's always the question of what causes what. But some other findings, cultural variations. Anthropologists have collected data from many different cultures. Somebody went through all the notes that they'd pooled together in these huge, they're called the human relations area files, so field notes on all the different cultures,

looking at adolescent sexual behavior—a much greater variation across different cultures in female than in male sexual activity and interests and so on.

- 6.53** There's even a couple of studies that looked at when people move from one culture to another; do they change? And the women change more. I think it was like, Latin Americans moving to Detroit, Michigan. And the women who did that, well, they adapted to the norms in Michigan, whereas the men tended to stay with what they had done with their home country. So again, cultural having more influence there.
- 6.54** Peer groups. Well, peer groups have some influence on both males and females, but it seems to be much stronger in the female. So, like a girl who has a best friend who is not—if she's a virgin, her best friend is not a virgin, and then they follow up a year later, she's six times more likely to have become not a virgin herself than if her best friend was a virgin, and it was two virgin friends.
- 6.55** So the influence of the peer group, and particularly what they think the peer group would approve—I mean, this had almost no effect on the boys, but the girls, yeah, the girls were really sensitive to what the peer group would approve. So again, external situational social factor, having a bigger influence on the females than on the males.
- 6.56** Parents. Parents is a little more complicated because you can't assume parents socialize their sons and daughters equally on sex. Parents want to make sure their daughter doesn't ruin her life by getting pregnant and so on. But still, even things like parents getting a divorce seem to have a bigger effect on the daughter's sexuality than on the son's. And we assume that isn't a factor motivating people to get a divorce.
- 6.57** Another way of even approaching the same problem is, instead of focusing on the social-cultural causes, what are the genetic causes? Because it's either the genes or the environment. So the best practice for this is to study twins. They study twins with identical genes, different environment, versus same environment, but different genes, so non-identical twins, and look for differences there. And that's a way of prising apart the environment versus the genetic factor.
- 6.58** And so a variety of things show the male is much more genetic, whether it's homosexuality. They can look, is the one twin gay, then is the other twin gay also? Other earlier analyses looked at how many of your cousins are gay, things like that—to look for, is there some of this genetic in there? Much bigger influence on males than on females.
- 6.59** It applies to other things, too. Like, one study looked at how old were they when they started having sex. And again, the identical male twins were much more close in age, doing it at about the same time, than the identical female twins. So again, if you apportion genes versus environment, the male is more genetic, the female is more environmental.

- 6.60 And last, how much do people feel that their sexuality is a result of their own personal choice? And that's much higher in women than in men. So again, they look at, are you gay or heterosexual? The men, both the gay and the straight, think, "Yeah, I had nothing to choose. I couldn't convince myself to do the other." Whereas some women say, "No, it was up to me. I chose to be a heterosexual or a homosexual."
- 6.61 There was a period during the women's movement when they were telling women, "Well, men are the enemy, so you shouldn't sleep with the enemy. So it's your political duty to become a lesbian." And some women actually did make a conscious change in their preferred sexual partner for political, ideological reasons.
- 6.62 For men, there's no sign of that. If the President and the chief of the top corporations or the top athletes all went on TV, holding hands and said, "Oh, guys, you should turn gay," I don't think it would make a bit of difference to most males. So again, there's a socio-cultural influence having a bigger effect on women than on men.
- 6.63 The third criteria and the third way of measuring this is looking at consistency. As I said, this should be lower among women than men. If women respond more to the immediate situation, then their behavior should more often deviate from what their general attitudes specify. So a variety of things on that. One thing is, they looked at people doing things they don't approve of. And women may have more of those, but men have plenty, too, that they don't approve of doing.
- 6.64 Anyway, women were much more likely to report having done, engaged in some sexual practice that they didn't really approve of. They didn't really think it was a good thing, but they got carried away at the moment or whatever and went ahead and did it. So again, the situation might have been stronger. Another, most people disapprove, if you're married, of having sex with someone else other than your spouse. Do they actually do it?
- 6.65 It turns out, it's hard to get data on that. I mean, there is some suggestive data that men do that more than women, but there are other possible explanations for that. But once, some researchers did, they found is a good way, instead of looking at being married and actually going all the way with someone else, they looked at dating couples, when you're in a serious relationship. Have you ever kissed and necked and hugged and not necessarily gone all the way, just had any low-level sexual activity with somebody else?
- 6.66 And there, they then did get enough variants to work with, to look at what predicted it. And so they measured their attitudes toward doing something like that. I mean, is it OK to have a 20-minute necking session with some hot guy, even though you have a steady boyfriend? Is it OK to do that or not? And have you actually done it or not? And again, the difference, the men's attitudes predicted their behavior much more consistently. The women are more likely to in either direction.

- 6.67 They say, "I wouldn't mind doing that, but I've never done it." Or, "I don't believe it. I don't approve of doing it. But yeah, a couple of times," and so on. Condom use. Sex researchers track that. And what are your attitudes? Most people say, if you're going out to have casual sex, you should use a condom, so most people believe it. A lot of people don't really keep to that and go ahead and have unprotected sex. But the difference is bigger for women and for men. It's substantial in both, but the difference is bigger for women than for men.
- 6.68 Having sex when you don't feel like it, that's an interesting case of it. Both genders do that, but women do it more. One study found that every woman past the age of 26 had, at least occasionally, had sex when she didn't really want to, usually because her partner wanted to, and she wasn't in the mood, but she wanted to be nice to him and take care of him. In that sense, again, it's more of a specific attitude, but again, more willing to act contrary, based again on the situation.
- 6.69 Same-gender sex. Well, that classic study from the '90s, the National Health and Social Life Survey, measured homosexuality in three ways. Do you identify as a homosexual? Do you feel desires to go to bed with someone of your own gender? And have you done it in the past year? Now, I'll set aside the identification one. But the other two, "Do you want to do it and have you done it in the past year?", for men, those are almost the same question.
- 6.70 If they said, "Yeah, I want to go have sex with men," 90% of them said, "Yeah, I've done it in the past year." For the women, they were completely uncorrelated, they were highly different. Lots of women said, "Yeah, I like the idea of going to bed with a woman, but never had the opportunity. Somehow it never happened." Others say, "No, I don't approve of that. I don't believe in homosexuality," but asked if they did it, "Well, there was this time, and she was kind of nice or whatever." So, again, bigger discrepancy between the attitude that I like doing this and the behavior of actually engaging in it.
- 6.71 At one point, I read the literature on sexual masochism of submissive sex acts. And it's a minority of both men and women, but again, a larger proportion of women than men have these fantasies of submissive fantasies. But a larger proportion of men than women have actually acted on them. So again, it's usually the case, the men who want to try this out find someone or some way to do it at some point. But for the women, it's a much bigger difference. They might have the fantasy or the interest, and never getting around to acting it out.
- 6.72 All right. Anyway, there's a little bit more evidence in the original paper, but you get the idea. I hope it's been somewhat convincing thus far, that this is a big difference. Now, of course, we can look for exceptions. Are there any cases in which this is not true? Well, there are some signs that early childhood experiences have a bigger effect on males than on females.

- 6.73** So, among other things, I mean, men have a lot more weird sexual tastes. The terms change; perversions is a traditional one. The sex researchers don't like to use the value-laden term perversions, so they call them fetishes or paraphilias, or sometimes the term. Anyway, men have more of these than women, but it didn't seem to be a plasticity thing. It didn't seem to be that they were flexible.
- 6.74** The man has a shoe fetish, and just a woman's shoe really turns him on. This is not something he decided. It's not something he can give up easily. If he gets married and his wife says, "Oh, don't be messing with my shoes," that stays there. Therapy for these people, the best goal that they aim for is to get them to be able to enjoy sex without the shoe, but they can't really decondition. He still likes the shoe, if it's a shoe fetish.
- 6.75** But these seem to have origins in childhood. It's very hard to know whether there's some genetic thing—why that would be is unclear—or some kind of early experience that got him excited about shoes or whatever the fetish is. So it seems to have origins in childhood. Once it gets to puberty, it seems to lock in. Basically, he's got the pattern for the rest of his life. It doesn't really go away, whatever the therapists try to do.
- 6.76** There's even some interesting data looking at the term sex abuse, sometimes referred to as any sexual activity in childhood. Sometimes it's limited to, clearly the child didn't want it. And the big arguments about—because children do fuss around and do play sex games, play doctor and things like that. Is that unhealthy? Is it bad for you? And so on.
- 6.77** But anyway, looking at children who had been victimized in sexual ways, and what was the correlation with the adult sexual health? I'm pretty sure, 9.5 out of 10 sex experts would have said it would have a bigger effect on the girls. But no, they found it had a bigger, more lasting effect on the boys. In fact, it could be argued that this is one reason for the greater plasticity among women, is if they're subject to these things, they need to get over it and be able to recover and have a healthy sex life as an adult.
- 6.78** But again, the childhood experiences seem to have a bigger, more lasting effect on the men. Sexual orientation, a lot of it seems to have a low level of plasticity, but more lack of plasticity among the males than the females. So anyway, yeah, I think the data are clear on higher erotic plasticity in adult human females, even probably starting from puberty.
- 6.79** But there may be a phase in childhood in which the male sex drive is open and vulnerable to social and situational influences. But once puberty comes in, it tends to be fixed. And as I said, once you become an adult, the man's sex desires remain about the same, other than dwindling in intensity across life, whereas women might come up with new ones.
- 6.80** All right, a couple of final questions. Why the difference? The difference, to me, is unmistakable in terms of the evidence. The reason for it is not. I sent this to psychology's, one of the top journals, and a viewer said, "Well, what about this, this,

this, and this?" And I looked up all those things, and they fit the pattern, too. So it would take an enormous amount of evidence.

- 6.81** I don't know where it would come from, to overturn the conclusion that the female sex drive responds more than the male to social and cultural and situational influences. The reason for it is much harder to say what it might be, and we can speculate a variety of things. One idea was that men have usually had greater social power, also had more physical power than women, so the woman has to adjust to the man.
- 6.82** So it would be useful to her, to be able to adjust and adapt in this way, and the man doesn't need to adapt as much to his partner. That's plausible. What that would suggest, if it's the societal thing or even a physical thing, that would be true not just in sex. That would be true in probably everything. So I contacted a bunch of researchers who studied attitude behavior consistency, because remember, it's lower in women on sex than on men.
- 6.83** So I said, "Well, when you study attitude behavior consistency on politics and religion and other things, is it lower among women?" And they said, "No, it seems to be the same." So, this plasticity seems to be fairly specific to sex. So that made me doubt this one, that it's an adaptation to male power. But again, we're not sure about this.
- 6.84** Another thought it had is that males maybe have stronger sexual desires, and that's harder to mold and shape, to change. It's a bit like having a house pet. You want house pets with very mild desires, like a dog or a cat who will learn what you want. A wild animal with very strong natural inclinations is not going to be as good a house pet. And that's more like what the male sexuality is. It's harder to tame and harder to culturally socialize and things like that.
- 6.85** For that, what I wanted to say, the test case would be, if we could find something where women want it more than men, is the male role more flexible? The only thing I could think of was the desire to raise children and to take care of them. Everybody says that is stronger among women, and probably for good long-standing biological reason.
- 6.86** And sure enough, the father role is much more variable culturally and historically than the mother role. Whenever cultures try to mess with the mother-child bond—I mean, the Soviet Union tried to get rid of it and have all children raised together because it was fair; that didn't work at all. No, no, no. The mother-child bond, a bunch of anthropologists, that's basically true everywhere.
- 6.87** The father-child bond varies in some cultures. It's not even the biological father. It's the mother's brother who does most of the the male providing and socializing and things. Even in my lifetime, just the change in the way fathers treat their sons, you see fathers kissing their sons. My father never kissed me, nor did any of my friends get kissed by their father. We got smacked right on the head if we did something wrong. You don't see that much either. Again, even historical change, the fathers were much more plastic.

- 6.88** That encourages me to think that that's the argument, that the stronger desire is less amenable to cultural influence. If nature makes you really want to do something, culture doesn't have as much room to play with. Now, it does raise the question, is the male sex drive indeed stronger than the female? I mentioned this in the paper, and a couple of viewers said, "Oh, you can't say that. That's not true." So we went in and did another research project. We said, "Let's go, all right, survey all the data and see, is it plausible that men are hornier than women?"
- 6.89** I remember mentioning this in the lab meeting, that we were doing this, and a woman, the postdoc, said, "Everyone who's ever had sex knows that." So, OK, everyone who's ever had sex, but not the authors of sex textbooks. We looked at the top books in the field. None of them said men have a stronger sex drive than women. I got a couple other colleagues to work on it.
- 6.90** One was been a young woman with a very strong feminist background, and she told me later, she hadn't said it at the time, but she said, "I was sure there was no difference, that men didn't have a stronger sex drive. But then I saw the evidence, and I just couldn't." So she changed her view. So the way we set it up is, imagine two women or two men, one of them has a stronger sex drive than the other. OK? Imagine two, one has a stronger sex drive.
- 6.91** What differences would you expect to see? Well, the one with a stronger sex drive, probably going to think about sex more often, have sexual desires more often, have sex more often, masturbate more often, have more different partners, have more fantasies, and stuff like that. So we made that list, and we then started looking through the published studies to see what they are. Basically, everything lined up. It's official, men are hornier than women.
- 6.92** Men think about sex much more frequently than women. Studies of sexual fantasies, men have more fantasies, more varieties, more fantasies with different partners. There's one study that asked these people, mostly asking college students, "So in your imagination, have you had sex with over a thousand different people?" I mean, but about 25% of the men said yes, that they had, and none of the women. I mean, this must be every woman he's met for the past five years. He's thought about having sex with her.
- 6.93** Anyway, a lot more fantasies there. Spontaneous arousal, how often you get turned on. Men report that more. Now it is a little more visible to the male. They know when they have an erection, where a woman might get aroused and not really realize it. So there's a little bit of an extra cue there. It's not totally convincing, but it does seem to fit the pattern.
- 6.94** Desired frequency of sex. I ask people, "How often do you want to have sex?" Men often report it more. There's a study of married couples. "How often do you have sex? How often do you want to have sex?" The women, the wives were having sex just

about the same amount as what they said they wanted. The men were wishing for twice as much.

- 6.95** Desired number of partners. There's a study of, I think, freshmen in college, so first-year college students. Asked them, "Well, if you didn't have to worry about disease or laws or social norms or anything, if you could just do whatever you wanted, how many people would you like to go to bed with over the rest of your life?" The average response from the women was two and a half. For the men, it was 64. Clearly, the men were wanting a much bigger variety there. Other studies find similar large differences.
- 6.96** I should mention, a lot of people in that study said one. Both men and women said they wanted to just have one. Now, these are probably people who had not had sex yet, so thinking they would like to have one, and that would be enough. But still, the range went way up for the men, where it didn't go nearly as far up for the women.
- 6.97** Masturbation is a pure measure of sexual desire because you don't need a partner or anything else. It's easy and safe and no risk of pregnancy. But all studies find men masturbate substantially more than women. I said, well, maybe women feel guilty, but no, it turns out, the men who don't masturbate are the ones who feel guilty about it for religious reasons or whatever. The women who don't do it mostly say, "I just really don't feel like it."
- 6.98** Willingness to live without sex. I mentioned the Kinsey Report, that sometimes women would lose their sex partner and just not have sex for a year or whatever, and seemed to be perfectly fine. They didn't miss it, whereas it bothered the men much more. The best data on this, I think, there's some studies of Catholic priests and nuns because there's no question of double standard. There's nothing else going on.
- 6.99** They both take a sacred vows that they will not have any sex, including masturbation, for the rest of their lives. That's who they are. The essential meaning of their lives is their spiritual vocation. So it's all the same thing. But all the evidence shows, the nuns can live up to their vows much more consistently and effectively than the men. One of the guys who studied extensively says about 2% of the male priests really are completely celibate, including no masturbation or anything.
- 6.100** And when they do do it, they do it, when they do have sex, in other words, they do it more or with more partners or repeatedly; the nun might have a lesbian thing one time or something like that, but the men do it much more. Anyway, it's much harder for the males to live up to the same ideal of celibacy than for the females.
- 6.101** My researcher came up with a list of 19 strategies, I think, were 10 ways to get sex, and 9 were ways to avoid having sex, and had people then rate them, which were more common among men or women. All 10 of the ways to get sex, they said, were more common things that men do than women do. And all nine of the ways to avoid having sex were things that women do more than men do. So again, men much more desperate to do it.

- 6.102** Also, studies, I don't think I have it up here, taking chances or expending resources or whatever for sex, men do a lot more for that as well. The study is asking, do you like your sex organs? How do you feel about your penis? And the men like their sex organs more than the women like theirs. Now you might think, well, maybe there's some difference there, but the men like the women's sex organs more than the women like the men's sex organs. So men like sex organs, both sexes, both genders, better than women do. So again, that's a sign of a greater sex drive.
- 6.103** And last, you had people rate, how strong is your sex drive? And the men rate their sex drive stronger than the women. So as I said, the evidence is overwhelming. Here and there, there would be a finding of no difference on something or other. But pretty much every study found some difference indicating stronger sex drive by men. And something on every measure we could think of, there were findings like that.
- 6.104** A couple things to point out about this. There's no clear value judgment. "Oh, wow, that's saying bad things about women, that they don't desire sex as much as men." Well, too much sexual desire is bad, too. And both men and women, on average at least, are in the middle range. So I don't think there's any value judgment to be drawn about the basic difference.
- 6.105** I mean, the problem there is, they have the opposite problems. More women suffer from, "I just don't want sex at all," and more men suffer from, "I can't stop thinking about sex," and doing bad or dangerous things because of it. But for the mainstream, for the average, they're both in the good healthy range. No value judgment there.
- 6.106** Evidence is mostly about frequency. How frequently do you want sex? How frequently do you feel sexual desire? How frequently do you think about it? The intensity, that's another way of looking at strength. Intensity, I'm not so clear about. I wouldn't make a strong decision there. The woman may have fewer desires, but they may be just as strong when she does have them.
- 6.107** And then the same goes in terms of, this is about wanting, in terms of liking. None of this means that women don't like sex or shouldn't like it or anything like that, and they may like it more. Certainly, women are multi-orgasmic in some cases, and even single-orgasm women seem to show levels of pleasure that go beyond what men might do. My guess is, there's greater variation in the amount of pleasure a woman gets per sex act than there is for the male.
- 6.108** It's pretty much always good for the male, good at about the same level for the female; sometimes it's not good at all, and sometimes it's really spectacular. But anyway, this has nothing to do with the enjoyment of it. This is about the basic desire, and so that appears to be stronger in the male. Again, this is descriptive, not prescriptive, nothing about what things should be.
- 6.109** And so what are the implications, in terms of understanding men and women for this? Well, there are a couple of things we could point out. For one thing, when society

changes, it'll be easier for the women to adapt than for the men. If suddenly sex were to become highly dangerous and we all had to stop, or we all had to start engaging in some other kind of sex act or whatever, women could make that transition more easily than the men. The male sex drive is fairly rigid and doesn't respond very well. So that's one advantage for the males. One advantage for the females, rather.

- 6.110** Advantage for the males is sexual self-knowledge. It's much harder for a woman to really come to an understanding of her own sexuality because it's more of a moving target. As we saw, it can change over time, even from '30s to '40s to '50s. You might develop new sexual interests or whatever. During the sexual revolution, I remember there were all these books for women, *Our Bodies, Ourselves*, and so on, to explain to them all these things. There weren't these books for men. There wasn't nearly as much mystery.
- 6.111** In terms of self-understanding, that's just easier for the male than for the female. Therapy, and therapy does seem to have gone this way, the pills, starting with Viagra and so on, treat a lot of male sexual problems very effectively, and that's a pure biochemical innovation. You take a pill and the problem goes away. Before that, men who couldn't get an erection, they would go to talk therapy and explain all their childhood and try all these things, and nothing seemed to work or help.
- 6.112** Anyway, we don't seem to have the same kind of pill available for women, but maybe that's—and certainly the market would be there. I mean, the drug companies would be glad to get a product like that. But it may just as well be that female sexuality is more about meaning, and so you need to talk about things, and what are your past experiences, and what are your expectations, and what do you think the norms are and your values, and re-examining all that.
- 6.113** Indeed, there's a huge meta-analysis done in England where they have extensive data about the people, and they looked at, did they have sexual problems or not? And then they do a giant factor analysis to see what that correlates with. And for men, the sexual problems were correlated with physical problems. And for women, the sexual problems were correlated with mental emotional problems, psychological problems. So that's another sign that the female sexuality is more about meaning, whereas the male's is more just a purely physical thing.
- 6.114** And then last, decision-making is going to be more complicated for women sexually, given the higher rate of plasticity. The man knows what he wants, and so he's obviously sure, yes or no, in each situation quite readily. Whereas the decision-making for the woman is going to be more complicated, changing in response to situation.
- 6.115** Anyway, as I said, the difference, I think, is overwhelmingly supported. The reason for it, my best guess, is that it's the strength of sex drive, but I wouldn't consider that a done deal yet. Again, I don't see that either is better than the other, but the difference, I think, is really useful and important to understanding people. OK, thank you.

Lecture 7: The Political Mind

At a glance — The political mind.

- 7.1 All right. So today we're going to tackle another interesting topic, politics. And a lot of people have strong opinions and views on politics, so let me state my position right at the start. I aspire to be neutral. I think politics corrupts social science, biases people toward certain conclusions. My goal in life is to try to get to the truth, and politics, you have to take sides, and I really want to understand both sides.
- 7.2 It's probably not possible to get rid of all one's biases and preferences and so on, but I do try to look hard at both sides. I stopped voting quite a while ago. Before that, I was a swing voter. I voted for plenty of Democrats and Republicans, some of each of which turned out great and some not so great. So that's my opinion. I'm going to try to look at both sides, and I'll tell you in a moment, I think the theory I came up with is one not too many people will automatically agree with, but maybe I can persuade you.
- 7.3 So let me say, too, politics and political behavior is very complex. Lots of things enter into it. This will be one theory among many. There are probably many true theories about why people vote the way they do or campaign the way they do. I don't see this as competing with, all the other theories are wrong, if my theory is right. Many of them can be correct.
- 7.4 So let me start saying a bit about prejudice. One of the classic theories of prejudice is it's based on ignorance, and that if people have distorted views of each other and stereotypes are wrong, because they just don't know each other. And so if people would get to have more contact with each other, they would get along better. The theory hasn't fared all that well.
- 7.5 For one thing, sometimes when people have more contact, they get more antagonistic. Sometimes if they're positive interactions, it can go in either direction, and as usual, the bad effect is stronger than the good effect. But another problem with that theory is that most of the stereotypes are fairly accurate. Some other scholars have gone through the data, compared people's, say, stereotypes of men and women with the traits that are actually measured in men and women.
- 7.6 And people are often really quite correct, not only in terms of what the differences are, but even statistically how big the differences are. So a lot of stereotypes are correct, but they found one big exception, which is political stereotypes. So in the U.S., Republicans and Democrats really distort how they see each other. They demonize the other. They take the most extreme examples of the other as typical for that.
- 7.7 Indeed, a few years ago, the Pew Foundation did this really nice study with a national sample. They asked Democrats and Republicans to rate their opinions on a whole bunch of issues, and they also asked them to rate, how do you think the average

Democrat or Republican will respond to these questionnaires? So they could compare, statistically, what Republicans think Democrats think, with what Democrats actually think, and vice versa.

- 7.8** And just some of it's right. Everybody distorts. They are off by a substantial amount. And so everybody distorts pretty much, but some more than others. And two groups stood out as the most distorted and most biased, in terms of the difference between what they thought the other side thinks and what the other side actually thinks. Specifically, one group, interesting one, people with high school education or less who vote Republican—they are mostly poor people.
- 7.9** Democrats would be their natural allies, but apparently they so misunderstand what the Democrats stand for that they vote against them. The other group with the maximum distortion was Democrats with advanced degrees. Now, that shook me up. These are my people. I'm not a Democrat or a Republican or anything, but I spent my life at universities and psychology departments, surrounded by people with advanced degrees who are almost all Democrats. Here and there, there's a Republican.
- 7.10** So I thought, we're social scientists. Yeah, we'll have preferences and biases, but we should be able to see the truth better than other people. But apparently, we distort more than other people. So that was disturbing, and I really set myself on a path to try to understand how at least the smart people on the political right think. Start reading Wall Street Journal editorials and things like that, because if you're surrounded by people who all think the same, then you think it's objectively true. And so you're not hearing other sorts of views.
- 7.11** So that's what I'm moving toward, trying to have an integrative understanding. I do think the one-sidedness is bad for our universities and education. Students should hear the conservative viewpoints about the positive aspects of capitalism, of two-parent marriage, and loving your country, and a few other things. It's not good for research to have everybody on one side, or for education.
- 7.12** Anyway, having half the country's views not represented in my social world of university culture was disturbing. So I really set out to try to understand both sides. And you want to understand both sides from the inside. In my career as a reviewer, I've come across many conflicts in different areas. And you have to be able not just to rely on one side's summary of how the opposition thinks, because, again, they caricature. And again, the political thinkers are among the most biased people in the population.
- 7.13** So how can you phrase their views in a way that the people who believe that would say, "Yes, that's what we believe"? So my position is, as I said, something that most people will disagree with, which, most people are Democrats or Republicans, and they kind of think the country would be better off if their party won all the elections. If you ask them, sometimes they say, "Well, maybe not all of them," but any specific election that they don't win, "Well, that was bad, a threat to democracy," and all these things.

- 7.14** So in contrast, I'm going to suggest that the formula for successful countries is to have alternation in power, between center left and center right. And most of the best countries in the world show that pattern. Now, when I say the best countries, some people say, "Well, all cultures are equal," and all that, but they're really not. There wouldn't be that much immigration if all countries are equally good. So you can look at the countries that are magnets for immigration.
- 7.15** I also looked at the United Nations Human Development Index, which ranks all the countries in the world from the best to the worst, based on pretty solid criteria, I'd say. It's economic standard of living, longevity, how long do people live, and how much education, on average, do people get? So if you look at the top ones there, a lot of them have fairly regular alternation in power, usually, again, between center left and center right.
- 7.16** The United States is a fine example. It's usually in the top 10 or 15 on those rankings, and obviously always a magnet for immigration as well. And past half-century, I think we've had six Republican presidents and four Democrat presidents. And half the time there, Congress was on the other side, too. So we really have input from both sides of the political spectrum in there.
- 7.17** And meanwhile, long-term one-party rule is generally a disaster. Some of the worst countries in the world, even things like Zaire, which has ceased to exist, Zimbabwe, North Korea, Soviet Union, and so on. There are a couple of exceptions. Singapore does pretty well, but Singapore is a sort of a glorified big city, split off from a poor country and has done quite well. And then China also had one-party rule, but they did change their economic system, from a pure socialist government-run one, to have a much more capitalist kind of marketplace.
- 7.18** I've heard a scholar being interviewed about China. The interviewer asked him, "Is the Chinese Communist Party today a Marxist organization?" And he said, "Well, yes, but kind of the same way that the Mafia is Catholic." They talk the talk, respect the symbols and all that, but in practice, they definitely depart from the doctrines. So alternation in power seems to produce benefits and seems to be a stable thing.
- 7.19** I mean, if one party had the right ideas, they would eventually win all the elections, and everybody would be happy about that, right? But maybe they both contribute something rather different, and so the alternation provides benefits. Now, you've heard me say that my understanding of the human mind is that we are cultural animals, that we evolved by nature to do culture, and it's worked very well, and that our human population is going up, even while most of the other mammal populations are declining.
- 7.20** So let's push that. Why does that work? What does a society have to do to enable a population to flourish like that? Well, culture does a lot of things, but among the most important, going back to cultural materialism, has to amass more resources, produce

more resources, than people could get working alone or that even people in an inferior system could get. And it has to share them to some degree through the group.

- 7.21 OK, it has to, I mean, if the society works hard and produces a lot of resources, but a few elite nobles hog them all, then the population doesn't thrive there either. So it doesn't have to share equally, of course. But it has to share enough so, again, that the population increases. So, again, we're measuring population and thriving of the society in terms of the population, either becoming larger or staying large.
- 7.22 Starting with food, have to be enough resources to take care of the population. It has to be—a population with not enough food is in danger of shrinking, of people moving out, or dying out or whatever. If there's a surplus, then the population is doing well and they can get larger. So they have to, again, do these two basic things, produce and store and protect the resources. I call it amassing the resources, versus sharing them through the group. So both are essential to the success of the society.
- 7.23 So the main idea that I'm going to present today is that the political left and the right specialize differently in terms of those two jobs. So the political right is focused on amassing the resources. The left is about sharing, redistributing them. The idea came partly from considering who votes on the right, and it's people who amass resources. Farmers have always been conservative, business people, managers, and so on, manufacturers, merchants who sell things, bankers who store the wealth, but who also enable growth by lending money for new projects.
- 7.24 And last, the military, of course, tends to be conservative, and their job is protecting the resources. At least that's what we call it now. In the past, the military was used for getting resources by conquering your neighbors and taking their stuff, making them pay tribute. So all the people involved in creating this stock of resources tend to lean to the right.
- 7.25 Meanwhile, the left started with the labor movement, which was redistributing profits from the owners to the workers. The supreme achievement of the left is the welfare state, which takes care of everybody from cradle to grave. That's all about redistribution, to take the money from the people who have more and give it to people who are in need and various kinds of crisis thing.
- 7.26 The left, more oriented toward helping the poor, especially by means of government. Affirmative action redistributes opportunities to other groups who haven't had as much and so on, generally favoring more government services and so on. So the left is needed for that. And so if you look at it this way—the left is about sharing the wealth, and the right is about creating the wealth—you can see both jobs make a valuable contribution to society.
- 7.27 And that could be a good reason why alternation in power is so useful in the most successful countries, because then both jobs get done. It's alternation in power or sharing power. Many governments have the parliamentary system, where everybody in

the government is supposedly on the same team. In the U.S., we don't, so we can have the President of one party and the Congress on the other side.

7.28 But either way, both continue to have input into the decision-making, and that will make sure that the resources are produced and are redistributed. And even just looking at recent history, in 2016, the Republicans took over the White House and Congress, and set about a whole bunch of business-friendly laws and policies and practices, and the economy started taking off.

7.29 And then four years later, the Democrats took over everything in 2020, and they set up a very ambitious redistribution agenda to help the people who didn't have as much and so on. As for what's happening more recently, I don't want to get into that. But you can see both jobs being done and producing a benefit for society. So again, both jobs are essential. They have to get done, one way or the other. A long-term one-party rule tends not to do it, to neglect essentially the other job.

7.30 Now, my argument would be more plausible, especially with the evolutionary basis, if I could say that, yes, human evolution, as compared to the other great apes, showed some substantial advances in both of these. Both we got better at, amassing resources and at sharing them, than our closest biological relatives, the other great apes. So I said, well, is this true? Can we find evidence like this?

7.31 Well, certainly humans came up with a lot of ways of producing resources that are very different from all the other chimpanzees and the gorillas and orangutans, things like that. First of all, we learned to hunt cooperatively, better than other animals. Some of them hunt in packs, but each one is really out for itself. It might do different things, but they don't really cooperate.

7.32 There's one style of hunting where one line of hunters will line up here and the other down here, this being downwind, and these guys will just make a bunch of noise and chase all the animals this way, where the others are waiting with spears and nets to capture them. That works. The gorillas, the chimpanzees, would never do that because the noisemakers wouldn't get anything to eat. They don't have the sense of sharing. But humans figured out, sure, we'll work together, we'll catch more prey, and then we'll share. And it gets, again, produces more resources.

7.33 Farming is another big thing. Almost no other animals do anything like farming. Here and there, there's a tiny bit, but farming occurs all over the world, and humans discovered that it's a better way of producing more food. And trade. Again, humans have trade. No other animals have any kind of marketplace or anything like that, but trade increases wealth. Economic history shows that very effectively. And again, the buyer wants to buy, the seller wants to sell. So if they can make a deal, they're both better off.

7.34 Humans have evolved new ways to protect resources as well, like more advanced military organization, things like that. But we don't need to go into that. So, anyway,

yes, human evolution did find a bunch of new ways of amassing resources. Meanwhile, did we get better at sharing? This is a remarkable paper by Sugiyama about 20 years ago that interviewed modern hunter-gatherers. The point of departure for it, he said, before human civilization arrived, just in the wild, something like 40% of human hunter-gatherers lived to be 50 years old, but only 9% of chimpanzees did.

- 7.35** Well, why is that? Are we somehow better? Do we have better hearts and lungs? Given we share 98-99% of our DNA with these apes, it's hard to argue we're just superior specimens of living things. Rather, look at how death happens. A chimpanzee takes care of itself, feeds itself. At some point, though, it might fall out of a tree and break its leg or get sick or something and be laid up, so it can't forage for itself. And so it dies. It starves to death or dies of thirst.
- 7.36** Humans have those same kind of mishaps, but when it happens, others share. And so, others will bring you food and water and nurse you. In Sugiyama's research, he was interviewing modern hunter-gatherers. He said half of them had had an interruption, an injury like that, for at least a month, so they couldn't forage. They wouldn't have been able to feed themselves for a month, but they were still alive because other people had shared with them.
- 7.37** So this is a huge advance in the psychology of humans. This is something to be proud of. We cooperate and share food to our benefit much better in a way that the other apes, our closest biological relatives, never catch on to. So if you think half the population of these groups would have been dead without the sharing, well, that effectively doubles the population in one generation. I mean, multiply that, you can see why there are 8 billion of us, as opposed to something like 300,000 chimpanzees left in the wild.
- 7.38** So, anyway, point is, yes, humans did evolve to share better. Another big innovation was getting the father to be a provider, to bring food for the children and for the mother of the children, so another important departure. So, yes, I would say it's clear that human evolution did make us better at both of these crucial tasks, better at getting resources, and better at sharing them.
- 7.39** All right, so let's move along to applying this to understanding politics. I'm going to use the terms the right and the left, just to go over terminology. The right and the left come from the seating at the Estates General in France in the 1700s, when the economy and the political situation was in trouble. So the king called together representatives of everybody, although I'm not sure the peasant farmers had a say.
- 7.40** And it's just the ones who wanted to keep things the way they were. Mostly the aristocrats sat on the right, and the people who wanted serious change sat on the left. And that at the time included the business people, because the business climate was extremely inhospitable and needed political reform. I mean, that particular historical

episode, of course, was a disastrous failure, soon followed by the French Revolution and the Terror. So it didn't solve the problem.

- 7.41** But the right and the left, that is stuck with the right being the people who are more conservative, "Keep things as they are," and the left saying, "No, no, no, we need serious change." The term liberal I won't use as much because it's really reversed its meaning in the last century or two. It means literally freedom, and it used to mean freedom from the government. So liberals were against that. And this is why, in a number of countries, the conservative party is called the liberal party.
- 7.42** I was just in Australia. The Australian liberal party is their conservative party. Netherlands, I think, is another one that the liberals are conservatives. So there's that confusion about that term. Progressive generally means more government intervention to make progress. And so there was a progressive movement in America around 1900, and then it somehow lost credibility. So it adopted the term liberal, but it's back now. And so liberal, progressive, is often what's that.
- 7.43** Socialism and capitalism. Socialism is basically the government should run the economy. Capitalism is basically the free market. Socialism, the extreme form is communism, in which there's no private property. But most socialists accept private property, and even the ostensibly communist countries generally do. And the last, the term fascist. As far as I can tell, nobody identifies with that. It's sort of a term of abuse that both the left and the right call the other fascists.
- 7.44** There's a book recently I read, looking at policies of the actual fascist governments in Italy and then in Germany, and said, well, they're kind of leftist policies compared to modern things. But somehow the left has come to make a claim that fascism is really something that's on the right, but it's not really there. I'm not going to talk about fascism at all. It's basically a term of abuse that nobody identifies with.
- 7.45** All right, so let's take this basic idea and apply it to a variety of political phenomena. I mean, for one thing, it's clear that the hostility between the left and the right has been increasing in the U.S. There's a paper in the journal *Science* by Eli Finkel and a dozen colleagues or something, where they tracked survey data over time, asking people, how much do like or love your own party, and how much do you hate or dislike the other party?
- 7.46** And over the decades, love for your own party has remained very stable. People feel about the same way toward their own political party as they did back in the '70s and '80s. But hatred for the other party has gone steadily up and surpassed it, and so now that's the stronger motive. Indeed, my impression of the last election, when I asked people why they were going to vote the way they did, they always started with how awful the other person was.
- 7.47** And so I had a Substack about this. "Whom did you vote against" was really the thing. The people I know didn't have a whole lot of enthusiasm for the candidates they were

voting for, but they had a lot of enthusiasm for not wanting the other person to get to the White House. And they both made good cases, I thought. It was not the best selection of candidates.

- 7.48** So the best system for sharing resources would be, everybody gets an equal share. Equality is a very high priority for people with leftist political views. And of course, equality means everybody gets an equal share. But for producing resources, well, then you need the free market. It seems to work better. And markets work by incentives, and incentives create inequality by definition. That's what an incentive is. It's a reward to somebody that other people don't get. That's why you exert yourself.
- 7.49** And so, that's not some solvable trick. That is basically a definitional fact. So rewarding merit, rewarding people who produce more and contribute more and so on, that's going to create inequality. And that's why the left isn't so crazy about that sort of thing. So that's one fundamental conflict between the left and the right, that if you push for more and more equality, you have to push against incentives and meritocracy, things like that. And indeed, the economy seems not to work as well.
- 7.50** There's an economist who said, "Really, there are three kinds of economies in the world— capitalism, socialism, and Argentina." I see, if you follow them, there's even Argentina is trying to get real these days. But setting aside the poor Argentines, capitalism believes in the market to allocate resources. Socialism allows the government to do it and enables more equality. I think capitalism clearly has a better success record in terms of producing economic growth, but in terms of, it also produces inequality, which is why the people on the left are more skeptical of it.
- 7.51** All right. Another reason for conflict is, back in the hunter-gatherers, most people participated in both jobs, both in amassing the resources and sharing them. They didn't have much wealth that they could carry along. There wasn't much private property at a few positions. And I keep coming back to them because, remember, that's what we evolved for. If human civilization has existed for 150,000 years, 140,000 of it, most of us were these nomadic hunters and gatherers.
- 7.52** So, yeah, there are a lot more of us now, and we've gotten away from that, and we don't have that lifestyle. But our psyche is designed for living in that, in that kind of system. And so everybody worked together to amass resources, whether it was hunting or gathering or cooperating in the group hunt. And then the fruits, the result were shared. There was no way to store the food anyway, so it was shared through the group, not necessarily equally, but close to equally.
- 7.53** So there, people could understand both jobs. But now, different people do the different jobs. And so you don't have as much direct contact with what the other side is doing, and you have less of an understanding of how it works. There's a paper a few years ago showing, how much do people understand about economics? It was published in the

stereotype and prejudice section of our top journal, Journal of Personality and Social Psych.

- 7.54** And they showed, interestingly, people have negative stereotypes of profits. Everybody can understand how quest for profits can motivate individuals or companies to do bad things, to deceptively advertise, to sell inferior goods, to exploit their workers, damage the environment, and so on. Well, people have a very hard time understanding that profits are what drive a market economy, that the basic way to make a profit is to make life better in society, to produce goods or services that people want and that they're willing to pay for.
- 7.55** So there is this widespread one-sided view, and it's unfortunate because people need to understand both, that profits are vital to enabling the system to work. If the company doesn't make a profit, it goes out of business, and whatever good or services it produced are no longer available. So yes, they really do improve society and make life better, although they do have the dark side as well and can motivate people to do underhanded things.
- 7.56** Privilege is a term that's been in the public discourse recently, and this is another basically unsolvable dilemma. And I looked on the origins of the idea of ownership. I did this when I was working on my book on the self, because you have to have a self to own something. It implies that there is a self there. Well, the original idea all over the world when ownership emerges, if you make something, it's yours. With the hunter-gathers, if you made a bow, say, for a bow and arrow, yeah, that would be yours. If you made three of them, somebody else would come and ask you for one, and you pretty much had to give it to them.
- 7.57** So there wasn't that. But still, the assumption was, if you create something, you grow it, you whatever, that it's yours, and that way you accumulate resources. So letting people keep some of the proceeds seems to be a vital incentive to drive society forward. Again, society has to produce resources. So, letting people keep the fruits of their labor, I mean, taxes take a substantial amount, but letting them keep some is a vital motivation. When you take that away, then the productivity falls off.
- 7.58** I relate, in the Soviet Union, ostensibly living by communism, they had a joke, "We pretend to work, and they pretend to pay us." But you didn't get any benefit from it. So it was hard to motivate people to work hard, whereas in capitalist societies, which America is still kind of one of the leading examples, people work really hard. And if you start taxing away, at times, certain tax rates have been 95%.
- 7.59** I don't know if you know the Beatles song, Taxman. "Here's one for you, 19 for me." Well, the Beatles quit being British citizens because of the 95% income tax, which they've, in the interim, have changed. So letting people keep some of the proceeds of their work is a vital motivation, and society doesn't work as well without it. But then

they pass their wealth onto their children after they die. Now, that's unfair. It gives some babies an unfair start over others. The baby hasn't done anything to earn it.

- 7.60** This is an interesting dilemma for me because I've had opinions on both sides. I went to a suburban public high school, and when I got to college, there were people from all kinds of different schools, and some had much better facilities, and some didn't. I thought, well, that's just really unfair. We're all just kids, and we all should have the same resources and the same opportunities and the same thing. I was ready back then to support the Soviet style, that all children should be raised in common, so that they all have the same opportunities and the same resources and the same educational stimulation and so on.
- 7.61** Then when my own daughter was born, I suddenly had this strong desire to take care of this little baby and give her the best start in the world that I could. And so the idea that, no, I shouldn't do anything for her that no other parent will do, or that any other parent will be unable to do, rather, that, well, I didn't care about that. I wanted to do the best for her. Comes up with over and over.
- 7.62** I mean, piano lessons. Piano lessons teach you self-discipline, which is a good trait in life. They increase your lifelong appreciation of music. Even if you don't become a musician, you understand it better. They even seem to make you smarter. Studies, they've tried to get rid of that, controlling for everything they can think of, but it still seems to be some effect that learning music is good, particularly, I think, for your mathematical intelligence.
- 7.63** So, sure, every kid should have piano lessons, but obviously some don't, and some parents can't afford it; some kids just don't want to take it and so on. So I had this personal crisis. I believed we should raise all kids the same with equal opportunity, but I also wanted to take care of this little girl, and I wanted to have her piano lessons. Well, she did have piano lessons. Everything else I could do for her. But I did have my own liberal conscience crisis in doing that for her.
- 7.64** And as Francis Fukuyama, I think one of the most brilliant political scientists, points out, the unfairness of passing wealth along increases over time. So maybe the kids of the first generation learn from their parents and will even enter the business and increase it or whatever. But by the time you get to three or four generations, these ones have nothing to do with it, and it's increasingly unfair.
- 7.65** So, some people say, well, we should just tax away, 100% inheritance tax. I heard a political scientist say, "You should tax inheritance at 125% because the kid has already benefited from the family wealth," but that's never going to happen, obviously. But the thing is, if you take that away, that also removes the incentive. Caring for your children is a deeply rooted motivation in psychology and wanting to do things. And it becomes a point of, why should I work this hard if it's not going to benefit my children?

- 7.66 So again, privilege, it just means you have parents who gave you a better start in life. And that's just going to happen in a large country with a free market and inheritable wealth and so on. Some kids are going to have an advantage over others. And again, the left is more uncomfortable with that because it produces inequality, and the right just shrugs and says, "Well, that's part of what motivates people to work hard and produce an advanced society."
- 7.67 Another issue is deserving this. Now, we think, again, it seems like Democrats and Republicans disagree fundamentally about a lot of things, but actually, they agree on a lot of things. If you ask them, they both say, yes, the government should help the truly needy victims of natural disasters. Well, of course, the government should come and take care of these people.
- 7.68 The left and the right also agree the government shouldn't really support just lazy freeloaders who don't want to work, who want to just live off the money made by others. Where they disagree is to what proportion of the welfare recipients are the truly needy versus the freeloaders. But the value approach is similar, and there's a lot of common ground there, which is often not noticed.
- 7.69 Personal responsibility. Well, some people's misfortunes are brought on by themselves. If they develop a gambling addiction and blow all their money, well, you can ask, is it the taxpayers' job to have, say, what you earned, go to rehabilitate these people who have gambled away their savings and gotten themselves into trouble? Anyway, some people, yes, make bad choices and get themselves into trouble. Others are totally random.
- 7.70 I mentioned the natural disasters and things like that that happen to people. Through no fault of their own, they find themselves in a desperate situation. So there is a difference. The political right emphasizes more personal responsibility, that people should take responsibility for their own lives and manage themselves and so on. But the left emphasizes more external causes. They think poor people are poor through no fault of their own. They're just victims of the system.
- 7.71 So the victim-of-pressure narrative is very popular on the left. The people at the bottom of society are just the result of social forces they couldn't do anything about. And both sides are right about many cases. It's hard to say exactly. There are a number of borderline ones and things like that, but again, both the left and the right have valid points. And so they may disagree about how money should be spent and redistributed.
- 7.72 And so the degree to which the poor are responsible for being poor is highly variable. And so it's hard to come up with a simple moral formula and say, "This is what needs to be done." And so the left is more sympathetic and says, "Well, it's mostly not their fault, so we should help them," and the right says, "Well, now they should get busy and work hard and take care of themselves."

- 7.73** So one of the basic differences again between left and right, going back to the original, the Estates General meeting in France, was attitudes toward inequality. The left is against inequality. You can always reduce inequality. You can always make more. Look at the groups who don't have the resources. How can we transfer more to them?
- 7.74** The right is more comfortable with inequalities. Well, it's an unavoidable result of an incentive system, of a free market and so on, and we need it because it makes people work harder. Performance gets better and the quality of life gets better when there's a competition. Again, both have valid points.
- 7.75** What happens with the one-party rule? Well, when the right takes over for a long time, there are not that many modern cases where you have a long-term political right on government, so they're a little bit hard to see. But what happens is, resources are produced, but an elite few hog them. The ancient empires were like this, where the royal courts lived off the taxes while most of the farmers were barely living at subsistence level. And what were they characterized? Lack of innovation.
- 7.76** The empires, the rulers, didn't really want things to change very much because that could be destabilizing. Even finding new products, new inventions, and so on, there was no interest. Some of them would regard that as threatening. And certainly for the farmers, there's no need to work harder. The taxman will just take more of your work, or if you came up with a new plan for irrigating the farm or something like that, again, there was no incentive.
- 7.77** So there is stagnation there and persistent inequality, which, as Fukuyama said, increasingly unfair over the time. So long-term one-party rule by the right is dysfunctional in a variety of ways. The left, we have a lot more cases of long-term leftist rule, even happening, what's happened in Venezuela in this century, rather rapidly shows a serious—I mean, Venezuela was a pretty rich country, but the left takes over, focuses on redistributing.
- 7.78** So when Chávez first took over there, everybody was happy. Well, everybody except the real rich, because suddenly everything was shared and so on. But they don't focus on creating more resources, and after a while, the shelves were empty. There was less stuff. People used to be moving into Venezuela, migrating in. Now they're migrating out. They're crossing the same border in the opposite direction. And Soviet Union, lots of other places show the same problem of long-term leftist rule.
- 7.79** All right. Attitudes toward capitalism, free market. Again, the left skeptical of all these things because they increase inequality, which they do. But they also increase resources, and that's why the right likes them. And they do that as well. Both points are valid. And what we end up getting with a more long-term leftist, socialistic government is more equality, although there's usually a very rich government; party members live well. But across the rest of the population, there's equality, but it's equality of being poor.

- 7.80** So there's that philosopher who said, well, we should morally evaluate a country by, how do the bottom 10% live? How well do the bottom 10%? I think that's a profound idea because, OK, so maybe there's huge inequality. Actually, I just read a book that the inequality in America is way overstated, as of the slippery statistics. But nevertheless, we have quite a bit of inequality. But our poor people still live pretty well, compared to even the dukes and earls of medieval Europe, who had no dental care and no television and no cars and everything else. So, the right's, again, more accepting of inequality.
- 7.81** A couple more things to look at. Time. On the left, the present is bad. It's got inequality, injustice, everything. It's just not fair that some people have more than others. The past is where the crimes occurred that produced the inequality of the present. And the future, we must move toward a very different future that will be much, much better. So ideally, this would be wonderful to create this utopian system, very different from what we have now.
- 7.82** In contrast, the right sees the present as pretty good. There are some curious data on happiness. The conservatives are generally a bit happier than the liberals. The right, people on the right, are happier than on the left. But you can understand this is why; it's more accepting of life is pretty good now versus life is terrible now. That may be a contributing factor. There may be other things going on, too, and it's not a huge effect.
- 7.83** But the right sees the past as where the systems were developed, and they realize it was hard to get to where we are now, which is pretty good. So the distant past was bad when there was chaos, and then we created these systems that are working well now. So don't mess with them because they fear we could go back to the chaotic past.
- 7.84** And so the future—it's not true that conservatives don't want any change. They mostly want incremental change, though. They think things are pretty good. We can tinker with it and make specific improvements here and there. But they see the future as basically a slightly improved version of what we have now. Whereas, again, the left is looking at the future as something very different.
- 7.85** Morality. Well, everybody believes in fairness, but there are different concepts of fairness. One is equality. Everybody gets the same. The other would be proportionality. This used to be called equity, but now the left has taken over the term and changed its meaning to mean more equality. To illustrate, imagine you and three friends agree to paint somebody's garage for a thousand dollars. And you're going to be there Saturday morning at 8:30.
- 7.86** And so 8:30 comes, morning, and you and one other person are there. The third person comes around 9:30, and the fourth doesn't get there till 11:00, and is sorry, and helps with painting, but then has to take a bunch of phone calls. And then at 3 o'clock, the garage is done, and the latecomer says, "I got to go," and leaves all the rest of you to clean up. Sowhat's fair? Should you each get \$250, a quarter of the amount, equal? Or should the ones who worked harder, who did most of the work, should they get more?

- 7.87** I mean, both of those qualify as a version of fairness. And we have both in our society. Families tend to be fair. So in the garage example, if it's you and your brothers or sisters, you're probably going to divide it equally. If it's a bunch of strangers, the ones who did most of the work are going to say, "Hey, you didn't earn a fair share," and they split it unequally there. So families share things equally. Nobody says the three-year-old shouldn't get any food because she didn't produce anything, contribute to the resources.
- 7.88** On the other hand, no corporation pays everybody the same salary. They all pay more, higher salaries to the more valuable workers who contribute more. In practice, both fall short of the ideals, but generally that is the pattern. So again, incentives drive competition, produce innovation, and so on, but they increase inequality.
- 7.89** Another aspect of morality is the difference between rights and duties. The people on the political right, it's a little ironic here because that's the left who favors rights, but the people on the right, well, let's start with people on the left—they want redistributing. They want resources to be moved from one to another. So they insist that people have rights, and the government should give these people these things. And we have a right to health care, and the list of inborn rights gets longer and longer.
- 7.90** De Blasio was a Democratic candidate for President. He had this great line. "There's plenty of money out there. It's just in the wrong hands." So that's a pure redistribution attitude toward things. Meanwhile, on the right, there's much more of an emphasis of doing your job, doing your duty. Everybody has to do their job in a system in order for it to work.
- 7.91** If you have an assembly line and one guy is out having a cigarette and not putting the wheel on, and you have cars come out without that wheel, that's not going to work. So, more emphasis on duties, and the researchers on morals have shown that people on the left and the right have somewhat different moral values. Everybody believes in fairness, but perhaps in slightly different terms.
- 7.92** When I was working on this, sometimes for fun, I go looking online at entertaining websites. And I saw this one of children's funny answers to test questions and assignments. I saw this one, I thought, "This kid's parents are Republicans." He said, "I earn money at home by—" and he's supposed to fill it in. He says, "I don't. I am a freeloader." I mean, it looks like the kid is about seven years old from the handwriting. Who teaches their kid that you're a freeloader, that you've got to grow up and do your duty and contribute?
- 7.93** Of course, I was hoping there'd be another one I would think the parents were Democrats. There was an assignment, "What do we call the science of classifying living things?" And from the handwriting, I guess it's a girl, "Racism." OK, her parents are Democrats. So, again, they both have positive values and so on. Again, the deserving

poor question is hard to settle down, what proportion are deserving, what proportion are freeloaders.

- 7.94** I would make a prediction from our theory that could be tested in the future. Who will be the biggest voters on the left? And it would be poor people in rich countries because they could benefit from both. If everybody else around you is rich and you're not, hey, come out and share the wealth. That's good. Poor people in poor countries can't ask for much because, well, there isn't much. And rich people, they may support redistribution for moral reasons or whatever, but it doesn't directly benefit them. So I would speculate that the poor people in rich countries would be the most reliable voters on the left.
- 7.95** A couple more things there. Pandering to voters. What do you do when you get desperate and you're afraid of losing the election? Well, the right say their focus is on amassing resources. "We're going to reduce taxes." So that's one of the perennial campaign slogans on the right, "so you can keep more of your own money rather than the government taking it." A few other things—we need a strong military to protect the resources and so on. The left, in other words, when they need to pander to people, more free stuff. The government will do more for you, increase the goods and services, expand the entitlements and so on.
- 7.96** Now, what is the sweet spot in policy? The left tends to think that the government should regulate more to prevent abuses of the environment and so on. And yes, the government has to play that role, and when the market is totally unregulated, you get all kinds of abuses. But then too much regulation really stifles productivity. Companies, I mean, if the company survives by selling toothpaste, making and selling toothpaste, well, the price of toothpaste has to go up to pay for all those people who are dealing with making reports to the government.
- 7.97** Many companies have buildings full of people who do nothing but fill out forms for the government to report on this or that. And all that has to come out of the profit. Otherwise, the, so that the price goes up. Meanwhile, on the right, they want to set the market free. The government should regulate less.
- 7.98** But then you start to have more abuses, of not getting away with paying people, minimum amounts of doing it, or abusing the environment, dumping your waste into the river, and things like that. So it's really hard to find the exact right spot there. And there might not be one. It might be a trade-off. And so going back and forth with alternation in government might get it as good as it's going to be.
- 7.99** What are people upset about? The left complaints that, "Well, the top 1 or 2% of the population own the majority of the wealth in the country. Clearly, that's unfair," and so on. But the right would come back and say, "Well, look, the top 1 or 2% pay most of the income taxes." A lot of people don't pay any income tax, and some don't pay very

much. And so in terms of the money the government's getting, it's coming from a small segment of the population. Again, both have valid points.

7.100 All right, let me wrap this up, why the misunderstanding increases. Again, hunter-gatherers didn't have wealth to preserve. They couldn't pass on things to their children. There wasn't any accumulating inequality or anything like that. But once you had farming, some farms worked better than others. Some farmers worked harder than others. And some people were able to amass and store up wealth more than others.

7.101 So modern society, again, not everyone works together. Stable arrangements, some groups depend on others for transferring resources. And the producers resent what they see as the free riders. Meanwhile, the low-status groups object to the producers, to justify transferring more resources to them; as elsewhere, both sides have valid points.

7.102 So to conclude, the right is about producing resources. The left is about sharing them. Both jobs are vital for a successful society. And their respective optimal policies and ways of thinking about morality and time and everything else are quite different, as being suited to the focus on creating and storing resources versus sharing them better throughout the group. The most successful societies, again, seem to alternate in power or share power, so that both of them gets done.

7.103 The misunderstandings have increased over time, as the redistributing and the producing tasks become more separate and less understood to each other, and also as the market systems create more inequality because of more wealth unevenly distributed. So final thoughts. Jonathan Haidt has written about that, and had a great line that "both the liberals and conservatives are right about the things they care about." So I'm echoing that both sides have valid points. They just have a different focus and emphasis.

7.104 And they may demonize the other as being against what they're in favor of, but they're not so much against what the other is doing, as in favor of what they are doing themselves. And each one promotes one of the two crucial jobs that a flourishing, successful society needs. I don't think the Democrats and Republicans are going to make friends and agree on everything, but perhaps they could agree to disagree respectfully and understand that, even though they're their opponents, they, too, have a valuable and important contribution to make.

7.105 Our elections in America are generally so close, nationally, that it can't be that half the country is bonkers. So this would be a way, at least, to respectfully disagree, see what they have to offer, and let the voters, which they generally seem to do, oscillate between the left and the right. Thank you.

7.106 So in your presentation or your lecture, it's mostly, you're pretty neutral, and it's just like, that's just seeking the truth. But in your expertise, what is the perfect middle that could last a lifetime and preserve, I guess, our country?

- 7.107** Yes. Well, we've been searching for the perfect system. I don't think there is another system. I mean, animal societies have fairly stable social arrangements with the alpha male on top, and then another one takes over and reproduces the system. We keep tinkering with our system, thinking to make it better. There aren't that many different systems. The idea that there's a better one that hasn't been tried yet, that nobody's thought about, seems rather implausible. So this is about as good as things are going to get.
- 7.108** Living here in Western civilization is very good. I mean, if you look at the history of the world, it's like winning the lottery. I sometimes say, anyone born in America after World War II should never complain about anything. The more I read about other places and especially other times in the past, I think just having a toothache for months with no way to treat it, that was a common feature. I think it was Ben Franklin said, "If they could ever solve the toothache, everyone would be happy ever after." And so we've solved the toothache, but we're not happy ever after.
- 7.109** You react to the things more locally. So, as far as I can tell empirically, the most flourishing countries have this regular alternation or sharing of power between center left and center right. And it's true in the U.S. and Great Britain, and Germany and France and Australia. So that seems, it's obviously an imperfect system, but I think it was Churchill said, "Well, it's a bad system, except compared to all the others that have ever been tried." So I don't really see a lot of room for things, for the system as a whole, to change, to get better.
- 7.110** And maybe we should just be comfortable with the alternation in power, and not every time the other side wins an election, say, "Oh, that's the end. Our country is going to be destroyed." America is going to be here in 100 years. We've had bad politicians before. We certainly have some now. But the system is strong enough that they don't really ruin things too badly.

Lecture 8: Free Will

At a glance — Free will.

- 8.1** All right. Well, one of the classic philosophical questions is, do people have free will? It's also debated in religious contexts and others. And if so, what does it mean? Now, over the years, I've talked to many scientists who say there could be no such thing as free will. They don't believe it at all. And I've talked to others who are fine with it, who accept the concept. And I find I agree with both of them because they're not talking about the same thing.
- 8.2** So I push them to say what they understand by it. Different concepts of free will emerge, and the ones that the naysayers object to, I also don't really go along with. And yet there are others that are more plausible, which means perhaps both sides have some valid points. Here's an example of a negative one, Montague, writing in one of the top biology journals.
- 8.3** "Free will is the idea that we make choices and have thoughts independent of anything remotely resembling a physical process. Free will is the close cousin to the idea of the soul." It has nothing to do with evolution or the brain or anything like that. Well, defined in that way, I don't really believe in that either.
- 8.4** On the other hand, there was an interdisciplinary panel test with defining a bunch of key terms for a grant competition. And they understand free will is basically the ability to act differently in the same situation. Well, that, I think, is much more scientifically viable, the idea that we could do things without involving the brain; that there would be psychological processes independent of the brain, very dubious.
- 8.5** But behavioral flexibility, being able to change your behavior to act differently, that's highly adaptive. Even it's a basis of learning, you could say, is that you can learn that one response is better than another. So how does behavior happen? We're talking about the system by which action is controlled. So the brain sends out nerve signals to tell the muscles how to move the bones. That's what behavior is. And then, of course, talking as well.
- 8.6** Flexibility in that system is adaptive. We'd be able to learn, explore, avoid being killed, and so on. Anyway, if there's any kind of free will, it comes just before that happens. It's some kind of mental process that changes what the brain tells the muscles to do to move the bones. So I said I've talked to scientists who believe in free will and others who don't, and you'd think they'd have very different psychological theories about how behavior happens. I mean, that's such a fundamental difference, right? But they don't.
- 8.7** They understand, they believe in the same theories about attitudes and behavior and decision processes and so on. So the difference is not as big as we thought. As I said, both sides agree roughly about how the behavior comes about as a result of mental

processes. And both sides agree, even more remarkably, that the human system by which action is controlled is qualitatively different from anything else in nature, that humans do things and make kinds of decisions and so on that we just don't see in other species.

- 8.8** So they agree there's this remarkably new mental system for causing behavior. They mainly disagree about whether it deserves to be called free will. OK, so for me, that's not the interesting problem. It's just an argument about terminology. The scientific challenge is to understand this remarkable new system and how it works. So I'm trying to take this ancient question of free will, take it instead of looking at it as a religious or a philosophical thing, just look at it as a scientific problem.
- 8.9** We're not going to do an experiment to prove whether free will is the right term or not; that's not going to happen. But we can do research to understand how human behavior is controlled and initiated by the mind and understand what is special about it. Again, even assuming it goes way beyond what is seen elsewhere in nature. So again, my take is that the scientific problem is to understand how this thing works.
- 8.10** If you want my own opinion on it, I think, if we discovered this today, we wouldn't call it free will. We'd probably come up with another term. But given that that term has been used for centuries, probably we should stick with it. For my money, the free part is less of a problem than the will part. Freedom is built into psychology, and I'll explain more about that in a moment. But we don't have a psychology of will.
- 8.11** I think William James, in his first psychology textbook in 1890, had a chapter on the will. But none of the modern textbooks have a chapter on the will. There are no theories about it and so on. So the will part is clearly metaphor, but the free part, I think, is real enough. One aspect of this is, some of the skeptics say, well, if it's going to be free will, it has to be independent of any cause, any external factor.
- 8.12** This is Robert Sapolsky, the biologist, had a recent book arguing against free will, and he defined free will that way. And hardly any philosophers, even the pro-free will ones, believe that. My view is, why would that even evolve to be able to act in a way that's independent of the environment? We evolved to be able to deal with the environment better. And my assumption, we're cultural animals, we've got to be extra responsive because culture is a very complicated environment. So free will is designed for that, not to act in ways that completely ignore and disregard what's going on in the world.
- 8.13** And so another point, a lot of the debate sometimes, free will, is it yes or no? Do you believe in it or not? Does it exist or not? So it's a dichotomous question. But we know in psychology, almost everything is on a continuum. The idea that there's an absolute yes or no probably doesn't make any sense. All we have to say is, some actions are freer than others. And certainly people have had that experience. I assume all of you had. You can remember an action you had where you were free and one where you were not so free, often as a result of the circumstances that you were in.

- 8.14** So all we have to assume is that some actions are freer than others. And then what are the mental processes that contribute to the freer actions? So we don't even need either extreme. It can be a continuum where there's no absolute total freedom and there's no absolute total un-freedom. I mean, even if somebody has a gun to your head, you can still say no. It would cost you. But you can argue there are no situations in which there absolutely is no freedom whatsoever, except maybe when you're unconscious.
- 8.15** So, all we need is the middle part of the continuum and say some actions are freer than others. So, oh, let's understand those. And freedom in terms of being free from external causes, I don't see any point in being free from all external causes and factors, but maybe you can override some particular causes. Like, someone trying to quit addiction, for example, there's a causal impulse, with the impulse to take the drug or gamble or drink or whatever. And the person can become freer to override and resist that.
- 8.16** So we definitely can stop some causes from influencing us. Why we would want to stop them all? I'm not sure. So think of it that way. Freedom is on a continuum, and we don't need either extreme. Why do people talk about it? Probably the biggest thing is moral responsibility. Freedom, concepts of freedom have been linked to morality throughout the history of philosophy.
- 8.17** Kant made this argument very strongly, that free action is essentially moral action because you're overcoming your animal nature and doing what, following a higher calling of right and wrong, which he believed was built into the mind. And so, sure enough, social systems are better when people behave morally and accountably, and they accept responsibility for their actions and follow the rules, and they get along.
- 8.18** Morality is essentially a set of rules to enable people to get along and live together and participate in the society that will have the kinds of benefits we've been talking about. There may also be a dignity aspect. People want to believe they're free, being told that, "Oh, everything you do, you're just programmed by nature and your genes and your environment, and you don't really have any choice."
- 8.19** So people don't like thinking that, well, I thought I was making a choice, but you weren't really. It's a bit insulting, but that's part of the appeal to the people who oppose it anyway. "You think you're making choices, but you're not really. It's all determined far in advance."
- 8.20** What more precisely does freedom mean? Well, we did a study some years ago where we asked people to tell a story from their life where they acted of their own free will and another story from their life when they didn't have their own free will. And so we took all these stories together and coded them for various aspects to look for what were the things that systematically differed between the two sets of stories.

- 8.21** So the free actions, the free will ones, had happier endings. They led to more positive things, which goes with the idea free will is adaptive. It's useful. It helps you get what you want and need. Likewise, reaching goals. There was more achieving goals in the free stories. It's an interesting and relevant point that these were the long-term goals, in particular. Short-term goals, doing what you wanted or needed to do in the immediate situation, those were found in both the free and the unfree stories.
- 8.22** But pursuing things over time, which is something, again, people do, and do much better than other animals; we have a sense of the present contributing to future things. We can make sacrifices in the present for the future in ways that animals, except for hardwired fixed-action patterns, don't really do. Free will benefits yourself. It advances your individual self-interest and also is better for the group that you belong in.
- 8.23** So the free stories were more associated with that. Lack of freedom meant doing things that are not so good for you. Conscious thinking was there in the free stories. They talked about—we didn't ask them about these things. This is what they spontaneously wrote in the story. So we just looked at whatever they mentioned. But thinking consciously about the alternatives and debating what to do, that was a part of free action and was missing from the unfree stories.
- 8.24** Morally good ones, again the connection to morality. When you had more freedom, you're more likely to do something morally positive. Resisting external forces and influences, obviously succumbing to external forces, as it makes you feel not so free; being told to do this, but instead you're going to do something else. That showed up in the stories about freedom.
- 8.25** And a relative absence of other powerful people. Obviously, powerful people tell you what to do, and then you don't have as much freedom. Those are some of the things that show up in free will. And this is in practice in people's own lives and in their accounts of what they recall as free actions. Now, so let's develop a scientific theory of free will. It's sometimes useful to think in advance, well, what would this theory look like? What should go into a theory, a scientific theory, about free will?
- 8.26** So the goal, again, is to explain this evolutionarily new kind of action control that humans have, that other animals might have limited degrees of, but basically, they don't have. So things will go into it. Conscious thinking, as I just mentioned. Logical reasoning, to figure out what's the best thing to do. Self-interested calculations, reasons, deliberate choice, considering moral economic factors, a lot of things like that.
- 8.27** Scientific theory is not going to have anything supernatural, so souls may or may not exist. And they may, for all I know, if they exist, may have influences on behavior and so on, but that's not part of a scientific theory. Scientific theories work with natural and, in social science, cultural factors. But it's not going to postulate any supernatural ability. As I said, I'm going to think of freedom on a continuum, not an absolute yes-or-no deal.

- 8.28** And some people say, well, freedom has to be, you're exempt from causality. No, scientific theories are causal theories. So free will is just going to be another kind of cause or probably a cluster of causes. All right, so that's kind of the description of the suspect we're looking for in building a scientific theory of free will. And one more point —has to be compatible with known facts, including, let's say, evolution.
- 8.29** I might ask then, OK, why would free will evolve? Like I said, Sapolsky's version that means you act independently of anything in the environment or any cause or any external factor, what would be the evolutionary basis for that? Why would that lead to someone surviving and reproducing better than others? That doesn't make any sense.
- 8.30** Well, in a more reasonable approach, the philosopher Dennett said, "Well, free will isn't worth having unless it helps you get what you want." I think that's a profound and basic point. It has to serve your wants and needs in some way, and in a way, presumably, that contributes to better survival, better reproduction. So that's what increases the quantity of life, and more recently, the quality of life as well.
- 8.31** Free will might evolve to help the animal to take care of itself in a complicated social system, especially as I said, culture gets more and more complex. Human societies are far more complicated than that of the other great apes or, really, of any other animals. So, maybe to deal with the complexity of society, you have to have more brain power and more capacity for steering your action in useful ways.
- 8.32** So in a sense then, the need for free will originates in the social system. It's not something that the individual mind particularly needs, but for the social system to work, again, people have to behave morally. The system only works if most of the people follow most of the rules most of the time. So they need to self-regulate their behavior, resist antisocial impulses, and so on. And human society, complex as well.
- 8.33** I've been talking to this guy, Mark Moffett, who's written a number of books about societies across all the animal kingdom and so on. He said, "Very few of them have groups larger than about 100 members." Maybe it's some ants, the few small bugs, have giant societies and humans. But we go about it differently. He said ant societies have a variety of interesting features. They have different roles and so on. But there are no bosses. No ant tells another ant what to do, whereas that's really built into human society.
- 8.34** And that's a place where a certain degree of free will would be useful, as the boss has to make the decision and then pass it down. Think of a military hierarchy chain of command. And so the middle management has to implement the rules, being instructed from above, but decide exactly how that should be done, so that they can tell the lower workers exactly what to do. So these cooperative hierarchies, again, pretty much unique to humans, but requiring a fair amount of autonomy to how to apply the rules and how to make use of them.

- 8.35** Now, I've spoken, you've heard me talk about the human being as a cultural animal. Our culture is our biological strategy to solve the problems of survival and reproduction. The essential human traits are the result of biological adaptations to make culture possible—communication, cooperation, being two of the biggest human innovations. And so free will, I would look in that context, this new action control system that we evolved that's so much more sophisticated than what the other apes have.
- 8.36** That's probably a result, again, to enable us to create and function in these advanced systems. So cultural systems benefit from them. What is it that then, you can think, what does the cultural system need individuals to be able to do? And my best summary of it is, it's two things. They need autonomy, which is self-government. You've got to guide yourself and make yourself get up and do things. And responsibility, in other words, accept the consequences of your actions.
- 8.37** So that also means probably being able to obey the laws, unless you are willing to accept the danger or the risk of the punishment or whatever that would be. So responsible autonomy, I would say, is one operational definition of free will. It's what we needed to evolve to make these systems work. I sometimes put up this slide, "Free will is for following rules." People are surprised by that because they think, well, freedom means being able to do whatever you want, with no rules. Wouldn't that be maximum freedom?
- 8.38** Well, I suppose so. But my point is, you don't need any special mental powers to do that. Any animal out in the woods is able to do whatever it wants with no rules. What you need a more advanced action control system for is to operate in a system that has lots of rules, and still take care of yourself and get what you want and need, while working within the system and, as I said, obeying the rules.
- 8.39** So what it might be, some of these features then, of this new capacity. Let me start with time. This is an important one. The humans have a much stronger and more extended sense of time than, really, any other creature. Sometimes people say, "Well, what about squirrels burying nuts? Aren't they planning for the future?" But the research suggests, no, they're not. They have a hardwired impulse that "this looks like a good place to bury a nut." And then, come midwinter, "oh, this looks like a good place to dig for a nut."
- 8.40** There are some studies where researchers went out, observed the squirrels, went out and dug up all their nuts and took them away. And the squirrels didn't change their action at all. They didn't understand. If somebody came into your place and took all the food out of your refrigerator, you would, "I got to go restock the food," otherwise, the future would be thrown off. But they don't really understand that. They have evolved natural impulses to bury nuts and to dig them up later.

- 8.41** Humans can base action on things in the distant past and the future. The human self—again, most animals live in the immediate present, in the environment where they are in the current place, the here and now, so to speak. But humans can behave in ways that extend far into the future and far into the past. And we understand our lives as a story, and that the present moment is a moment within the story, a result of things that happened in the past and moving towards something in the future.
- 8.42** So this extension across time, it helps make the social system stable, also enables the individual to benefit. Delaying gratification, something very hard for other animals, but humans do this all the time. I mean, even getting an education is usually not done for the pleasure of the moment, but because it pays off in the long run, that you'll be better down the road and you make sacrifices in order to achieve that, saving money as well.
- 8.43** Moral reputation, also very important to living in society, that other people won't trust you if you do bad things, and you'll lose partners, and end up alone, and in various kind of trouble. So often you're tempted to do something that's morally wrong, and so people will not do it, will hold themselves back from doing it. That's a good application of free will. You don't have to act on every impulse. If it's morally wrong, you'll get in trouble, damage your reputation, you'll be sorry in the long run.
- 8.44** Humans thinking about the future alters the behavior in present. We're able to think about, "Well, if I do this, then this will lead to this and so on, and down the road it will produce this result. And is this what I want? If it's not what I want, then I shouldn't do the action in the present." So you simulate the future, bring the result of the simulation back into the present, and use it to guide behavior, especially if you think of several different options and which one will lead you down the road to where you want to be.
- 8.45** So, being able to envision multiple futures, this is something humans do pretty well, and use that information to guide present behavior. Arguably, that makes us freer than animals who cannot do that, so that's one reason I think the term free, of the free will part, is at least a reasonable scientific designation. So again, the time extension is an important first step. That's one piece of the theory.
- 8.46** Before I go on to others, there are two side issues I want to address briefly before I'll continue about those. First, concretely, what are we thinking of? What kinds of things are in practice, where an agent needs this new power of free will? So I'll mention a couple.
- 8.47** One is a moral agent. Again, you're making actions that have moral implications, taking responsibility. Hopefully, you're doing the right thing. But if you're not doing the right thing, you're understanding why it's immoral and understanding the risk that you're taking, which might be legal or reputational or whatever. But moral choices, again, are one important human version of free will, of which we see is not completely absent in other animals, but much is diminished compared to humans.

- 8.48** Another, economic agent. Humans have marketplaces. I think I've mentioned Matt Ridley's point that trade goes back over 100,000 years in human history. We've been trading for a long time. No other animals do that. But for trade, you need an advanced agentic power of the brain. You have to calculate, how many of this will I trade for that? In the last few thousand years that we have money, but it's a recent invention to improve trade. But swapping stuff is age-old. Again, you have to calculate your self-interest, what is good for you, what is worth it, and make the deal, and make the other person trust you.
- 8.49** And third is an information agent. And this one isn't as intuitively obvious, but we spend a lot of our time trafficking in information. One thing we do in society, it's what we're doing right now, so people talk to each other. They share information. The information agent is not just passing things along, it's evaluating it, it's criticizing it, it's arguing about things. Sometimes it's spreading false information. For some reason, people do that, too, of course.
- 8.50** So, deciding what to communicate, how to deal with information that comes into you, and that you may or may not pass on to others in some form, this is a big part of human activity, largely absent from other animals. Indeed, a recent review article somebody did that intentional teaching is pretty much specific to humans. Now, animals have a lot of learning, and they model things, so they copy what others, but the learning is mostly by imitation, the social learning.
- 8.51** Deliberate teaching, there are one or two borderline cases, but basically it doesn't happen. Whereas all human societies, the adults teach the children things. And most countries in the world have a lot of schools where this is done systematically. But even the ones who haven't gotten to that point, simpler societies, there's plenty of teaching that goes on. So again, very important to humans and very important for culture. This is how we do things, learning how to live and get along within the culture.
- 8.52** So all these three help the system to flourish and help the individual to flourish within the system. OK? They're all communication and cooperation. And the other side issue I need to address here to forestall it, because it comes up over and over again in scientific debates, is free will random action. I think that's what Sapolsky was talking about, that you could act in a way independent of any cause. Well, that would be essentially random action.
- 8.53** Scientists think that might be because they're looking for causal patterns in behavior. And so, well, if the cause doesn't apply, it would show up in your data as random action, as error variance. They think that's essentially what it would mean, but it doesn't make any sense. There are even studies showing that people see a robot acting randomly, they'll think it's freer than the one that follows the same path all the time. So there is some conflation of randomness with being free in the sense that it's being unpredictable.

- 8.54** But there's very little payoff, biologically, evolutionarily, for a random action. Culture, in particular, does not welcome randomness. I mean, would you hire a plumber who does random things to your plumbing? Or would you want to have a surgeon do random things, operating on your child? No, you want them to do their job and do it exactly the way they're supposed to. So the organized system, which is what culture is building, it's the opposite of randomness.
- 8.55** Again, free will is for following rules and following best practices and proper procedures, and not for acting in a random system. Indeed, going back to what I put up a moment ago, the three examples of agents, the moral agent who did random things would soon be in a lot of trouble, right? Because you'd be doing some immoral actions to mess up your reputation. The economic agent who made random decisions, well, you're going to lose money pretty fast. Others will take advantage of you.
- 8.56** The information agent who'd spread random information would soon not be trusted. So randomness is destructive. Randomness is not the model for a scientific theory of free will. It's not for that. It's for understanding a complex social environment and choosing the best option that will lead you. Enlightened self-interest is what's best for you down the road. All right.
- 8.57** OK, so let's get back to building the theory of free will. Motivation is one of the foundations of psychology. And so I'd say, wanting come first. Again, mentioning Dennett's point, free will is only worth having if it helps you get what you want and/or need. Motivation, I think, is the beginning of the psyche because it's the closest to survival and reproduction. You want things that help you survive and reproduce, and thinking and emotions and so on are often there to serve motivations, to help you get what you want.
- 8.58** People want food and safety and sleep, and then sex and connection, need to belong, things like that. So the agent is there to enact motivations. Without motivations, you wouldn't have it. One of my favorite lines in one of Steven Pinker's books, as he was talking about back in the late 20th century, there was a whole series of movies about computers taking over the world, the Terminator movies with Schwarzenegger in there. Anyway, there are a whole bunch of these.
- 8.59** So apparently, it was a real fear, and it was something concerned, and the movies made a lot of money. But Pinker says, "Well, now we got computers all over the place, but they don't show any sign of wanting to take over." I have probably more than a dozen computers in my house, if you count the phones and things. But they're not taking over. They're not oppressing me. But Pinker says, "Why not? What would they want? More surge protectors?"
- 8.60** Not being alive, they don't have desires. They don't have motivations. They're not concerned with surviving or reproducing. Computers don't even mind when you shut them off or even throw them away. So it's being alive, not wanting to continue being

alive, wanting to be comfortable, wanting other things—that's what gives rise to the need for agency. And people try to take over power over others because it will help them get the things that they want. But inanimate objects without desires, there's no point in having free will.

- 8.61** Next point is, from being a social psychologist, we tend to look at the situation, whereas the philosophers and theologians tended to look at the person. They go, "Free will must be some kind of power inside the person," but the multiple options are there in the situation. And this simplifies somewhat the theory. You don't have to think, well, if your brain in its current state is the result of causal processes, how can it steer those into different directions? How can it split the stream of causality in different ways?
- 8.62** It doesn't have to have this metaphysical power to do that. It just has to recognize the situation offers multiple opportunities, and then be able to size them up and pick the best one. So the possibilities in an important sense are out there, and the free will capacity is just the ability to see them and act on them, capitalize on them for advantage. And that, I think, simplifies the scientific explanation quite a bit.
- 8.63** So the multiple possibilities are there. You can call it the matrix of maybes, the multi-maybe future. And so the mind recognizes them and capitalizes them. And this seems to be something that's very specific to humans. There was a cool experiment done a couple of years ago, where they had a tube, and they would put a treat into the top of it. It would fall down the tube, and fall out, and fall into a hole, and be gone unless you caught it. And so the subject would have to catch it.
- 8.64** Now, the subjects in this were two-year-old humans, three-year-olds and four-year-olds, and then adults, chimpanzees, gorillas, and orangutans—so our smartest biological relatives against young human kids. Anyway, everybody could learn to do this, could put out their hand. Well, then they put out another tube that had two openings on the bottom. So randomly, it would come out this side or that side. Well, then, if you put out your hand, you're only going to get it half the time, unless you do what this girl does, is prepare for both.
- 8.65** Now, all you have to do is understand that it could come out this side or that side, so I prepare for both. The two-year-old children couldn't do it. They just used one hand, so they only got half the treats. But the three-year-olds got it, and the four-year-olds got it right away and got it every time. There were 20 trials that they did it. So somewhere between two and three, which is usually around the age, developmentalists say, that that's when children really take a big leap beyond other animals. They were suddenly able to do this.
- 8.66** None of the apes could get it. These are the smartest biological relatives we have. None of them could figure out the idea of just using both hands. And in fact, they even said, out of the 20 trials, once in a while, an ape would put out both hands and get the treat,

so it solved the problem. But then it would go back to guessing with one. It didn't realize that it had solved the problem, even though it had stumbled on the solution.

8.67 To me, this is a profound finding—the authors are Redshaw and Suddendorf—because it means the capacity to understand multiple possibilities in the same situation. This is a big cognitive advance that humans have over other animals. And of course, it's a real important component of free will. If you don't understand that there are different possibilities to choose to act on, you can't act nearly as effectively.

8.68 What else do we have to say? So one more point, as social systems become more and more complex, they do create more and more possibilities. So it's not just the physical environment offers multiple possibilities. As we use our free will to create more complicated social systems, we create more possibilities.

8.69 I think a good example is the university course catalog. So the fall semester, there's a list of courses that a committee has worked on coordinating, many suggestions from many different people about the courses they want to teach. And the whole purpose is to allow different students to select different courses, because they don't want everybody to take the same thing.

8.70 So it's a giant matrix of maybe, created, and it's a very adaptive one. It's very useful because different students at the university need different courses depending on their different career goals and interests and ambitions and so on. Anyway, this is something my philosopher friends say that they haven't really devoted a lot of thought to yet, and this might be something they want to explore more. But our complicated social systems do create more possibilities.

8.71 Consciousness. So conscious thoughts. In the second course, I'll have a whole lecture focused on that. But consciousness is this feeling that you have when you're awake and you lose when you're asleep. And people assume free will involves conscious thought, conscious choices. There are basically no theories of unconscious free will. So insofar as we have free will, it's something that the conscious mind does.

8.72 And as I said, that's true among the philosophers. It's true among ordinary people's impressions, too. They think free will has to involve conscious thoughts. Now, that does raise the question, which has been disputed for a long time, do conscious thoughts cause behavior? This is something that's been argued for a long time, and there are people who say, no, consciousness just follows along and watches what happens. But it's the unconscious doing all the activity.

8.73 I don't really agree with that. The skeptics are really impressed by these studies by Libet and his people, showing that you're supposed to make an arbitrary random motion, like moving your finger or your wrist, and you have a clock. And so you know exactly when you decided to make that movement. And then they had brain scans, and they noticed that the brain scan started as activity. The brain activity started before the time the person said was when I made the decision.

- 8.74** So the skeptics said, "Aha, the brain is already doing it. The conscious decision is after the fact." Others of us just looked at that and said, "Well, duh. The brain has to do something to create a conscious experience. So of course, it has to get started earlier." And Libet himself didn't really—he thought the idea might be initiated, but then you could still change your mind and not act on it. And they didn't really explore that very much. Anyway, those studies have been discussed and, to some extent, debunked, but they certainly don't argue at all about free will.
- 8.75** So with a couple of colleagues, they said, "Well, then that leaves open the scientific question, do conscious thoughts cause behavior?" So what kind of evidence would we do? Well, the logic of experimental design, if you randomly assign positions on the independent variable, which would be some conscious thought—tell some people to have this thought, other people to have that thought, or don't tell them anything, or this conscious feeling or that conscious feeling—then you measure a behavior.
- 8.76** If the behavior changes, then, QED, the conscious thoughts have caused behavior. That's how the logic of experiments work. Experiment with random assignment is the only real reliable way to show causation. So we set out to see, is there evidence in psychology? Have there been experiments that manipulated conscious thoughts and had some people have this one, some people have that thought? And did they then behave differently as a result?
- 8.77** And the answer was yes. All kinds of them. I don't want to spend a lot of time on that today. But mentally simulating, thinking about doing something makes people more likely to do it. Mental practice has worked in just about every sport that it's been tried, which is several dozen. It's no substitute for physical practice, but mentally, practice does really help you get it better.
- 8.78** Anticipating, thinking, planning—people who do that, their behavior is different from the people who didn't do that. Thinking about the past, the studies have people go over something that just happened and think about it in a different way, revise it or whatever. Again, that changes their behavior. Logical reasoning depends on conscious thoughts, and people who engage in conscious thoughts then can behave more adaptively and more differently, based on the reasoning.
- 8.79** And I did some of those experiments, too, where we did in our lab, that the unconscious can fake it. It wants to be logical, but it can't really do it. If you're distracted, your response to logic problems is really just chance, whereas thinking consciously enables people to use logical reasoning. Think about other people's perspectives, that changes behavior. Like in negotiation, they encourage people, OK, imagine what the other person wants and try to understand their perspective, leads to a better result in negotiation.
- 8.80** Thinking positive things about yourself. Some studies do that. Just what's important to you or why are you a good person? Things like that. That reduces defensiveness,

changes behavior in a variety of ways. So this goes on and on. We have a longer list of those. But anyway, the important point is, yes, I think it's clear beyond a doubt, that conscious thoughts do have significant causal effect on behavior.

- 8.81** Now we want to go, how does this happen? We went back and looked at all the findings there. So one thing to notice is that the effects are indirect. I think it's fair to think that the unconscious is really what executes it. Consciousness thinks what should be done and then hands off to the automatic system. I mean, most obviously, when you move your arm, the proximal cause is the nerve impulses traveling down your arm from your brain. You're not aware of any of that. So that all happens unconsciously.
- 8.82** But thinking about the future, thinking about what your opponent is doing on the other side of the woods or whatever, those things do influence behavior. But again, this is thinking away from the immediate control. Direct control of muscle movements is not the function of consciousness. We had no proof that even making the choice is conscious. I think it probably is. But it could be that you review all the things consciously, and then the unconscious picks the best answer, and then launches it into consciousness so you know what you're doing.
- 8.83** But it could be. And conscious and unconscious are generally working together. They're not competing systems, as people sometimes think. Also, the impulse to action. There's very little sign that that originates in consciousness, other than somebody telling you what to do because that comes in consciously, and then you do it.
- 8.84** But a lot of impulses, you want to go get a drink of water, that originates in the body somewhere and comes up then into the mind, "Oh, I'm thirsty. I should go get a drink of water." And then you act on it. So it passes through consciousness on the way; it doesn't necessarily originate there. Even so, many diverse and empirically strong evidence of consciousness-causing behavior.
- 8.85** Meaning. All right, for this, I could go—I have a slide with a lot of stuff, but let me tell a story, it's a long lecture, and this will sort of illustrate it. Religion is an important part of life for many people and an important activity that they engage in. So I'll tell the story of my father's religion. He was born in Germany and grew up Catholic. Everybody in his town was Catholic. There may have been a few Jews or Protestants, but basically everybody was Catholic. So he was, too. I don't know that he thought that deeply about it or cared much about it.
- 8.86** Anyway, and then his timing was unfortunate. He got caught up, ended up as one of Hitler's teenage soldiers, sent into battle, and had his 18th birthday in a Russian prison camp. So he wasn't really feeling so good about the society that he'd grown up in. And after the war, went to the university and met this young American woman who was visiting. And they decided to get married and move to the U.S., but she was a Protestant, and his parents didn't want him to leave the Catholic Church because they were committed to it.

- 8.87** But my mother mostly did whatever he said, but she really thought they should be Protestants, not Catholics. I think what swayed him is, he finally got a job at the Standard Oil Company, and he noticed that the guys working in the gas stations were Catholics, but the top management, they were Protestants. So he wanted to be at the top. I don't know how strong he felt about religion, but he did want money. And so that made him think, "Yeah, I want to be one of these guys."
- 8.88** So he went along with my mother's suggestion, and they became Protestants, sent their children, my sister and me, to Sunday school at Presbyterian, painfully boring dumb things, which I gradually rebelled and said, "This doesn't make sense," and so on. And so we started having discussions in the family, and my mother stuck up for Christianity, but my father defected, and was joining me in pointing out all the problems and inconsistencies that we found in the doctrines were being taught.
- 8.89** Now, notice already, he switched from Catholic to Protestant, not because he had some new insight into the Trinity or the virgin birth or any doctrine or something. It was just, these were the kind of people that he wanted to be. But he did make a choice. He didn't go to church much, mainly Christmas and Easter, which was the best chance to meet other big shots from the company he worked for, because they would go on those days, too.
- 8.90** And so we left it there, and I went off to college. And for a while, we had a couple of fights and everything. We didn't talk for several years. Then we kind of made up and made friends again. And I was home visiting one time and I was surprised to discover that he'd become a minister. "My goodness, why are you a minister?" And he hadn't really gone to divinity school or something. It was a mail-in deal. I mean, today it would be like doing it online.
- 8.91** And he'd gotten his doctor of divinity from some cheap place where he just paid a fee and took a test, which had a few questions about religion, but also a lot of questions about the tax laws, because there are big advantages to having your own church. So he got my mother to join in, and they founded a church, which meant that all kinds of their expenses were tax-deductible.
- 8.92** So religious organizations enjoy this because they like to drink wine. I mean, they weren't drunk or anything, but they liked to have good wine. And wine is a legitimate religious expense; the Catholic Church uses wine in its sacraments. So I mean, if the government then is, if it's tax-deductible, it means the government's subsidizing your wine, paying half the thing, you might as well get the good stuff. So, always importing good wine.
- 8.93** They like to go on vacations and travel, and they realized, well, if we printed up some brochures about our church and just handed them out here and there, then the trip was tax-deductible. Anyway, there was a whole bunch of things, saved enormous

amounts of money. Again, not because he'd had some religious revelation or anything, but it was because the tax laws provided advantage, and money was his big goal.

- 8.94** Anyway, eventually, he gave that up. I don't know quite what the ending was, but my point is, he's making these meaningful changes in his life, but based on what other things going on in the social system, what kind of people he wants to be with, how he can save money instead of giving it to the government, and making decisions in that way.
- 8.95** Anyway, so the point from meeting is basically, ideas can cause behavior. And again, this is a pretty distinctly human thing. All kinds of complicated ideas, like religion, like certain political movements, like laws and other things, can then affect how people decide to act. And all this gets incorporated into the brain processes before the brain tells the muscles how to move the bones.
- 8.96** A couple more points. Free will is no free lunch. I've spoken about ego depletion and limited amount of energy, that exerting conscious control takes more energy, takes a certain amount of willpower. It's extra work that the brain is doing to enact free will, as opposed to just responding habitually or automatically. But that means that free will, you're not going to be free all the time.
- 8.97** Jean-Paul Sartre, in his youth, said something about, people are condemned to freedom. Everything they do is always free. I'm told later on, he looked back at that and said, "I don't know what I was thinking when I said that." But the way I would look at things, no, it can't possibly be all the time. A lot of actions are habitual, are automatic, are just following along with the crowd or whatever. So, really making a free decision takes more energy, takes more willpower. And so you can't do that indefinitely.
- 8.98** So, free will is going to be a sometime thing. It's a capacity that we have, that we're not making use of all the time. The same thing goes with learning a new task and automatization. The mind is designed, you learn something consciously, and then the unconscious gradually learns to perform that same action. I'm giving the example of learning to play tennis. You have to pay a lot of conscious attention to how to move your feet and hold the racket and all that. But after a while, that becomes habitual.
- 8.99** Same with, say, learning to play the piano. You have to pay attention exactly how to use your hands, where are the keys, which ones produce what. But once you become an expert, the automatic system does that, and you don't even have to pay attention. In fact, if you start to pay attention, often it messes it up.
- 8.100** In terms of operation of free will in day-to-day processes, we've got a couple lists of specific kinds of things. Now, I have to point out, psychology likes to make lists, but we have no way of assessing whether the lists are complete. Back when I was going to be a philosophy major, I took a couple semesters on symbolic logic, and the second one was all about consistency and completeness of systems. I'm not sure I could reproduce the symbolic arguments today, but I've always thought about that pair.

- 8.101** And in psychology, we're pretty good with consistency. We can say, well, this finding contradicts that finding, or this idea is not compatible with that. But completeness, we have almost no idea. I was, for a time, giving lectures or giving talks at some of the same conferences with the great Albert Bandura. So I went to hear his talks, and his talks were lists. He'd say, "There are four causes of moral disengagement." And he'd list the four and talk about them. How does he know there are just four? How could he say that there isn't another one?
- 8.102** So I'm going to give some lists, but I have no argument that these lists are exhaustive. So we have two lists for free will. One from the self-regulation, ego depletion work. We're going to get the processes there. The other is executive function, which our colleagues in cognitive psychology have looked at things as well. And just briefly to go through these, the four main phenomena have been connected with ego depletion, so self-control. And this is actually how I originally got into topic of free will.
- 8.103** I'd been studying self-control, and when then it started to show that the same processes applied to self-control as decision-making, then I thought, "OK, well, we're on to a bigger phenomenon there. It's not just self-control. And well, maybe that's what they're talking about, is free will." And sure enough, philosophy books that talk about free will, they use examples of self-control and rational choice. They just didn't have any way of showing that those two are really related.
- 8.104** And that's where the psychologists can come in and say, "Well, yes, we can show that, after people exert self-control, then their choosing is different, or after they make a lot of choices, then their self-control is weakened." The other two, planning and initiative—to take, respond actively, rather than passively—those would be there. Those four, I mean, that's a pretty good quartet of psychological processes. So if you want to delve into how free will works in practice, those would be a good place to start.
- 8.105** Meanwhile, executive function, there are different lists depending on which theorists you read, but they're all rather similar. And so, they would add a few other things like switching back and forth among tasks. Animals aren't that good at doing that, but humans will do it. They'll work on something and leave it and go back and work on it tomorrow, and in the meantime, do other things.
- 8.106** So switching among tasks, that's an executive function. Working memory is keeping things in your memory, in your mind at present. They talk about flexible thinking. Again, that goes with our notions of free will, managing time to get things done, on the clock. That's another. Anyway, I think all these could go as in-practice versions of free will.
- 8.107** All right, I'm coming to the end of our time here. Let me talk about one or two things. I was sitting in my office some years ago, and I got this email from this student in Germany, saying he'd been doing these experiments on free will and free choice, and his advisor didn't really know the literature, and would I help him write it up? Now, as a

professor, you get these crackpot things coming in periodically. Somebody says, "Listen, I've solved the world's problems. I need you to help me share the information." So I said, "OK, well, we'll see."

- 8.108** But I took a look, and what he had done was really quite brilliant. He had had people, so he was looking at, when do people feel free versus not free? And he had had them make mostly hypothetical choices, but highly involving ones, and he varied features of it, like, you should be freer. So here was kind of the general theory that came up, borrowing from what the philosophers and others had said. So if you have more options, obviously you're freer. If you have one option, you're not free at all. Two options is somewhat freer.
- 8.109** So he had people make decisions with a lot of options or only a few. Or there are multiple reasons to choose this or that. A deadline is constrained on freedom. So he had some make a decision with a deadline, others with no deadline. Anyway, this is a very elegant theory. It's always nice to have two competing theories. So the other one we could come up with, a kind of very simple-minded one. Since this is the process, we thought of the outcome theory, that maybe people just feel free when they get a good result and don't have to work very hard.
- 8.110** So looking at his findings, though, the outcome model won every time. So people report feeling free when they get a good result. All those process things, having more options made it more difficult. Now, we came to think people just don't really understand what freedom is. It's not that the philosophers are wrong. But again, people have the assumption freedom should enable you to get what you want. And so if you get what you want and it's easy, that's better.
- 8.111** There's even one choice where you had to choose between two equal candidates or two unequal candidates. If they're unequal, then they felt free. If they're equal, they felt less free. But two good ones versus two bad ones, it's basically the same dilemma. But they felt a lot freer choosing between two good ones than two bad ones—the only difference being, you're going to get a good person if you're choosing between good ones than if you're choosing between bad ones.
- 8.112** So, more uncertainty, conflict, difficulty, made people feel less free. And I had one where they played an online soccer game and pretended—he was running these in Germany— and pretended they were the coach, choosing the players for the team and playing against their arch-enemy, the British. And then they won or they lost by random assignment. And then they decide whether to change or keep a different lineup for the rematch, because in soccer, you have a home and away pair.
- 8.113** And they felt freer if their team won, even though that was random, than if they lost; and how much effort they put into and stuff like that, that didn't really get it. Free will is about the psychological capacity to act differently in the same situation. That is a

scientifically viable definition, and again, one that would be, fit with evolution. It's very adaptive to do that.

- 8.114** Also, freedom is relative. We can think of freedom on a continuum. We don't have to have absolute complete freedom as being independent of everything else in the world or exempt from causality or anything like that. It evolved, presumably, like all the other distinctively human traits, to enable us to participate in culture. And so the moral agent is good for culture. The economic agent is a big part of culture. Information, culture is full of information, so how you handle information, that's there, too.
- 8.115** There may be more, but those are at least three important examples. It's not about being able to act randomly. Culture doesn't need that. That's not all that adaptive. There might be one or two things that randomness is occasionally useful for, but it's clearly not the main purpose of human free will. I mentioned contributing issues, that this marvelous new mental capacity is able to integrate across time, to act based on calculating what the future will be like with this or that and comparing different alternatives.
- 8.116** Incorporate meaningful ideas into the causation of behavior, whether it's moral rules or religious rules or just the laws, in fact, or even economic possibilities and so on. Free will depends on conscious thought. There's no unconscious free will, but conscious thoughts do have powerful causal impact on behavior. It is costly in terms of, it uses energy to use your free will. But that's really powerful and advantageous for helping you get what you want. But it just means we don't use free will all the time. We save it for when we really need it. All right. Thank you.
- 8.117** Something I was thinking about with yesterday's lecture and then listening to a lecture on free will today, you were talking about guilt as an emotion, but it's also something that we learn from. It's a mechanism of learning. You're discussing free will today. I'm sure you've heard a lot of talk about AI. Is there, because machines don't have the ability to feel emotion of guilt, right, and that's a mechanism that we learn—
- 8.118** That's probably true. Yes. They do apologize. I haven't used AI that much, but I know other colleagues who are doing it a lot. I want to say, they ask it for a bunch of information about some topic because it's a really good research assistant. And then they looked up the references and they didn't exist.
- 8.119** They said to the AI, "Hey, you made this up. You lied to me." And it's, "Oh, I'm so sorry." And then it gave up new responses. But it made up those, too. So it didn't really learn from feeling guilty. It didn't really feel guilty. It apologized the second time as well, when she complained to it. But yeah, there's no emotion there, and certainly the constructive benefits of guilt won't be there.
- 8.120** So the psychology of our mind doesn't really translate to the AI that we created, if that makes sense?

8.121 Yeah, well, not entirely. Again, artificial intelligence doesn't have any connection to the real world. It just looks at what people are saying and makes up stuff that looks like it would fit in with the conversations for what people are having, which is why, again, it will make up references. I had the same experience with a colleague. We were trying to get a bunch of references for a point for a paper we were writing. And she said, "Yeah, you'd better look them up." And they didn't really exist. It gave us very precise references that were plausible, but it didn't exist.

8.122 So, yeah, there's a lot to happen there. Before we were studying, before we can start to say it would have free will, I mean, it's an interesting question. What would it have to be able to do for us to say it has free will? It would at least have to first be able to reprogram itself, right? Now, there's some steps in that direction. That could be moving there. Now, whether that's the same thing as humans do, that's a harder argument.